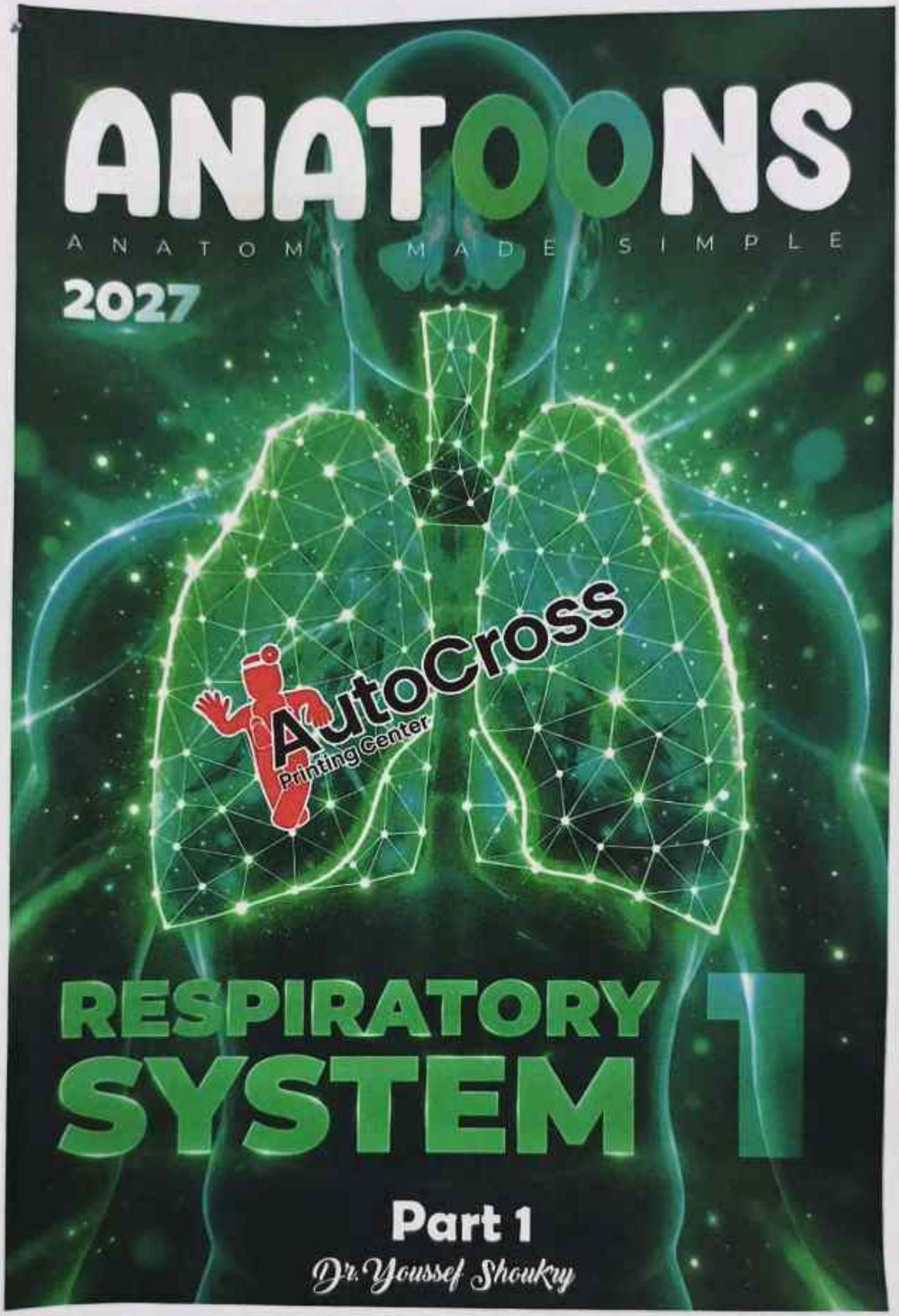
ANATOONS

ANATOMY MADE SIMPLE

2027



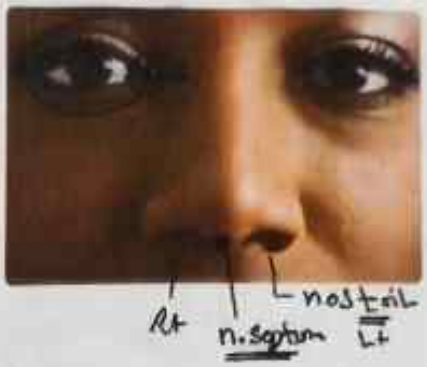
RESPIRATORY SYSTEM 1

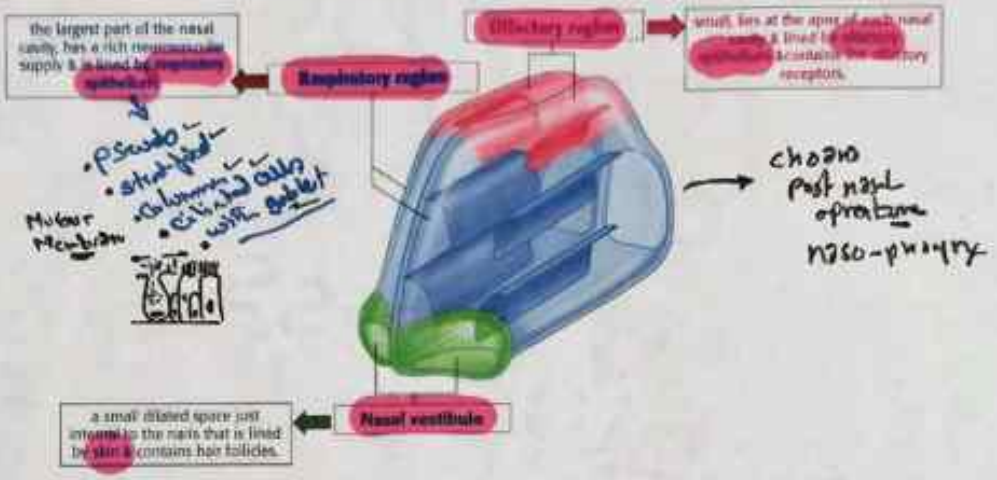
Part 1

Dr. Youssef Shoukry

The Nose

- The Nasal Cavity
  - 2 — Rt, Lt
  - 2 op. — E — choana → naso-ph.
  - 3 Regions — Resp, olf-ct.
  - 4 walls
- Orientation





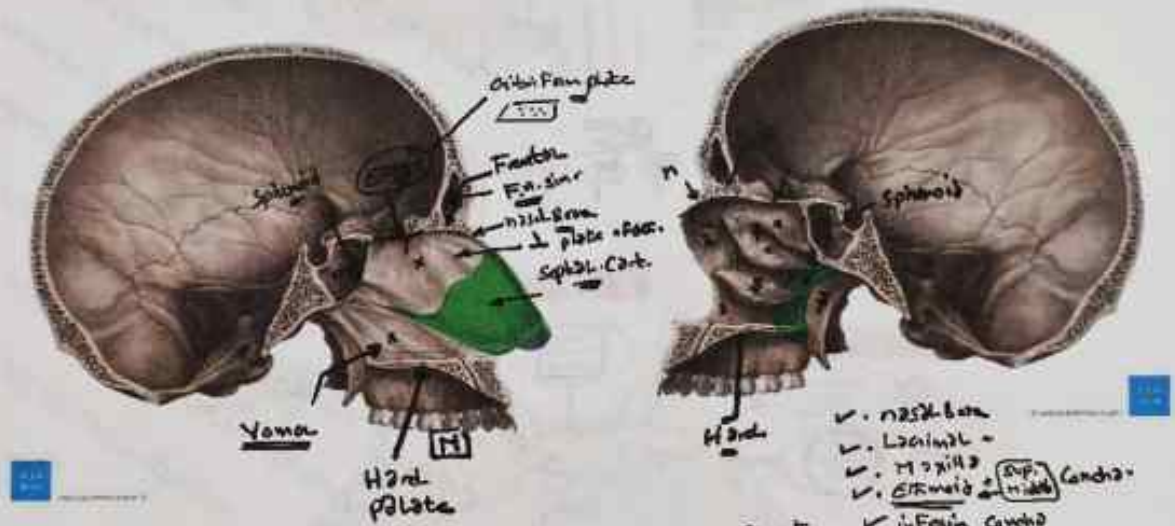


Boundaries : 2 openings + 4 walls

nostril
choana

- **Roof**
  - Frontal
  - Ethmoid • Cribriform plate of...
  - sphenoid
  - nasal
- **Floor** = Hard palate
- **Medial wall** (nasal Septum)
  - ⊥ plate
  - Vomer
- **Lateral wall**
  - nasal
  - Lacrimal
  - Maxilla
  - Ethmoid
  - inF
  - palatine
  - pterygoid

Bones



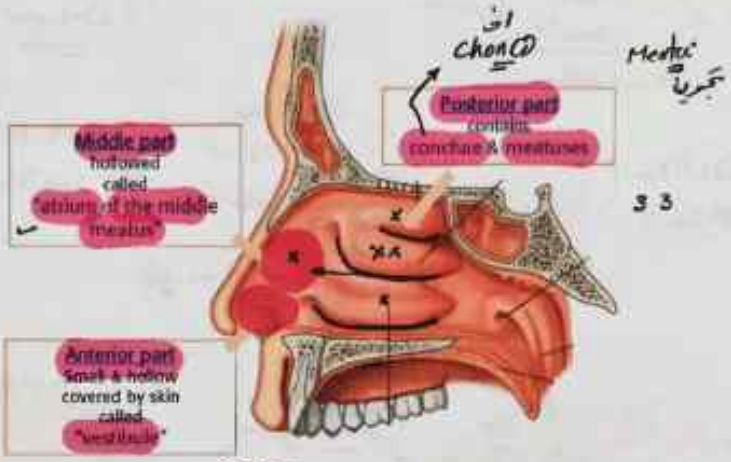
- nasal Bone
- Lacrimal
- Maxilla
- Ethmoid + Sup. Middle Concha

separate
- inFerior concha
- palatine
- pterygoid plate

Lateral wall

Essay map

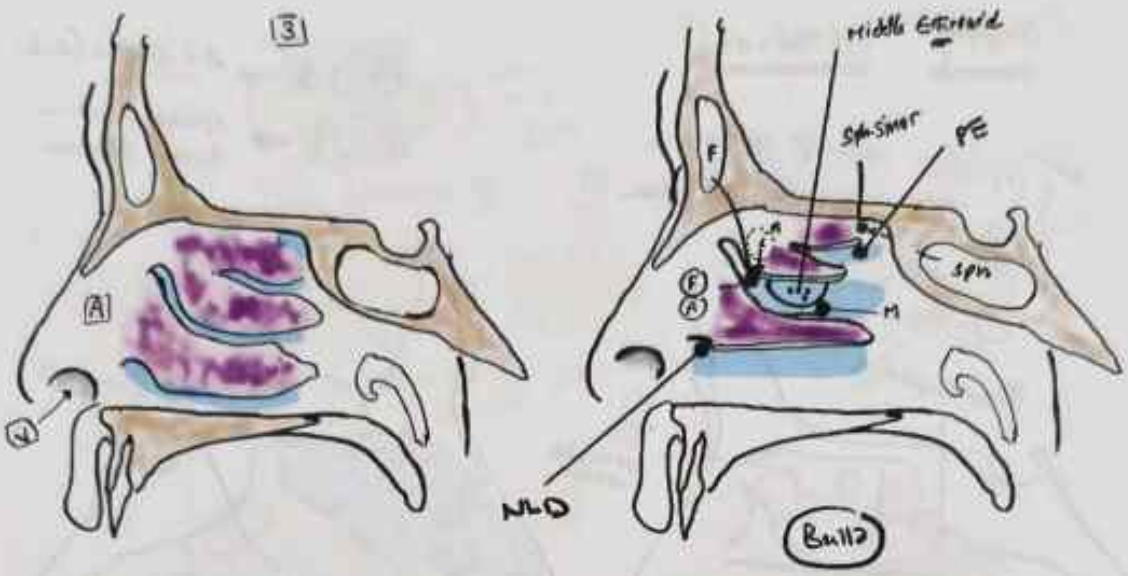
3 — A) V
M) Atrium — m. n
P) — 3 × 3



Middle Meatus
Superior Meatus
Sphenoethmoidal recess
inferior Meatus
at junction of its anterior 1/3 & posterior 2/3

Middle Meatus

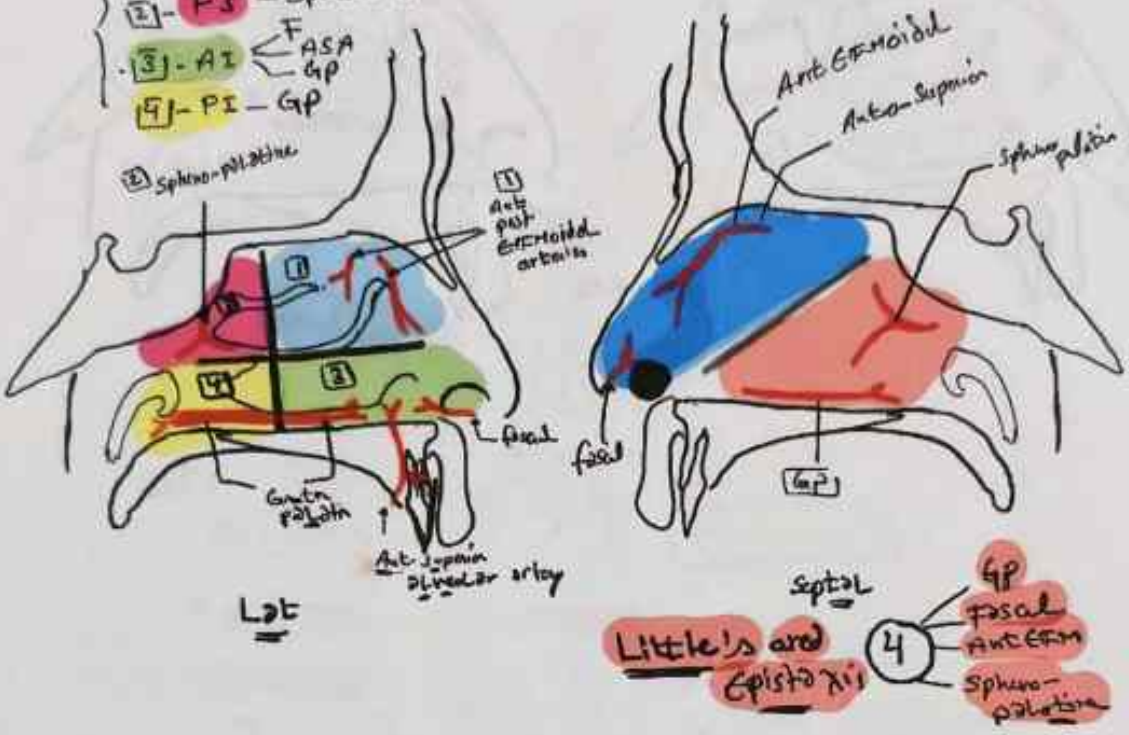
Sup — Post-Eth
inf — N.L.D
Sph Ethmoid Recess — Sphenoid


3
A
4
Middle Turbinate
Sphenoid
PE
F
A
M
Sphen
NLD
Bulla

Supply ARTERIAL

- [1] – A & – A+P Ethmoid
- [2] – PS – Spheno-palatine
- [3] – AI ← F, ASA, GP
- [4] – PI – GP

[AS] ½ → Ant Eth + facial
[PI] ½ → Spheno-palatine, Greater palatine



VENOUS

Facial v → pterygoid venous → pharyngeal v. plexus

Lymphatic

→ Submandibular LN → Upper Deep Cx. LN.

AutoCross

Supply NERVOUS

GENERAL — pain, temp, Touch.

5th trigeminal

(1) AS → Ant. Ethmoid N (5th N)

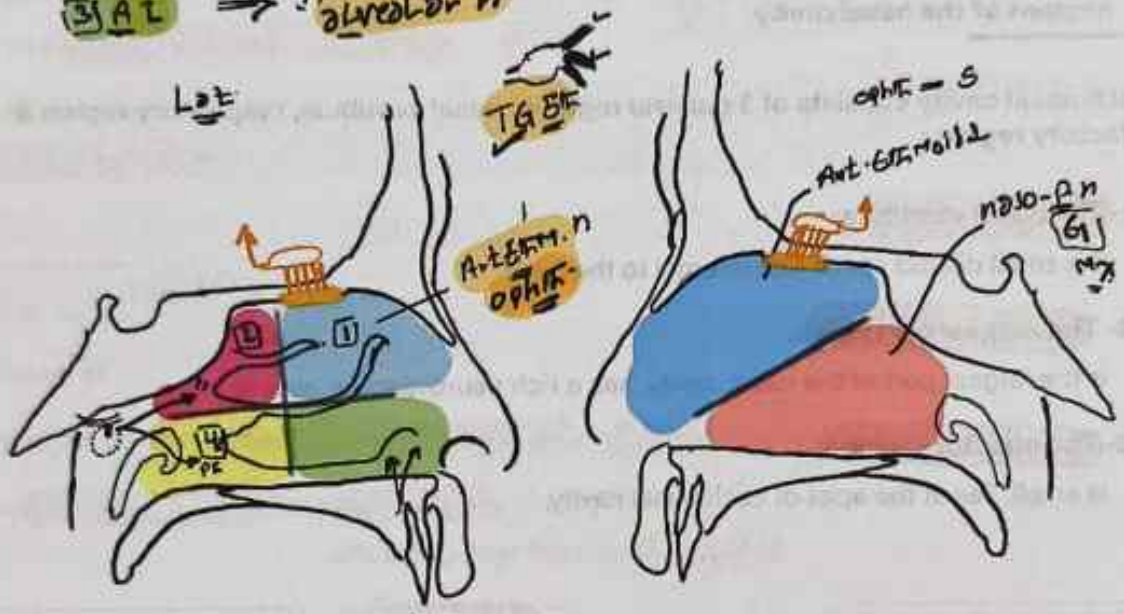
(2) PS → Lat. P.S.N → pterygo-palatine G

(4) PI → Lat. P.I.N → pterygo-palatine G (5th N)

(3) AL → Ant. sup. alveolar n

AS = ½ → Ant. ethmoidal

PI = ½ → naso-palatine



SPECIAL = small ? olfactory

The Nasal Cavity

The nasal cavity is the posterior wide part of the nose.

- **It extends** from the nostrils in front to the posterior nasal apertures or choanae behind, where the nose opens into the **Nasopharynx**.
- **It is divided** into 2 halves by the **nasal septum**, which is often deviated from the median plane making the two nasal cavities, unequal in size.
- **It is located** between the two orbits and below the middle part of the anterior cranialfossa.
- **It is separated** from the oral cavity below by the **hard palate**

Regions of the nasal cavity

Each nasal cavity consists of 3 general regions: nasal vestibule, respiratory region & olfactory region.

1- The nasal vestibule:

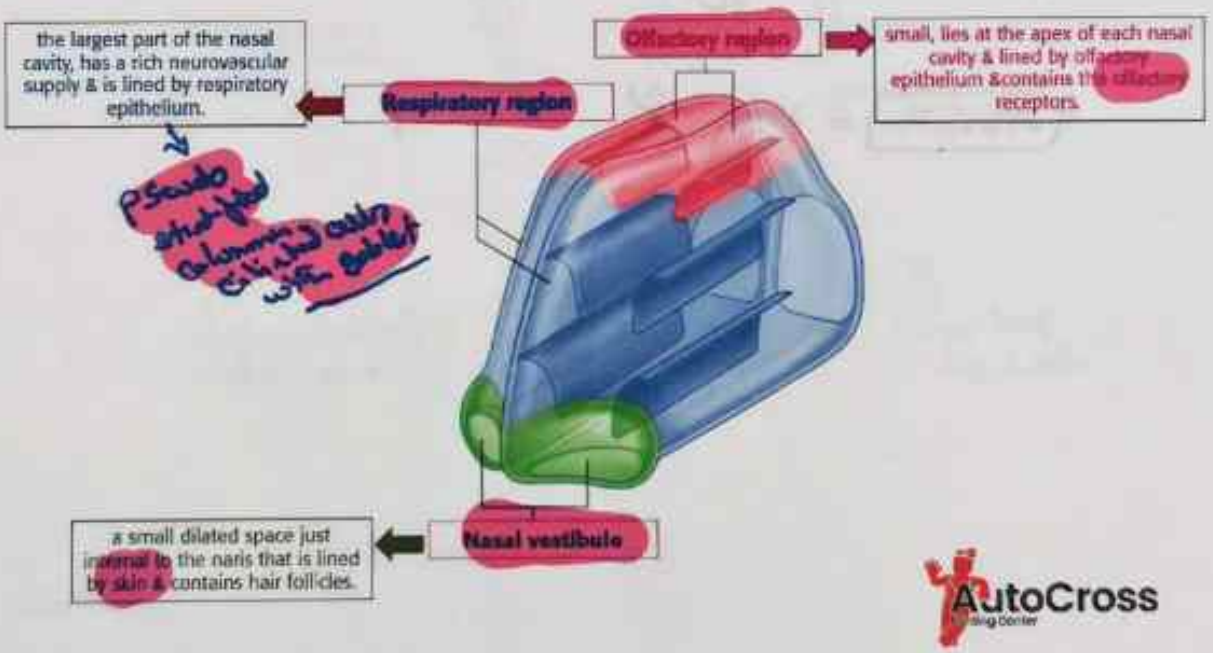
- is a small dilated space just internal to the naris.

2- The respiratory region:

- is the largest part of the nasal cavity, has a rich neurovascular supply.

3- The olfactory region:

- is small, lies at the apex of each nasal cavity.



Boundaries of the nasal cavity

- Each nasal cavity has a floor, roof, medial wall, and lateral wall.

Septum

1- The Roof:

Its length is about 3 inches

It is formed of 3 parts:

Anterior sloping part.

- Nasal bone & nasal process of frontal bone

Middle part.

- The Cribriform plate of ethmoid form

Posterior sloping part.

- Body of sphenoid

AutoCross

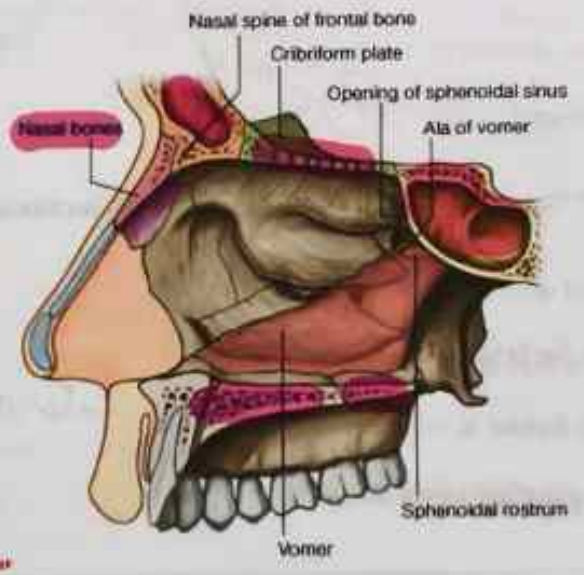
2- The Floor:

is formed of

the upper surface of the palatine process of the maxilla

the horizontal plate of the palatine bone

which together form the hard palate



AutoCross

3- The Medial wall:

- The medial wall of each nasal cavity is the mucosa-covered surface of the thin nasal septum
- It is Oriented vertically in the median Sagittal plane
- It separates the right and left nasal cavities from each other.

The nasal septum consists of:

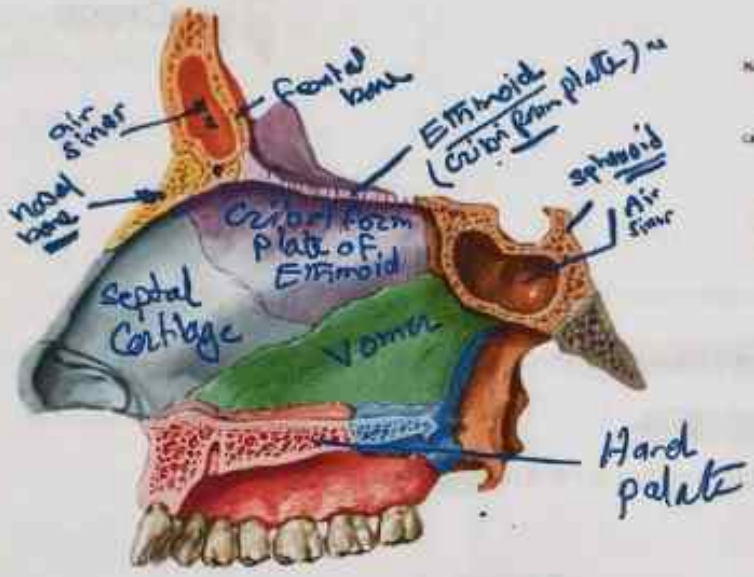
a. Bony element:

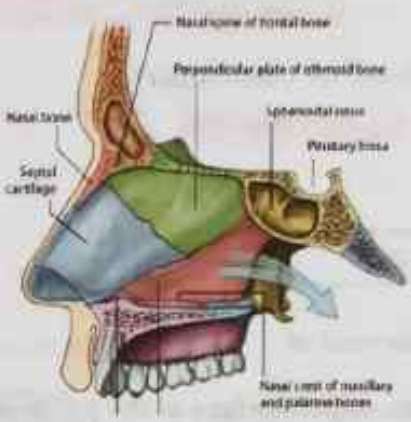
- Vomer bone (posterior)

b. Perpendicular plate of ethmoid bone (above)

c. Cartilaginous element:

- Septal cartilage (anterior).



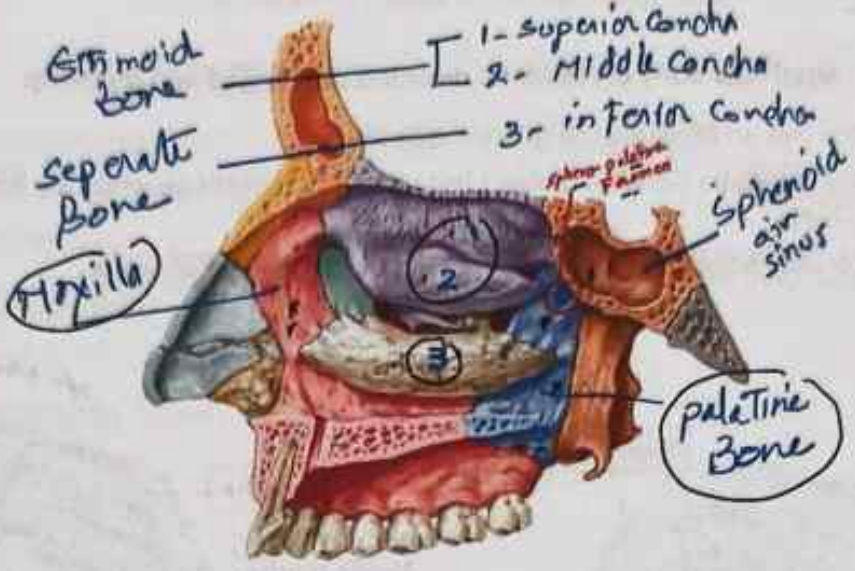


4- The Lateral wall:

An irregular surface due to presence of 3 shelf-like bony projections to increase the surface area of the nose.

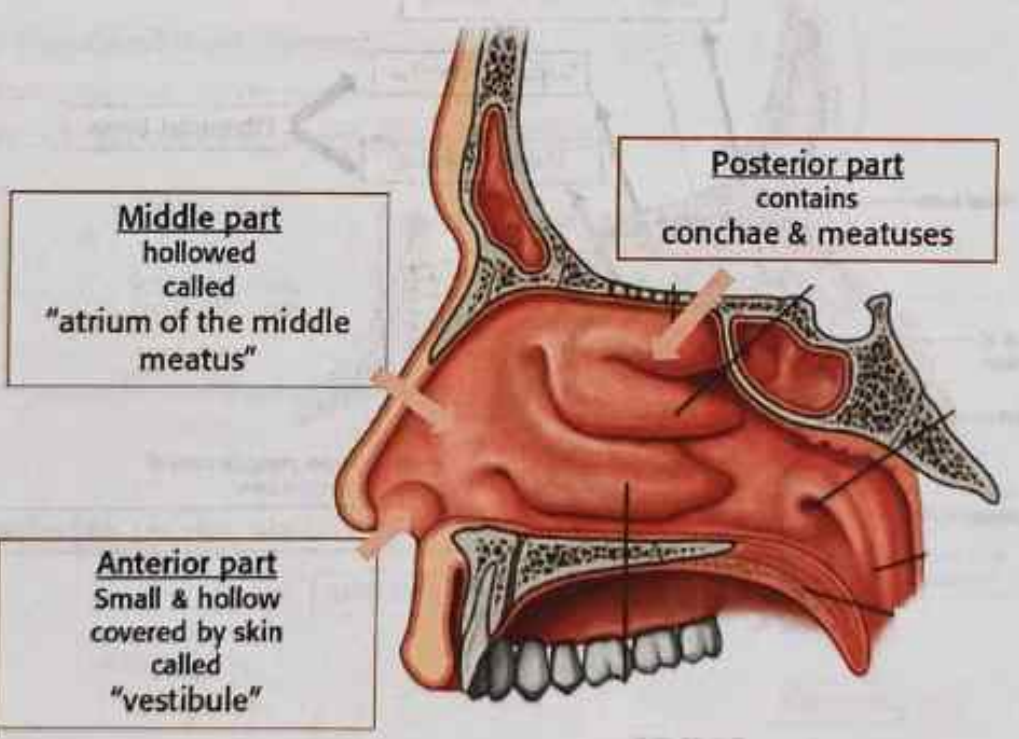
Bones forming the lateral wall

- Nasal aspect of ethmoidal labyrinth above.
- Nasal aspect of maxilla below & in front.
- Perpendicular plate of palatine bone behind.



The lateral wall is divided Into 3 parts

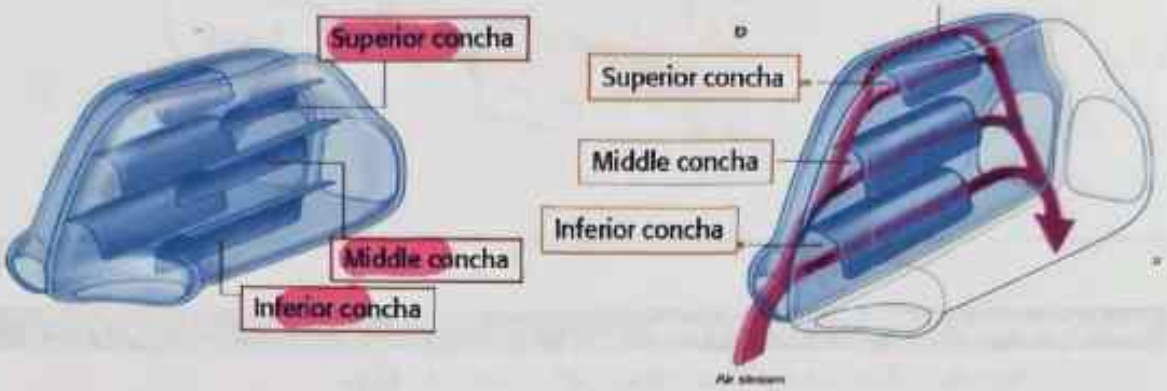
- **Small anterior hollow part** called "vestibule" covered by skin.
- **Middle hollowed part** called "atrium of the middl meatus"
- **Posterior part** containing the Conchae & meatuses

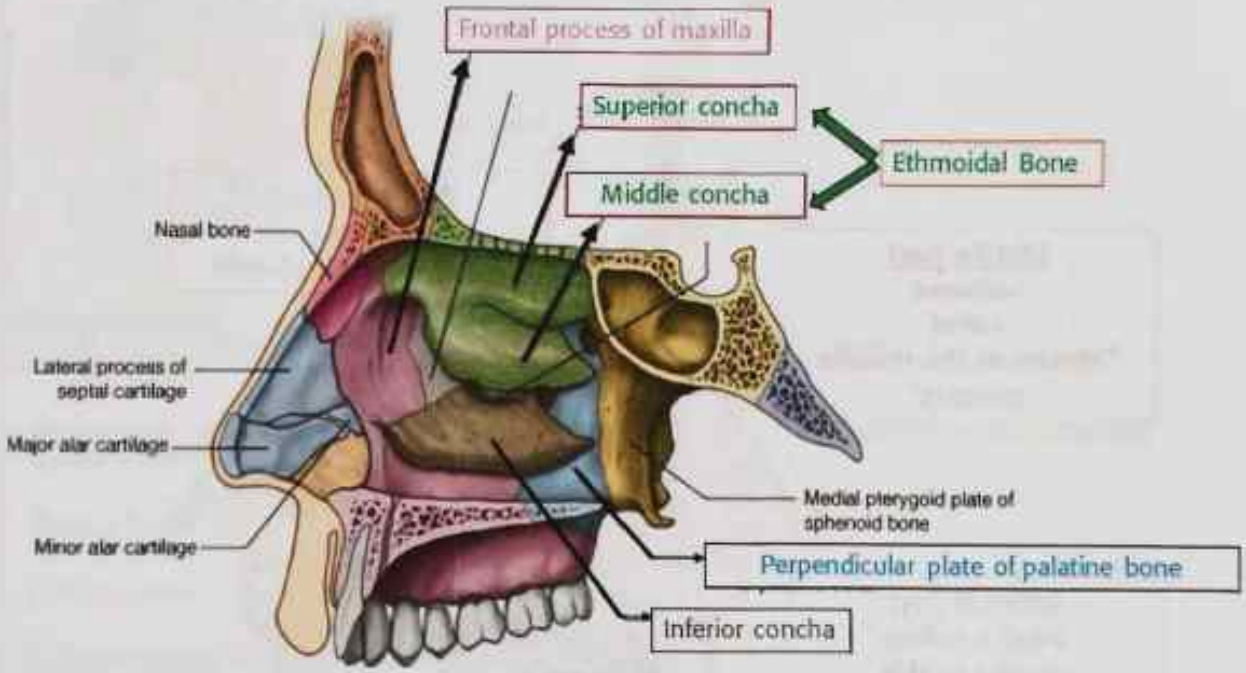


Nasal Conchae or Turbinates: (Superior, middle & inferior):

Shelf-like bony projections directed downward and medially.

- Each concha has an attached margin and free margin.
- The superior & middle Conchae project from the ethmoidal bone while the inferior one is a separate bone.
- Superior concha is the smallest while the inferior concha is the largest.





Nasal meatuses

They are passages beneath the Conchae.

I- The superior meatus:

- the shortest and shallowest one
- Receiving the opening of the posterior ethmoidal sinuses.

II- The middle meatus:

has 3 features:

1. Bulla ethmoidalis:

- rounded elevation produced by the middle ethmoidal sinus (open on it).

2. Hiatus semi-lunaris:

deep semicircular groove below the bulla has:

- **Anterior end,** receives the opening of frontal air sinus & anterior ethmoidal air sinus
- **Posterior end,** receives the opening the maxillary air sinus.

3. Infundibulum:

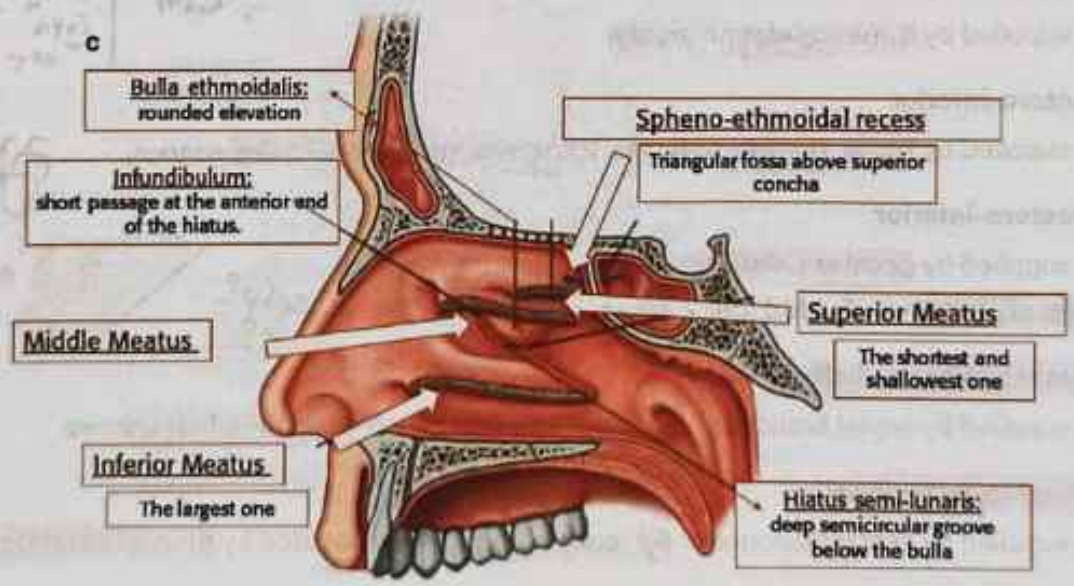
- short passage at the anterior end of the hiatus.

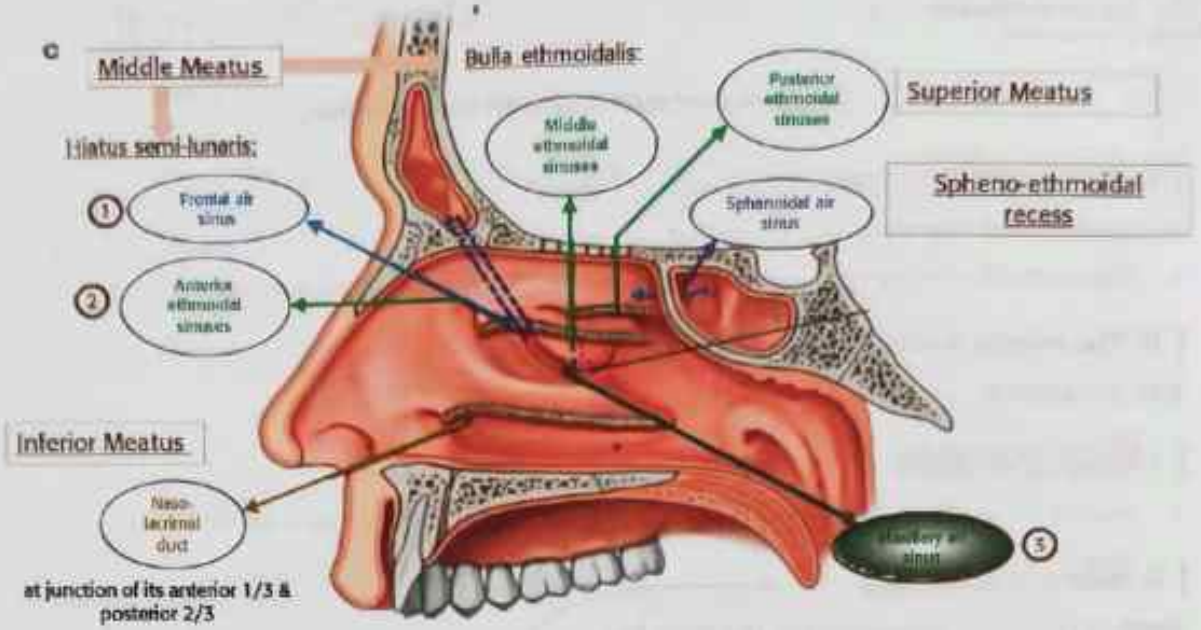
III- Inferior meatus:

- the largest one, at junction of its anterior 1/3 & posterior 2/3, naso-lacrimal duct opens.

IV- Spheno-ethmoidal recess:

- triangular fossa above superior concha
- Receiving the opening of Sphenoidal air sinus.



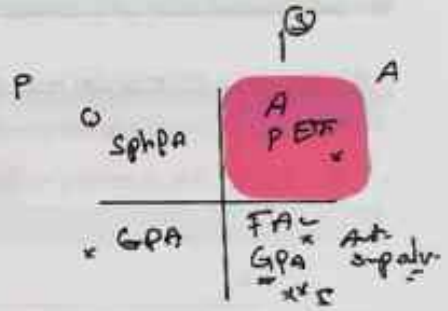


Blood supply

- The nose is rich in blood supply to warm air entering the nose.

Arteries of the nasal cavity

The lateral wall of the nose is divided into 4 areas:



a. Antero-superior

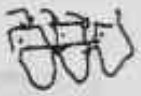
- is supplied by anterior & posterior ethmoidal arteries

b. Postero -superior

- is supplied by Sphenopalatine artery.

c. Antero-inferior

- is supplied by facial, greater palatine & anterior superior alveolar arteries.



d. Postero-inferior

- is supplied by greater palatine artery.



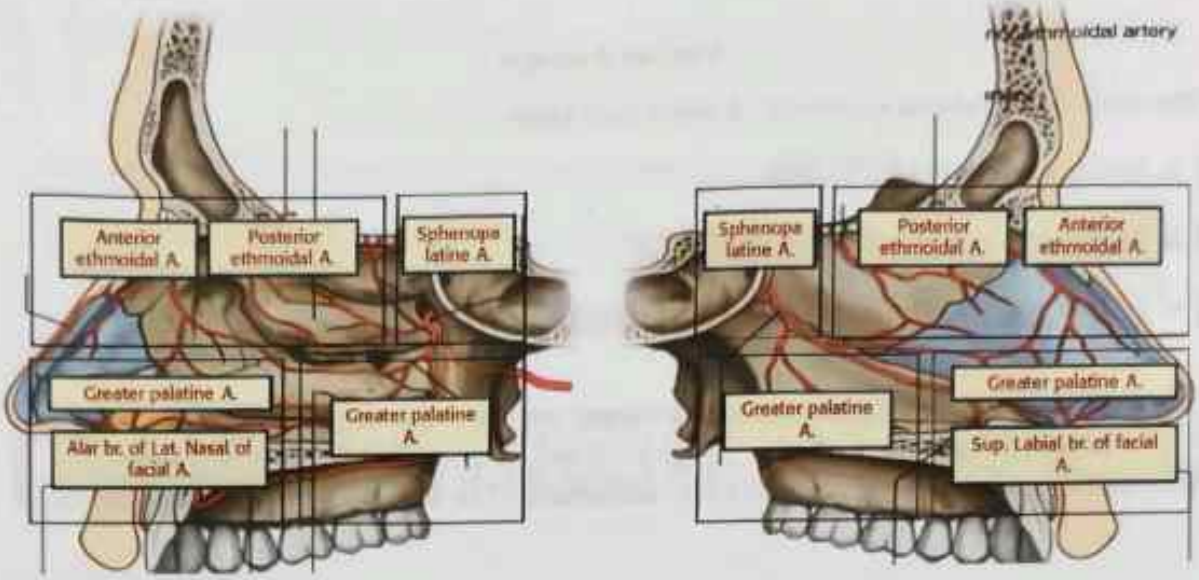
The nasal septum is divided into 2 areas:

a. Antero-superior half

- is supplied by septal branches of anterior ethmoidal and superior labial arteries.

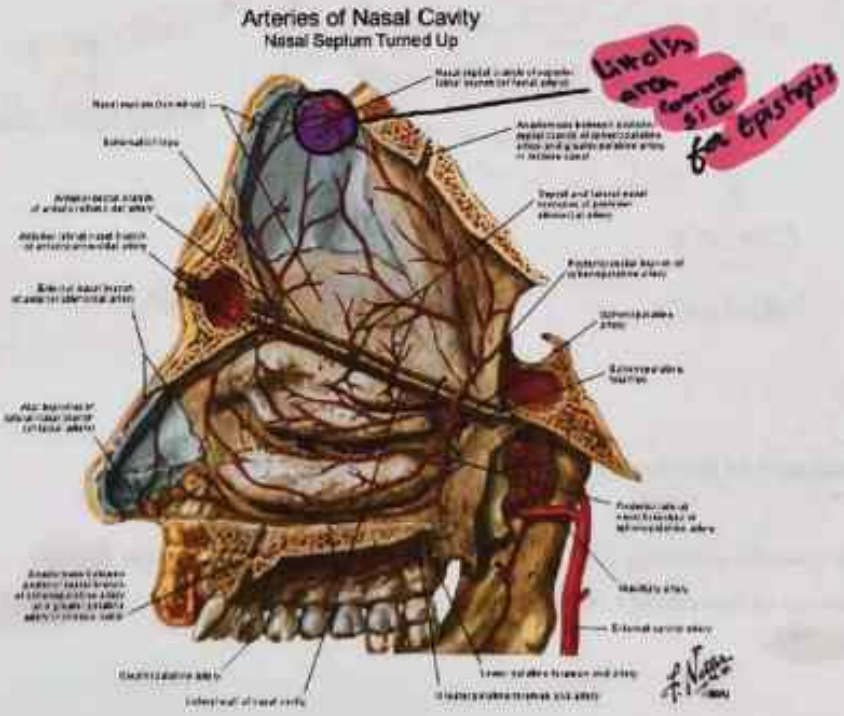
b. Postero-inferior half

- is supplied by septal branches of Sphenopalatine artery assisted by greater palatine a.



N.B.

- Arteries supplying the nasal cavities form extensive anastomoses with each other.
- This is particularly evident in the anterior region of the medial wall where there are anastomoses between branches of the greater palatine, sphenopalatine, superior labial, and anterior ethmoidal arteries, and where the vessels are relatively close to the surface.
- This area is the major site of 'nose bleeds' or epistaxis.



Venous drainage

The veins form submucus plexus, drains into 3 sets:

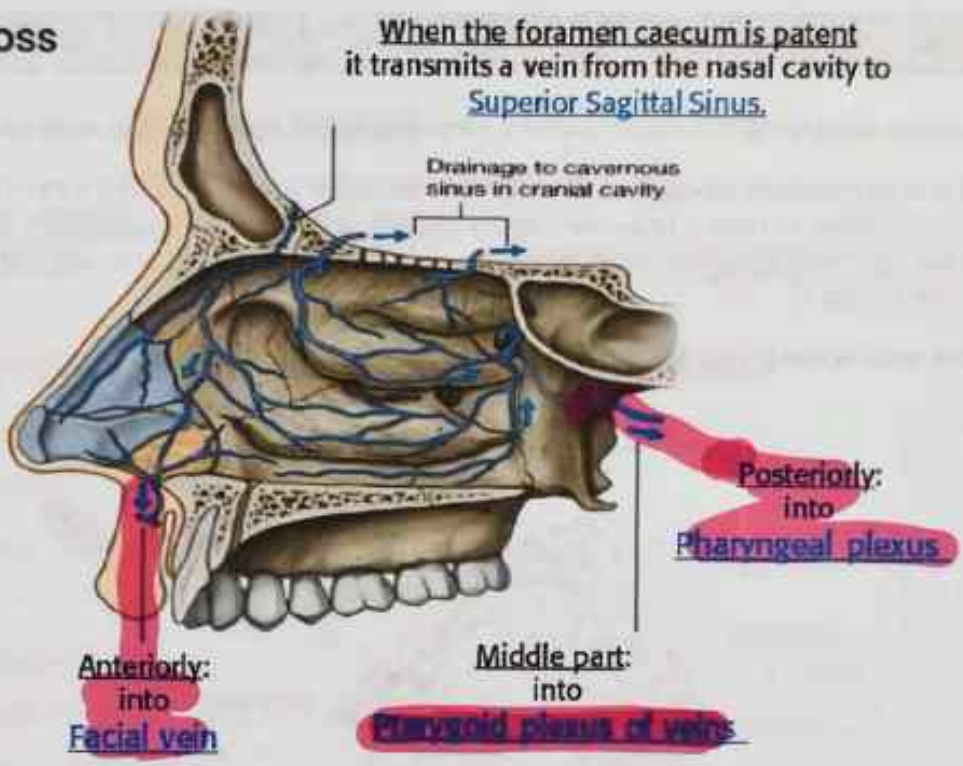
a. Anteriorly into the facial vein.

b. Posteriorly into the pharyngeal plexus.

c. Middle part into the pterygoid plexus of veins.

- N,B; If the foramen caecum is patent it transmits a vein from the nasal cavity to the superior sagittal sinus.

AutoCross



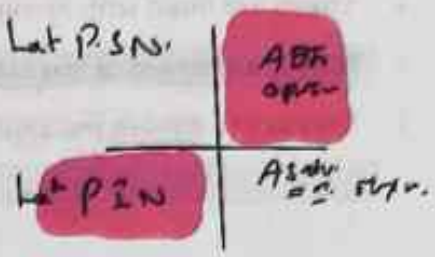
Lymph Drainage of the nasal cavity

- The lymph vessels draining the vestibule end in the submandibular nodes.
- The remainder of the nasal cavity is drained by vessels that pass to the upper deep cervical nodes.

Nerve supply of the nasal cavity

A) General sensation: (from trigeminal)

The lateral wall of the nose is divided into 4 areas:



a. Anterosuperior part

- is supplied by anterior ethmoidal n.

b. Antero inferior part

- is supplied by anterior superior alveolar n.

c. Postero-superior part

- is supplied by lateral posterior superior nasal n. of pterygopalatine ganglion.

d. Postero-inferior part

- is supplied by lateral posterior inferior nasal n. of greater palatine from the pterygopalatine ganglion.

The nasal septum is divided into 2 areas:

a. Antero-superior half

- is supplied by septal branches of anterior ethmoidal nerve.

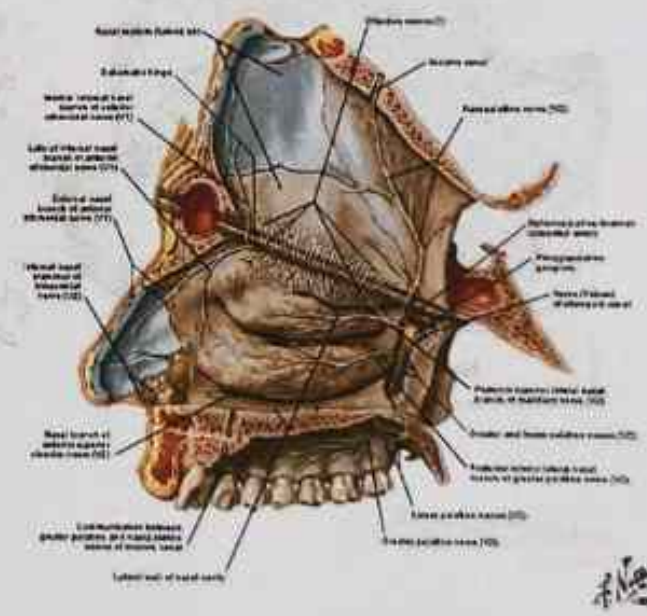
b. Postero-inferior half

- is supplied by septal branches of nasopalatine branch of pterygopalatine ganglion.
- It supplies the septum and then descends to the hard palate via the incisive foramen.

B) Special sensation:

Olfactory nerve

- carries smell sensation
- is formed by the central processes of the bipolar neurons which collect into 20 filaments that traverse the cribriform plate of ethmoid bone to end in the olfactory bulb.



Paranasal Air Sinuses

- Air filled cavities present in the maxilla, frontal, ethmoid and sphenoid bones.
- Theyb are lined with respiratory mucosa and all drain in the nasal cavities.
- They are abscent or rudimnentary at birth increasing in size until adolescence.
- They act to lighten the skull and add resonance to the voice.

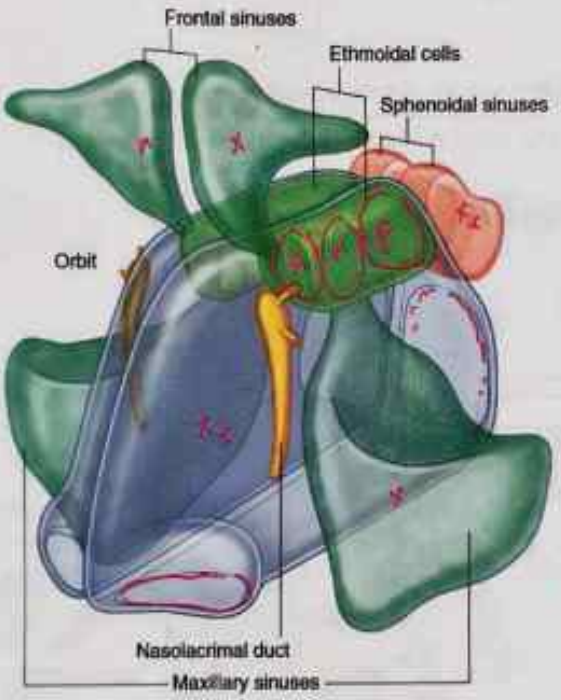
They are divided into 2 groups:

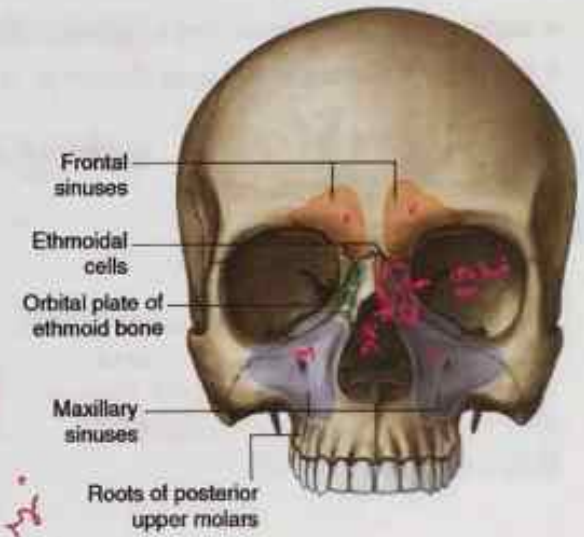
i. Anterior group:

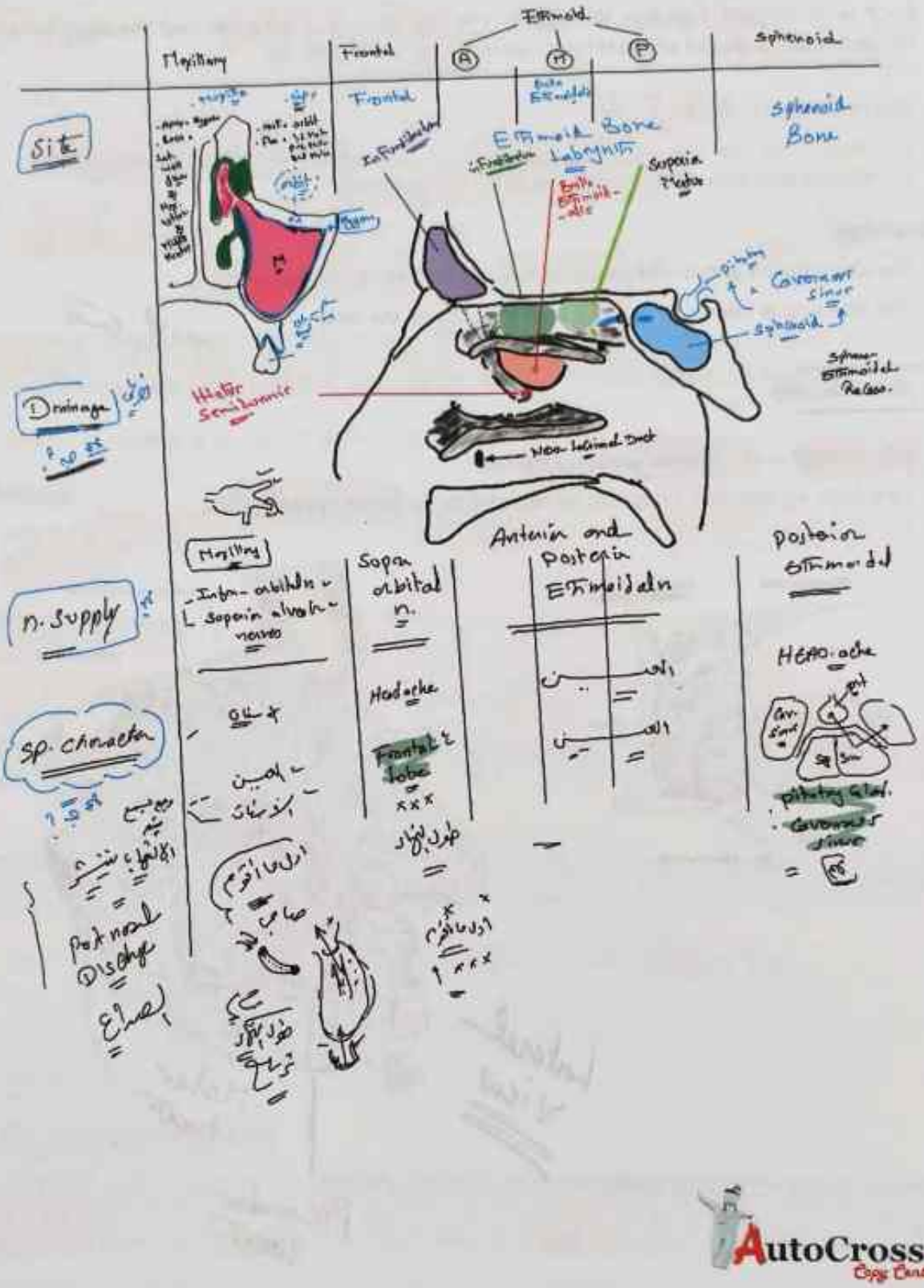
- frontal, maxillary, anterior & middle ethmoidal (all open in the middle meatus).

ii. Posterior group:

- Sphenoidal opens in spheno-ethmoidal recess
- Posterior ethmoidal opens in superior meatus.







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1- Maxillary Air Sinus

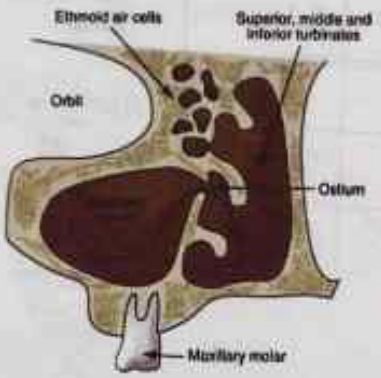
- It occupies most of the body of the maxilla.
- Each is pyramidal in shape, the base is the lateral wall of the nose and the apex extends to zygomatic process of maxilla or even the zygomatic bone.
- The roof is the floor of the orbit.
- The floor is the maxilla's alveolar process at the roots of the first and second molars, sometimes also including the premolars and third molar.

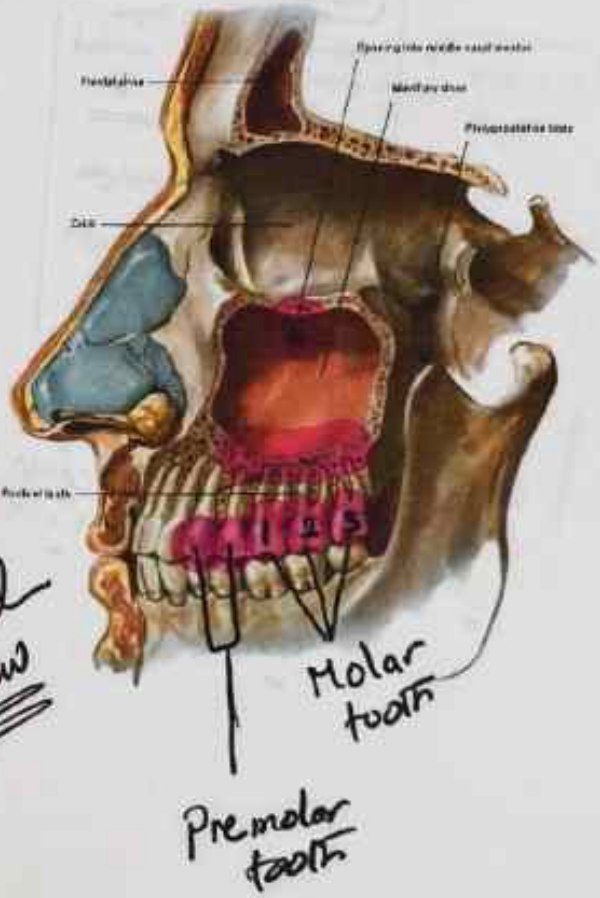
Drainage

- The sinus opens into the hiatus semilunaris by one or two openings.
- This opening is nearer to the roof than the floor of the sinus.

Nerve Supply

- infraorbital and superior alveolar nerves.
- Pain from an infected sinus can be referred to the teeth (upper jaw).

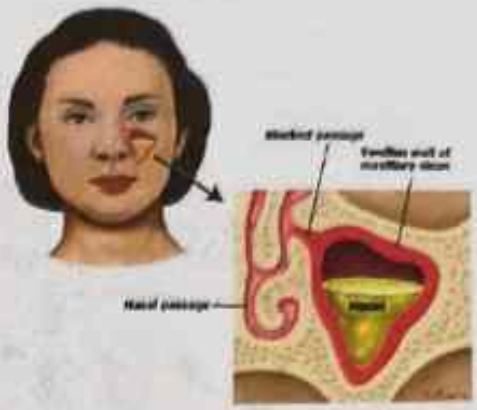




N.B.

The maxillary air sinus is commonly infected (maxillary sinusitis) because:

- **Its drainage is difficult (its drainage orifice lies near the roof) and discharge from an infected frontal or anterior ethmoidal air sinus can pass to the sinus.**
- **It is closely related to teeth so spread of infection from tooth to the sinus can occur.**



2- Frontal Air Sinuses

- The two sinuses are usually unequal in size.

Drainage

- into the **middle meatus** of the nose by the **Infundibulum** in the **anterior part of the hiatus semilunaris.**

Nerve Supply

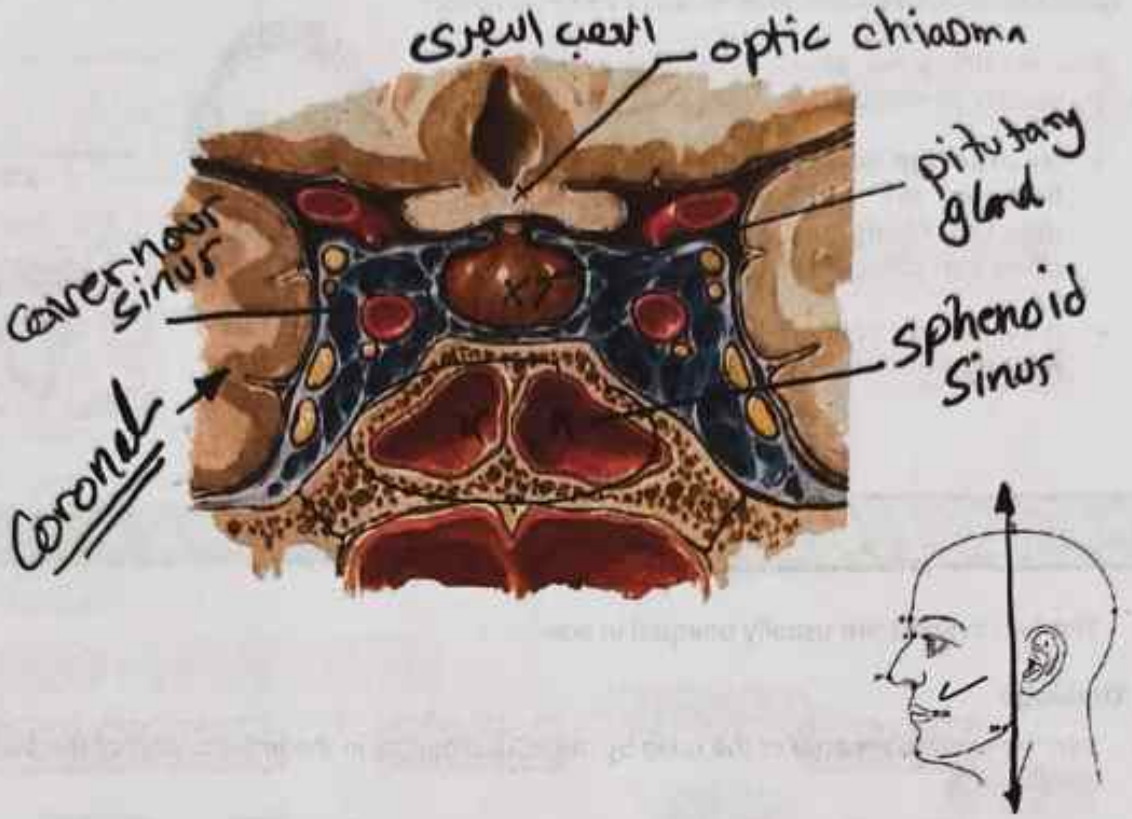
- **Supraorbital nerve,** so pain from an infected sinus can be referred to the forehead.
- N,B,: Infection of frontal air sinus can spread to frontal lobe of brain.

3- Sphenoidal Air Sinuses

- They are two right and left separated by a septum inside body of sphenoid.
- They drain into **Spheno-ethmoidal recess.**

Nerve Supply

- **Posterior ethmoidal nerve**
- The sphenoidal air sinuses are related superiorly to pituitary gland & optic chiasma and laterally to cavernous venous sinus & its contents.



4- Ethmoidal Air Sinuses

- They are present in the ethmoid bone arranged into: anterior, middle and posterior ethmoidal air sinuses.
- They lie between the nasal cavity and orbit.
- Infection in these sinuses can spread to the **orbit**.

a) Anterior ethmoidal sinuses

- drain into **infundibulum** at anterior end of **hiatus semilunaris** of the **middle meatus**.

b) Middle ethmoidal sinuses

- drain by an opening on top of **bulla ethmoidalis**.

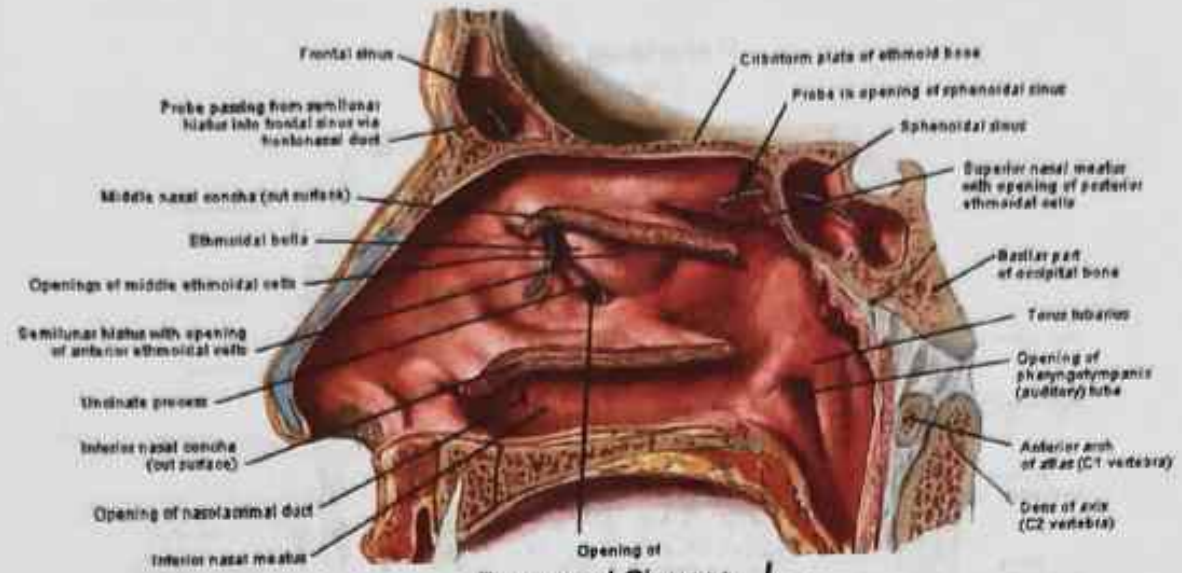
c) Posterior ethmoidal sinuses

- drain into **superior meatus**.

Nerve Supply

- They are supplied by **anterior and posterior ethmoidal nerves**.

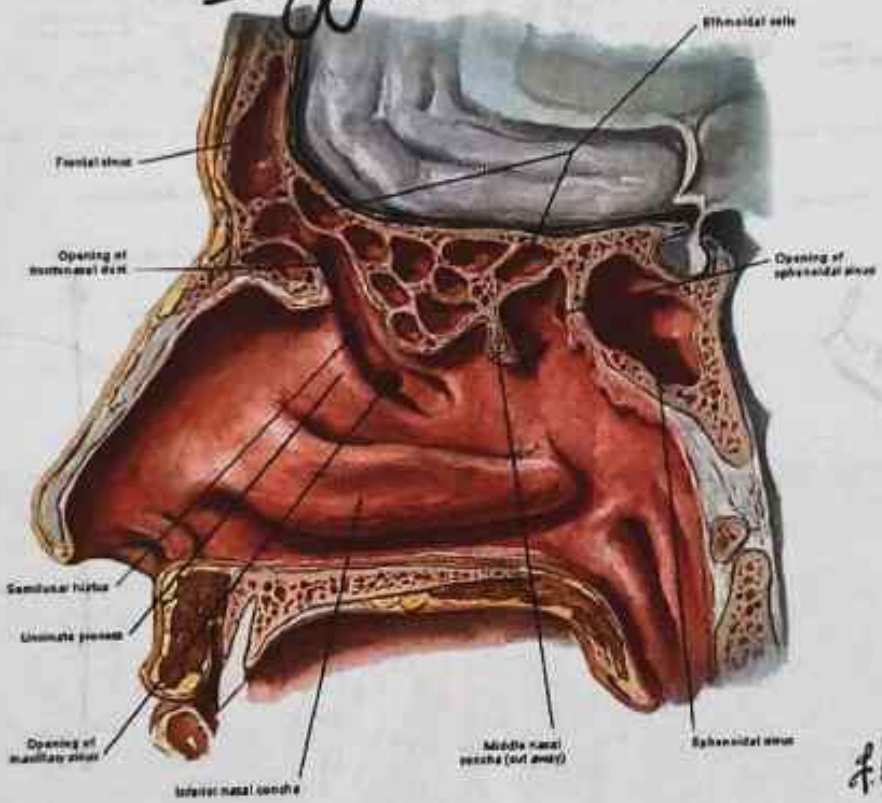
AutoCross Printing Center



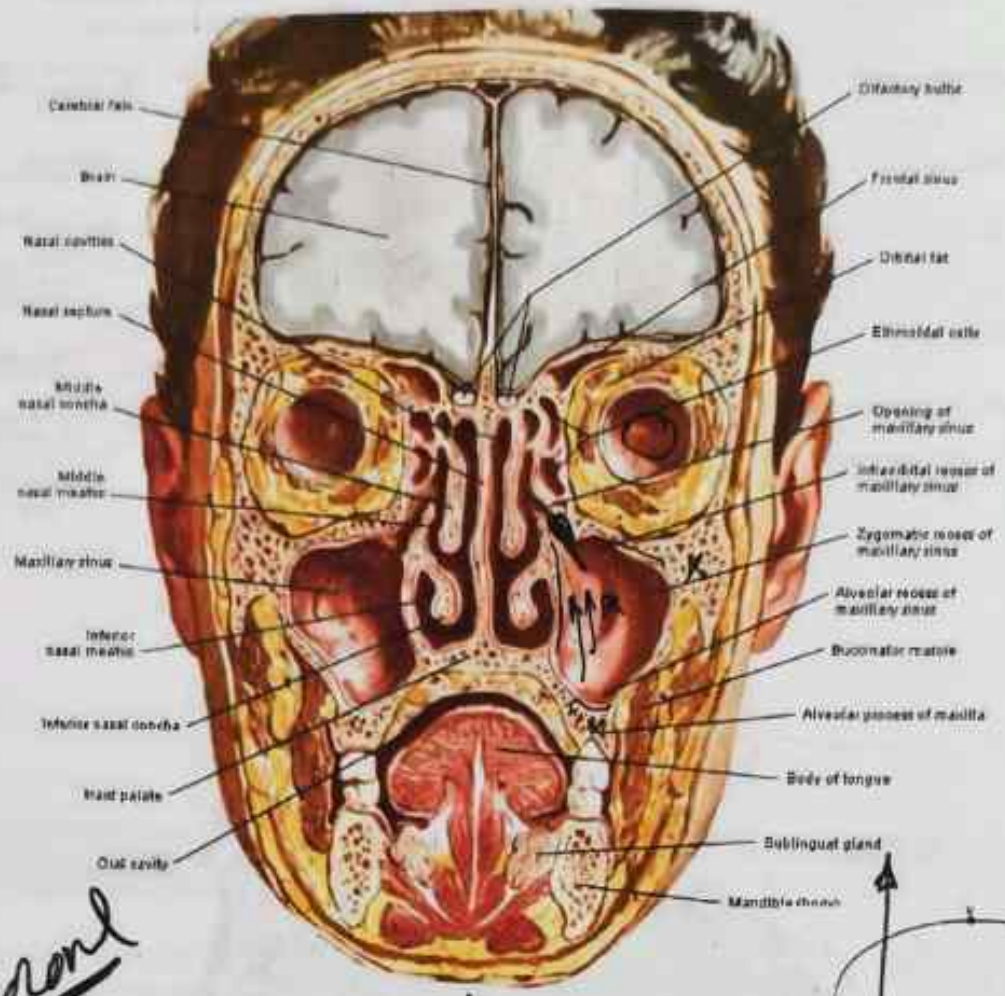
Paranasal Sinuses

Sagittal Section

Saggittal section

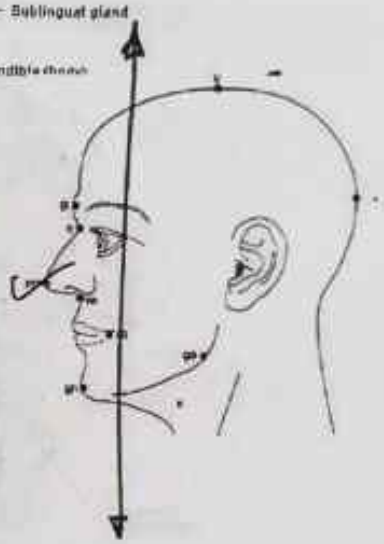


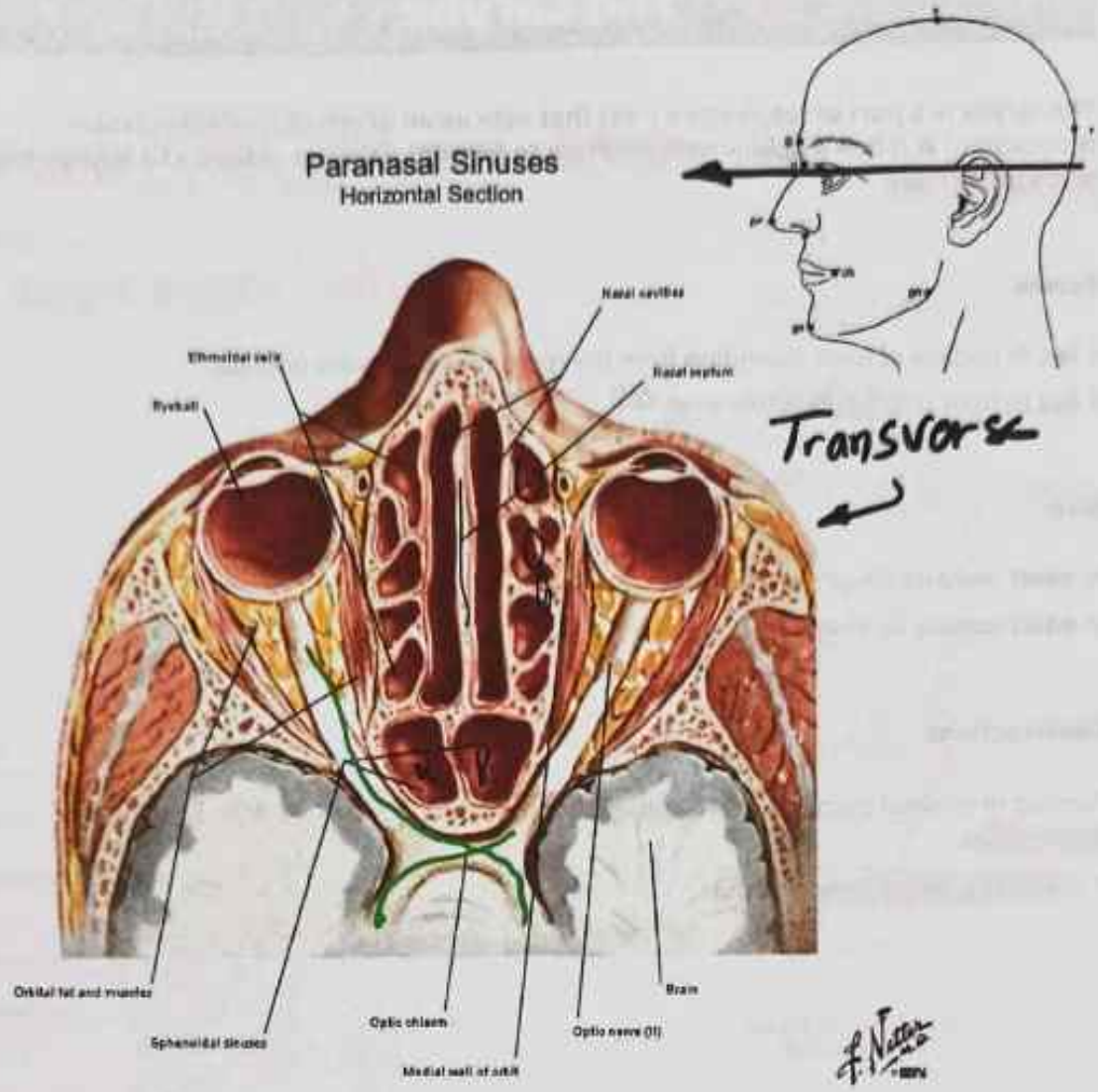
Paranasal Sinuses
Coronal Section



coronal

Coronal section







Identify the marked structure ①,②

Diagnosis

The Larynx

- **The larynx is a part of respiratory tract that acts as an organ of phonation (voice production) & It has a sphincteric function to prevent passage of food and foreign bodies through its inlet.**

Extent

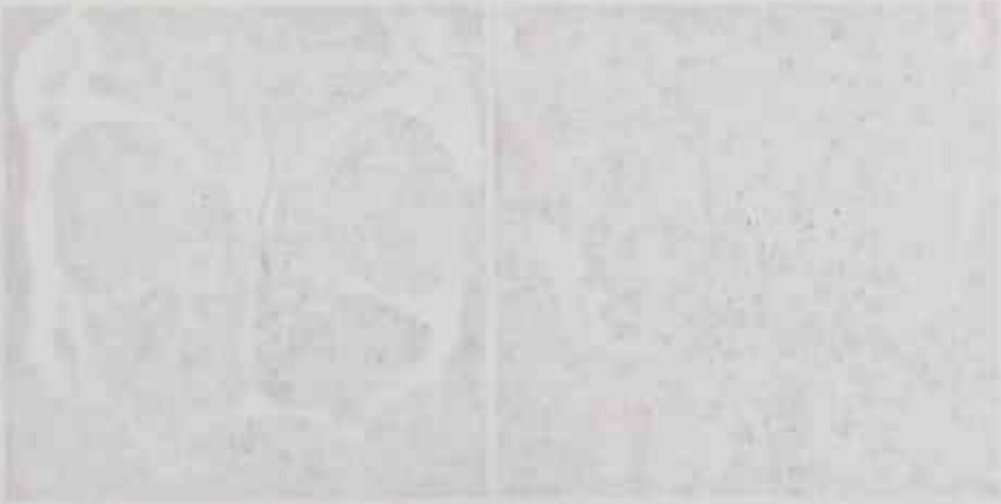
- it lies fn midline of neck extending from the root of tongue to the trachea.
- It lies tn front of C3-C6 vertebrae

Size

- In adult male its length Is 44 mm.
- In adult female its length Is 38 mm.

Constructions

- Formed of skeletal framework of cartilages connected together by joints, ligaments & membranes.
- It is lined by mucous membrane.



Cartilages of Larynx

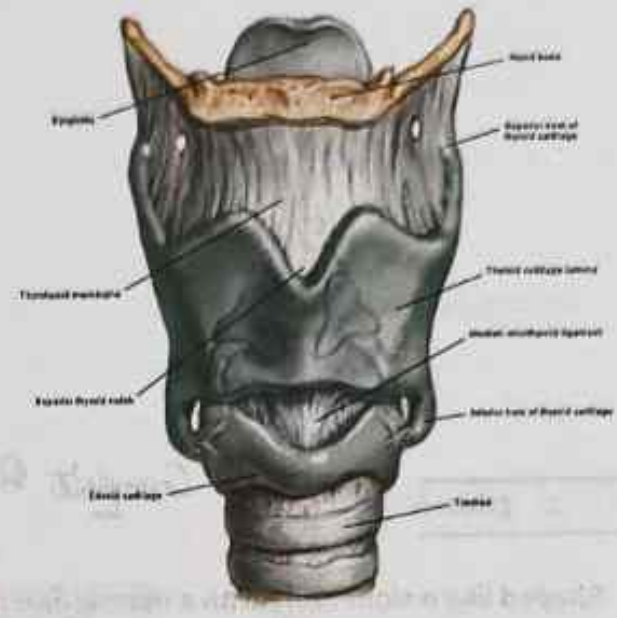
Single:

- Thyroid, cricoid & epiglottis.

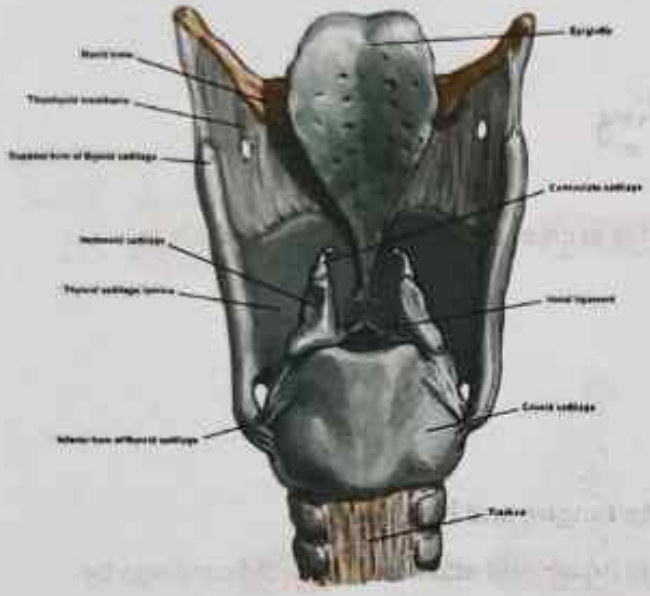
Paired:

- Arytenoid, Comiculate and cuneiform.

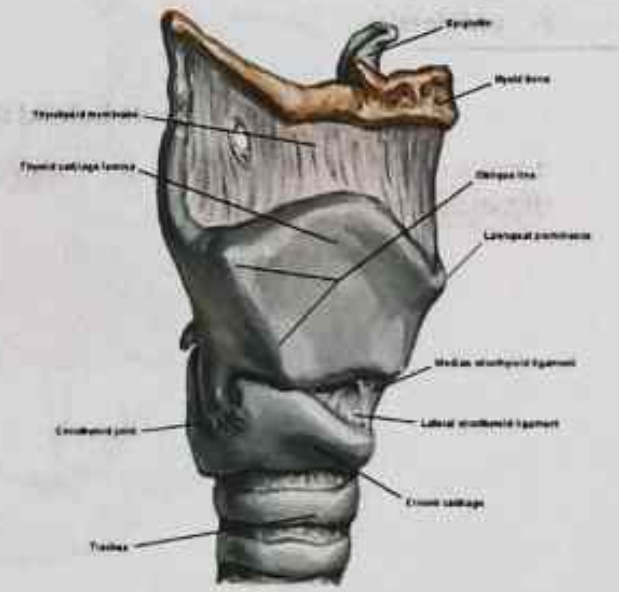
Cartilages of Larynx
Anterior View



Cartilages of Larynx
Posterior View



Cartilages of Larynx
Right Lateral View



1- Thyroid cartilage:

- Consists of **two laminae** united anteriorly but separated posteriorly.
- The posterior border shows **superior and inferior horns.** M cq
- The anterior border forms the laryngeal prominence **"Adam's apple"** in the male.

2- Cricoid:

M cq Complete Ring

Shaped like a signet ring with a narrow anterior arch and a broad posterior lamina.

3- Epiglottis:

Leaf like, projects behind the tongue and hyoid bone.

- It has an **upper free broad end** and a **narrow lower end** attached to thyroid cartilage by **thyro-epiglottic ligament**.

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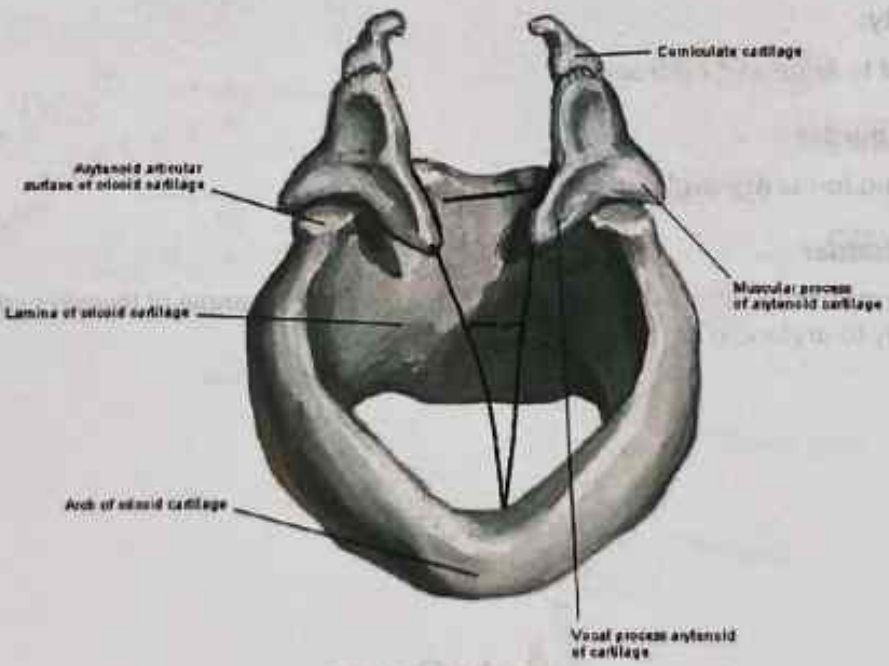
4- Arytenoid:

Pyramidal in shape with an apex and a base having two processes: muscular process and vocal process.

5- Corniculate:

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6- Cuneiform



Laryngeal ligaments & membranes

A) Extrinsic ligaments:

1- Thyrohyoid membrane:

- Extends from upper border of hyoid to upper border of thyroid cartilage.
- It is pierced by the internal laryngeal nerve & superior laryngeal artery.

2- Cricotracheal ligament :

- Connects lower border of cricoid cartilage & 1st tracheal ring

3- Glosso-epiglottic:

- connecting tongue to epiglottis.

B) Intrinsic ligaments:

1- Quadrangular membrane

it is a fibro-elastic membrane that lies beneath the mucous membrane of vestibule of larynx.

Anteriorly:

- attached to side of epiglottis.

Posteriorly:

- attached to arytenoid cartilage.

Its upper border

- is free and forms aryepiglottic fold.

Its lower border

- is free and forms vestibular ligament which extends from angle of thyroid cartilage anteriorly to arytenoid cartilage posteriorly.

2- Cricothyroid ligament Conus Elasticus

a fibro-elastic membrane that lies beneath mucous membrane of lower part of larynx.

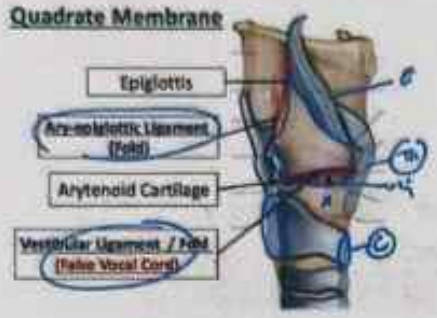
Its anterior part

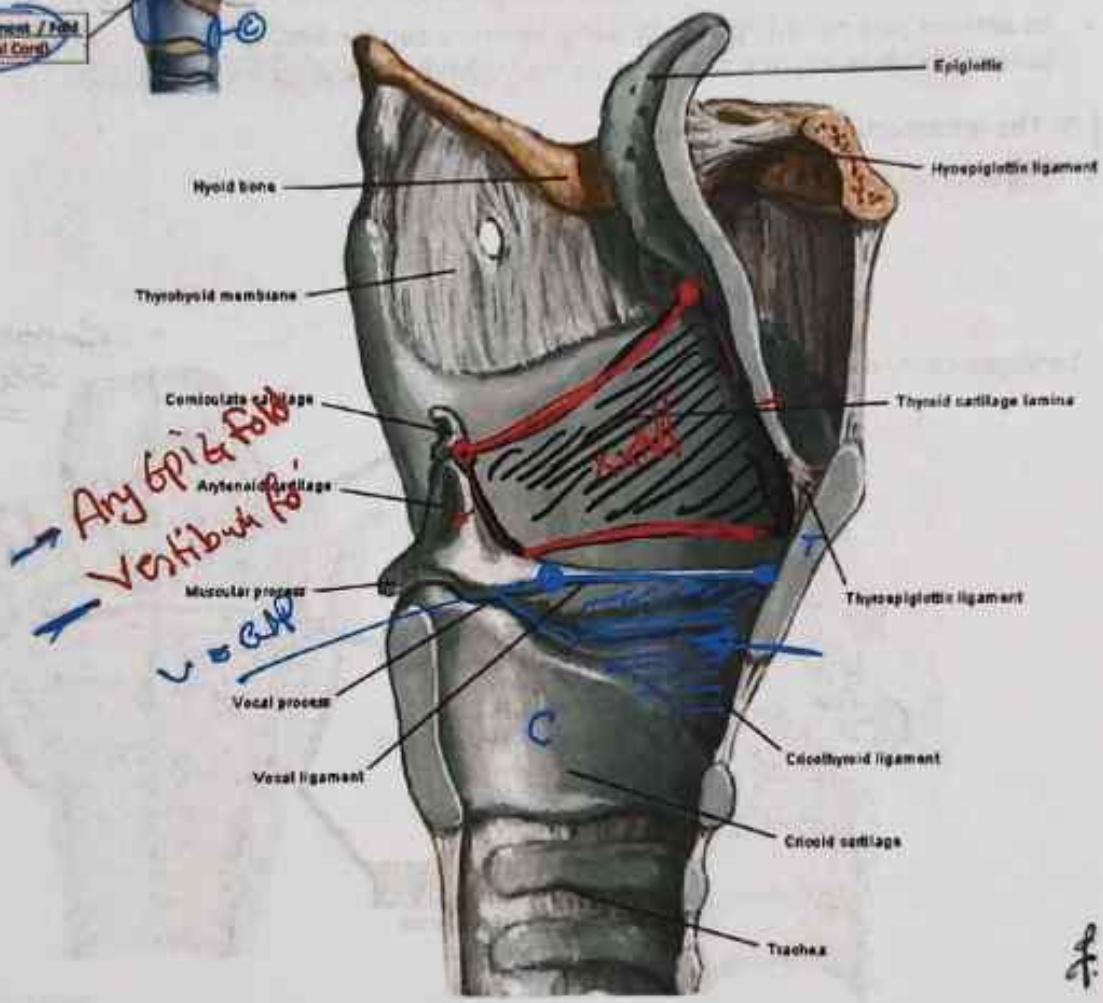
- is thick & connects the adjacent sides of cricoid and thyroid cartilages

Its lateral part

- has an upper free border that forms vocal ligaments which extends from thyroid angles anteriorly to vocal process of arytenoid posteriorly.







Laryngeal cavity:

- The central cavity of the larynx is tubular in shape and is lined by mucosa.
- Its architectural support is provided by the fibre-elastic membrane of larynx and by the laryngeal cartilages to which it is attached.
- It extends from laryngeal inlet to the trachea.

It is partially divided into three areas by two folds of mucosa.

- The upper folds: vestibular folds
- and the lower folds: the vocal folds.

1- The vestibule

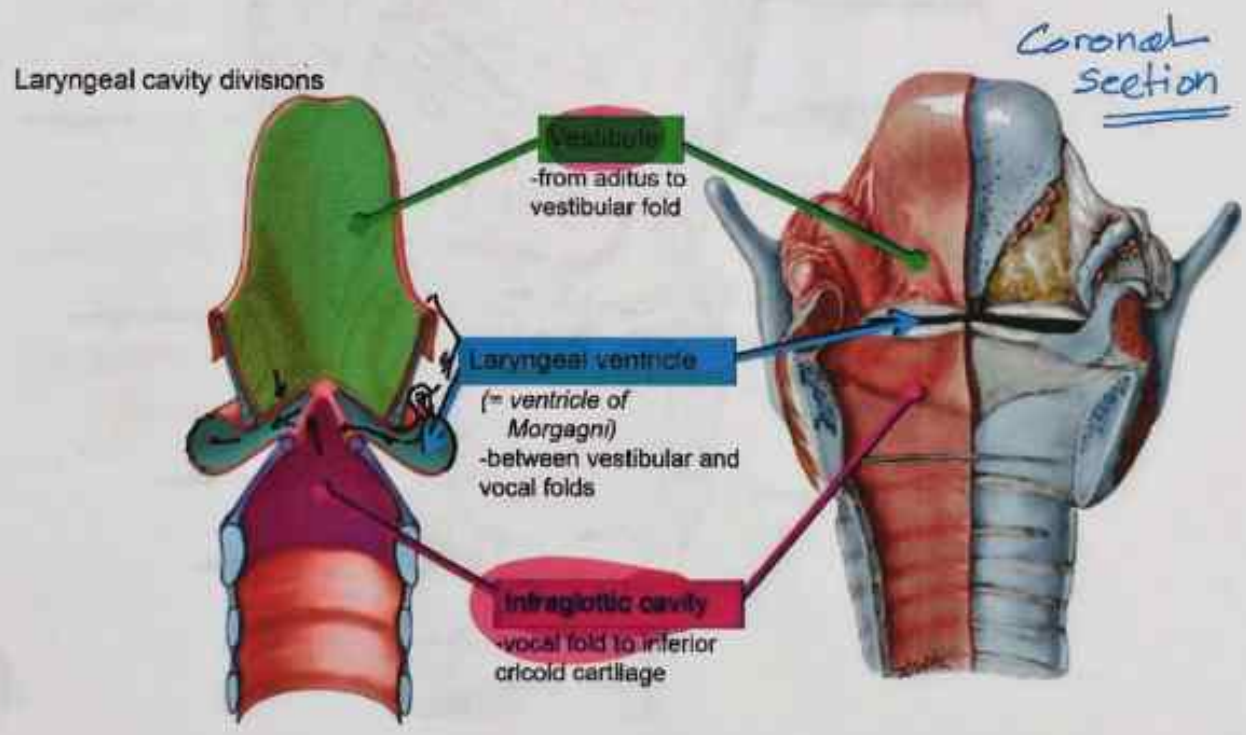
- lies between inlet and vestibular folds.

2- The laryngeal ventricle (sinus)

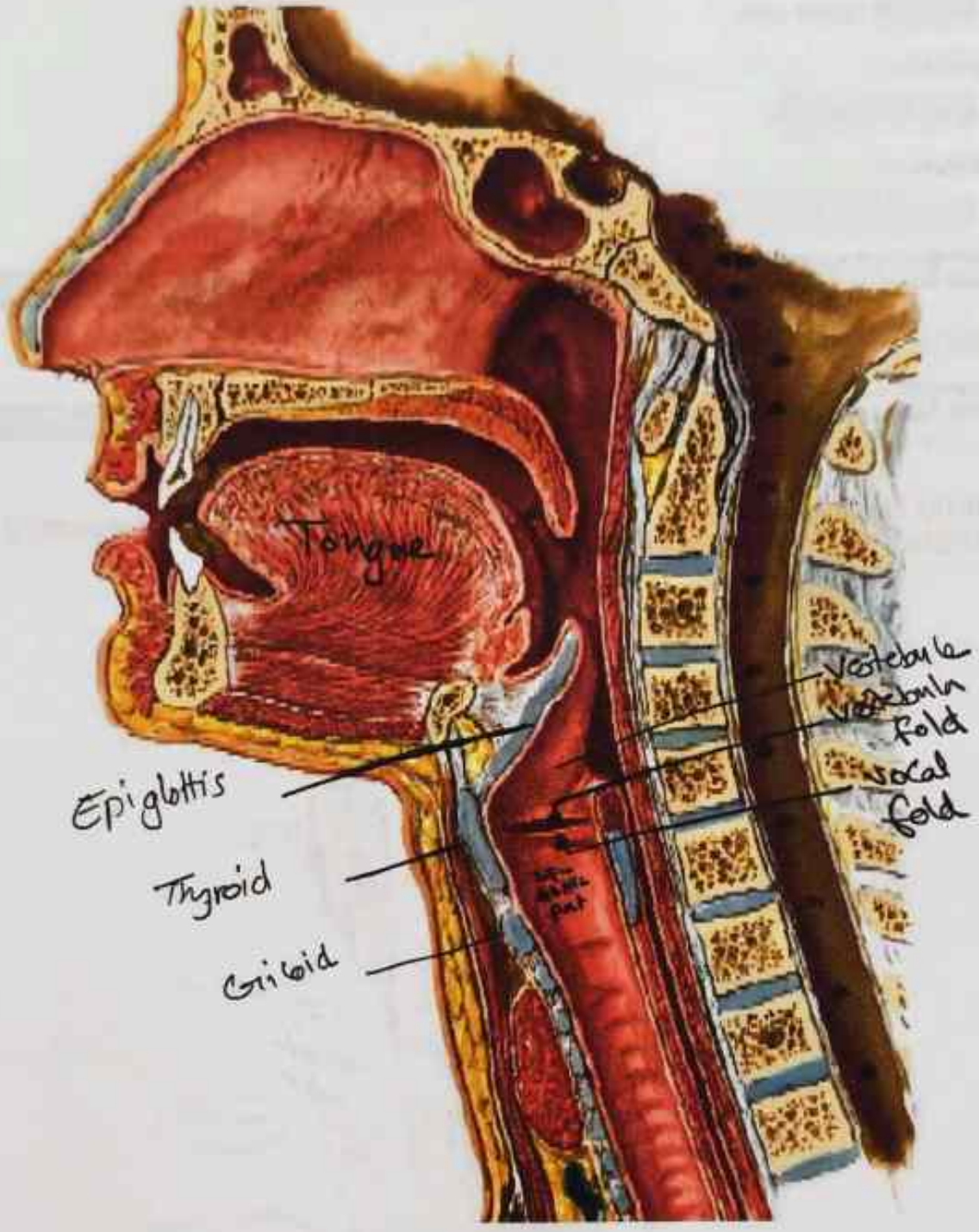
- is the depression between the vestibular folds above and the vocal folds below.
- Its anterior part has an upward prolongation between the vestibular fold & thyroid lamina called saccule of larynx it has many mucous glands lubricating the vocal cords.

3- The infraglottic compartment

- The area below the vocal folds is.



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Boundaries of the Laryngeal Inlet

Anteriorly:

- Epiglottis (upper end).

Laterally:

- Aryepigfottic folds.

Posteriorly:

- Mucosa covering the arytenoids.

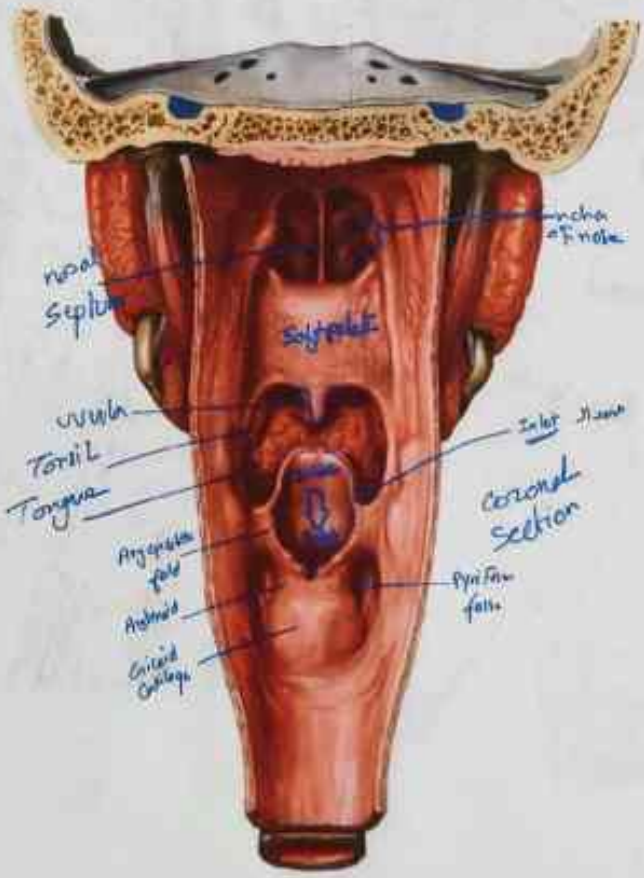
Rima Glottidis:

- it is the fissure between the right and left vocal folds (anteriorly) and between the arytenoids (Posteriorly).

Rima vestibule:

- it is the space between the 2 vestibular folds.

Both the rima glottidis and the rima vestibuli can be opened and closed by movement of the arytenoids and associated fibro-elastic membranes.



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Laryngeal muscles

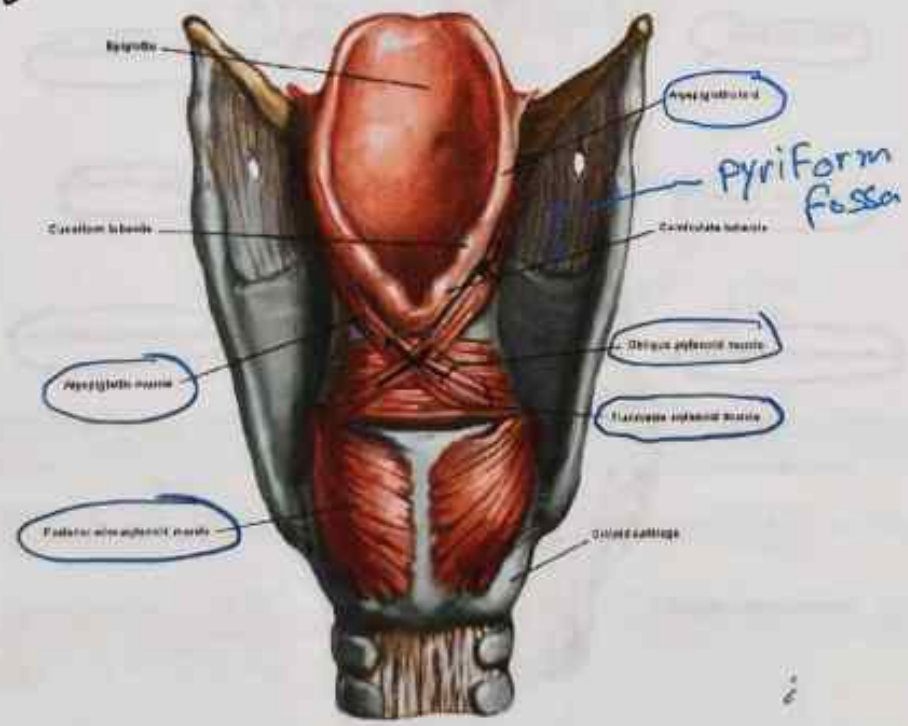
A) Extrinsic muscles:

- Infra-hyoid muscles
- Inferior constrictor
- stylopharyngeus
- Palatopharyngeus.

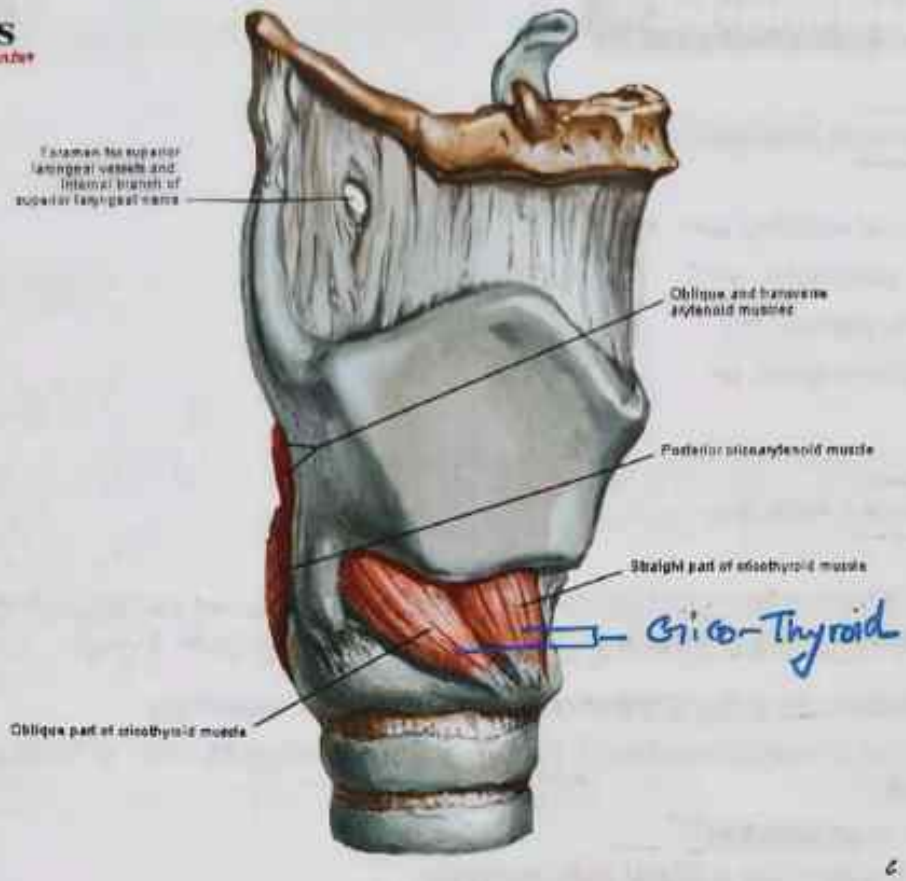
B) Intrinsic muscles:

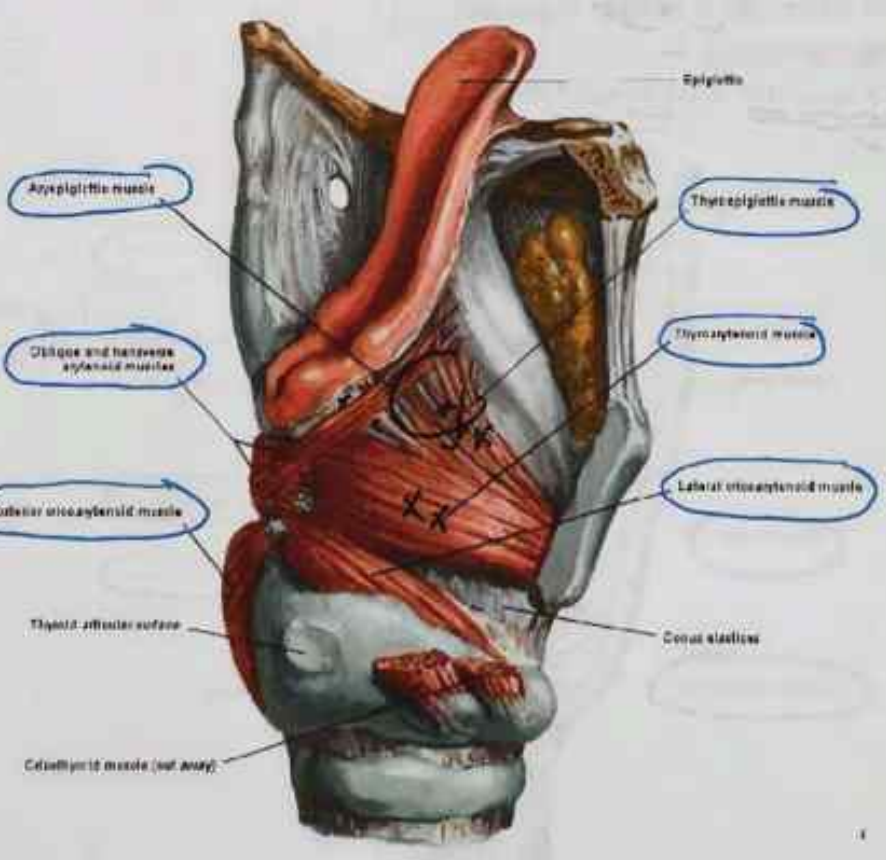
They have no extra laryngeal attachment (connecting different cartilages together, so most of them are paired except transverse arytenoid muscles).

- **All of them lie from inside except cricothyroid that lies externally**
- All of them are developed from **6th arch except cricothyroid** which is developed from **4th arch**.
  - 1. The **cricothyroid**.
  - 2. The **posterior** & **lateral crico-arytenoid**.
  - 3. The **transverse** & **oblique arytenoid**.
  - 4. **Thyro-arytenoid**.
  - 5. Ary-epiglottic & thyro-epiglottic



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Action of laryngeal muscles

1) **Muscles controlling laryngeal orifice:**
   - Closing the inlet: aryepiglottic & interarytenoid muscles
   - Widening the laryngeal inlet: thyro-arytenoid muscle

2) **Muscles stretch the vocal cords:**
   - Cricothyroid muscle

3) **Muscles relax the vocal cords:**
   - Thyro-arytenoids.

4) **Muscles abduct the vocal cord:**
   - Posterior crico-arytenoid.

5) **Muscles adduct the vocal cords:**
   - Transverse arytenoid & lateral crico-arytenoid.



N.B.

- All intrinsic laryngeal muscles are supplied by recurrent laryngeal n. Branch of vagus
- except cricothyroid which is supplied by external laryngeal n. Branch of superior laryngeal nerve of vagus

Joints of the larynx

1- **crico-thyroid joint:**

- synovial joint between inferior horn of thyroid cartilage and side of cricoid cartilage.

2- **Crico-arytenoid joint:**

- Synovial joint between upper border of cricoid lamina and base of arytenoid cartilage.

Supply of the larynx

Above vocal cords	Below vocal cords

Arterial

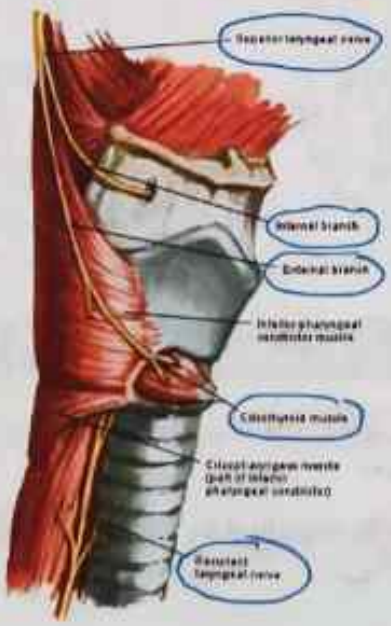
Superior laryngeal artery (Branch of superior thyroid)	Inferior laryngeal artery (Branch of inferior thyroid)

Lymphatics

Upper deep cervical LNs	Lower deep cervical LNs (and some in prelaryngeal LNs)

Sensory nerve supply

Internal laryngeal nerve	Recurrent laryngeal nerve



The Trachea

- The trachea is a fibromuscular tube 10-12 cm long.
- It is kept patent by the presence of C-shaped bars (rings) of hyaline cartilage embedded in its wall.
- It is not quite cylindrical but is flattened posteriorly, So that in transverse section its lumen is like a letter D

Length

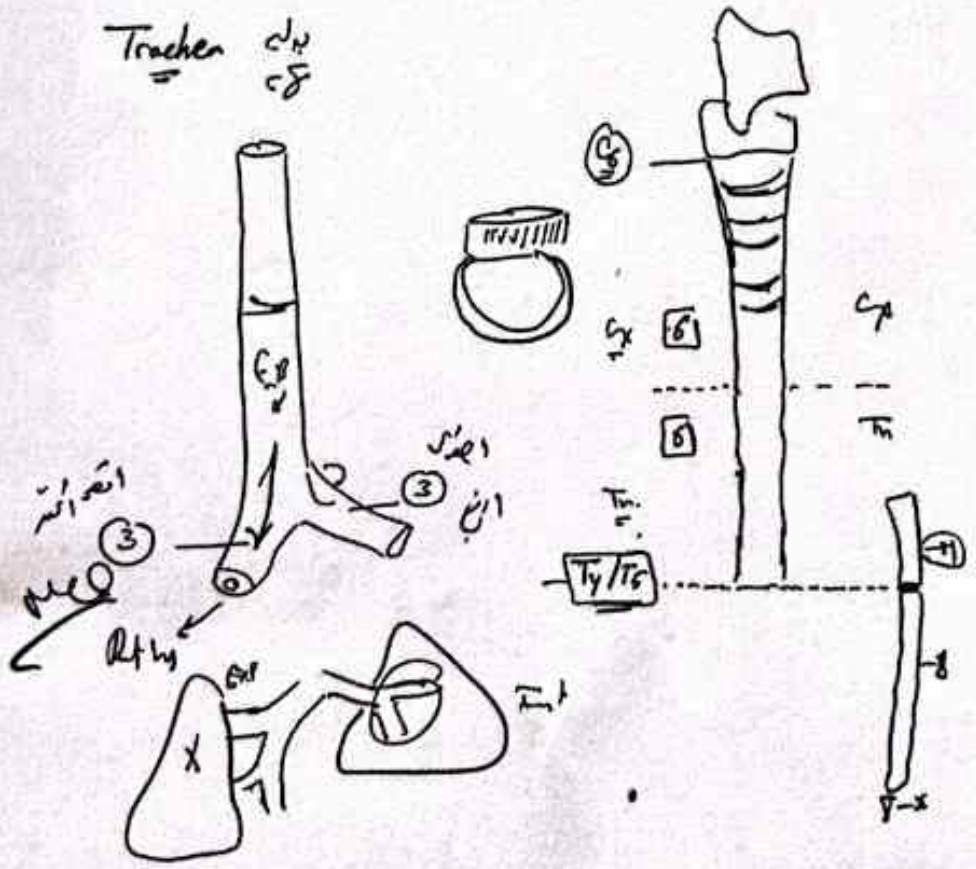
- is about 10-12 cm.

Beginning

- It begins in the neck as a continuation of the larynx at the level of the sixth cervical vertebra.

Termination

- In the thorax, the trachea ends at the level of the sternal angle by dividing into the right and left main bronchi.
- The tracheal bifurcation (carina) is usually a little to the right of the medianplane.
- During deep inspiration, the bifurcation of the trachea may descend to T6 vertebra.



Relations of the trachea

Cervical part of the trachea:



Anteriorly:

- Thyroid isthmus covers 2, 3 & 4 tracheal rings.
- Sternohyoid & sternothyroid muscles.
- Inferior thyroid veins descend over its lower part.

On either side:

- Lobe of thyroid gland
- CCA
- Brachiocephalic artery on the right side.



Posteriorly:

- Esophagus
- Recurrent laryngeal nerve ascending in the trachea-esophageal groove.

Thoracic part of the trachea:



Anteriorly:

- Deep part of cardiac plexus and lymph nodes on the tracheal bifurcation.
- Arch of aorta and beginning of brachiocephalic and left common carotid arteries.
- Inferior thyroid and left brachiocephalic veins.
- Remnants of thymus.
- Origin of sternohyoid and sternothyroid muscles.
- Manubrium sterni.

On the right side:

- Right vagus
- Arch of azygos.
- Right brachiocephalic vein and SVC.
- Right pleura and lung.



On the left side:

- Arch of aorta.
- Left common carotid & left subclavian artery.
- Left recurrent laryngeal nerve: is at first situated between the trachea and arch of aorta, then in the groove between the trachea and the esophagus.

Posteriorly:

- The esophagus separating it from the vertebral column.

Supply of trachea

Arterial

- Inferior thyroid artery
- bronchial arteries.

Venous drainage

- Left brachiocephalic veins.

Lymphatic drainage:

- Pretracheal and paratracheal lymph nodes.

Nerve supply:

Parasympathetic

- from the vagi and recurrent laryngeal nerves

sympathetic

- from sympathetic trunks.

The bronchi

- The right and left principal bronchi start at the tracheal bifurcation opposite the lower border of T4 vertebra.
- Each bronchus passes obliquely lying in the middle mediastinum to enter the hilum of the corresponding lung and divides into lobar bronchi which give rise to segmental bronchi.

N.B.

- **A foreign body which may enter the trachea, will pass usually in the right bronchus because it is wider and more in line with the trachea.**

	Right bronchus	Left bronchus
Length	Short (2.5 cm)	Long (5 cm)
Diameter	Wide	Narrow
Course	Vertical, in line with Trachea	Oblique or horizontal
Division into lobar Bronchi	Extra pulmonary	intrapulmonary

Joints of the Thoracic Wall

joints of The sternum

The manubriosternal joint

- a cartilaginous joint between the manubrium and the body of the sternum. T4/T5 2nd CC
- A small amount of angular movement is possible during respiration.

The xiphisternal joint

- a cartilaginous joint between the xiphoid process and the body of the sternum. T9/T6 7th CC
- The xiphoid process usually fuses with the body of the sternum during middle age.







Costovertebral joints:

Joints of the heads of the ribs:

The head of each rib from the 2nd to the 9th ribs

- Articulates by means of a synovial joint with the costal facet of the corresponding vertebral body and that of the vertebra above it (e.g. the head of the 6th rib articulates with the superior part of the body of T6 the inferior part of T5 and the intervertebral disc between these vertebrae).

The 1st rib and the 10th, 11th, and 12th ribs

- Have a single synovial joint with their corresponding vertebral body. Radiate fig.
- **There is a strong intra-articular ligament that connects the head to the intervertebral disc.**
- **Only slight gliding movements occur at the demifacets, pivoting around the intra-articular ligament of the joints of the heads of ribs.**

Costotransverse joints (joints of the tubercles of the ribs):

- The tubercle of a rib articulates by means of a synovial joint with the transverse process of the corresponding vertebra.
- This joint is absent on the 11th and 12th ribs

Supporting ligaments:

Costotransverse ligament

- passing from the neck of the rib to the transverse process

Lateral costotransverse ligament

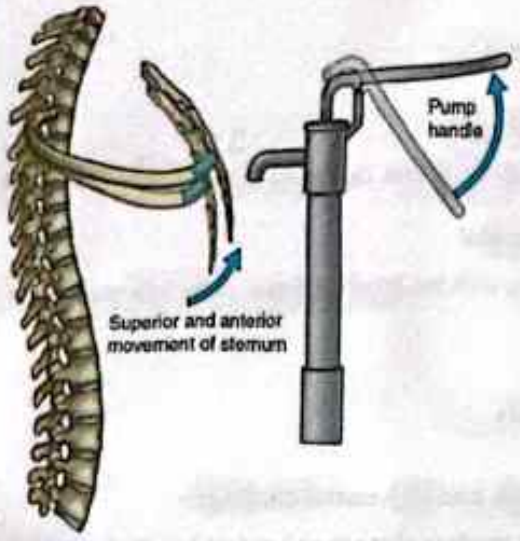
- passing from the tubercle of the rib to the tip of the transverse process

Superior costotransverse ligament

- is a broad band that joins the crest of the neck of the rib to the transverse process superior to it

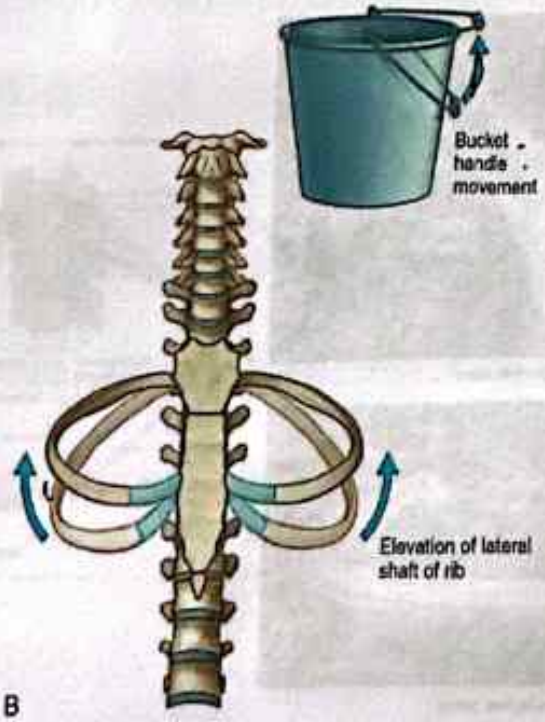
Rotation movement

- Occurs around a transverse axis that traverses the head and neck of the upper 6 ribs
- this results in elevation and depression movements of the sternal ends of the ribs and the sternum causing the pump-handle movement.



Gliding movement

- Of the joints of the 7th -10th ribs result in elevation and depression of the lateral-most portions of these rib in the transverse plane causing the bucket-handle movement



B

Costochondral joints (joints of The r ibs and costal cartilages)

- These joints are cartilaginous joints.
- No movement normally occurs at these joints.

Sternocostal joints

The first costal cartilages

- articulate with the manubrium, by cartilaginous joints that permit No movement.

The 2nd to the 7th costal

- cartilages articulate with the lateral border of the sternum by synovial joints.

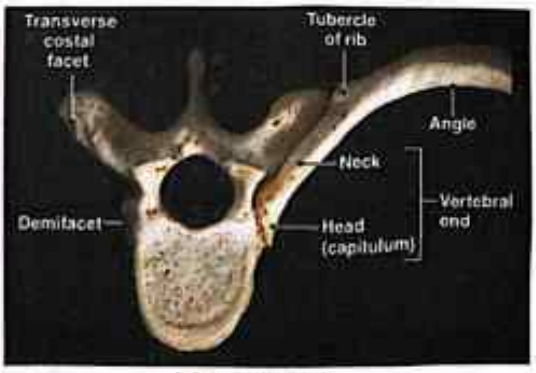
Interchondral joints

The 6th , 7th, 8th, 9th, and 10th costal cartilages

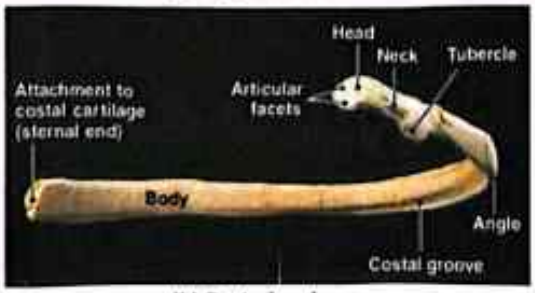
- articulate with one another along their borders by small synovial joints.

The cartilages of the 11 th and 12th ribs

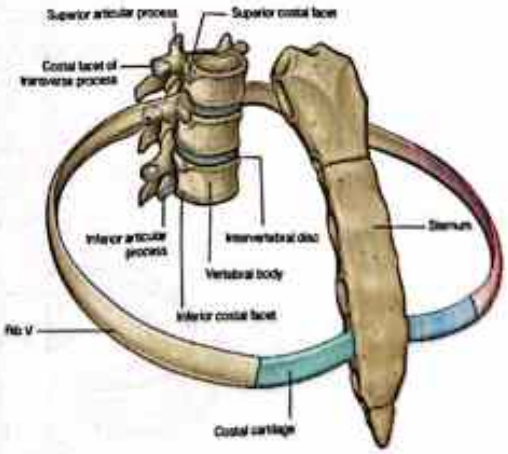
- are embedded in the abdominal musculature.



(a) Superior view



(b) Posterior view



The intercostal muscles

Connect adjacent ribs and are arranged into 3 layers from out inward:

1. External intercostal muscles:

Extent:

- from the tubercle of the ribs posteriorly to the costochondral junction anteriorly where it is replaced by an aponeurosis, the anterior (external) intercostal membrane.

Attachments:

- Each muscle passes from the lower border of one rib to the upper border of the rib below.

Direction of fibers:

- downwards & forwards (as one putting his hand in his pocket).

2. Internal intercostal muscles:

Forms the intermediate layer:

Extent:

- from the sternum infront to the angle of the rib behind where each is replaced by internal (posterior) intercostal membrane.

Attachments:

- Each muscle descends from the floor of the costal groove of one rib to the upper border of the rib below.

Direction of fibers:

- downwards, backwards i.e. at right angles to those of the external intercostal muscle.



3. Innermost intercostal muscles:

Forms the deepest layer:

It is considered to be the deeper part of the internal intercostal muscle which is split off by the intercostal vessels & nerves.

Int / Inn

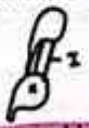


Extent:

- occupy the middle 2/4 of the intercostal spaces.

Attachments:

- Each muscle is attached to internal aspects of two adjoining ribs (from the upper border of the costal groove of one rib to the upper border of the rib below).



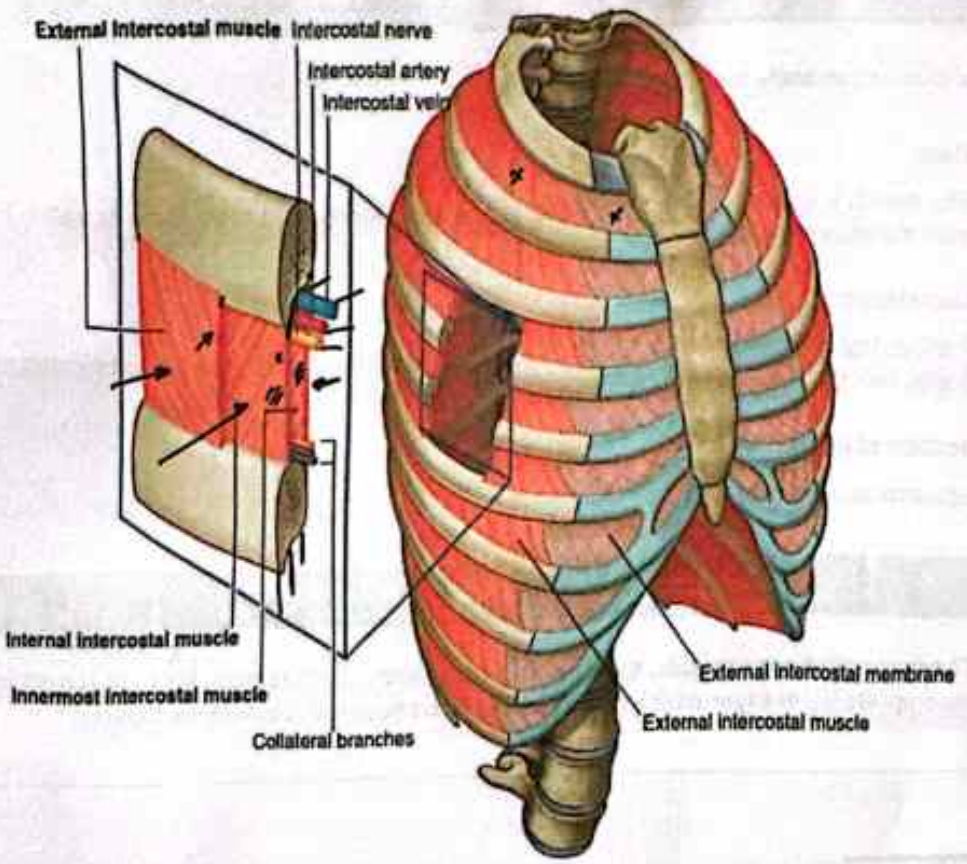
Direction of fibers:

- As internal intercostal; actually it is considered to be a part of the internal intercostal which is split off by the intercostal nerves and vessels.

Note

- **The innermost intercostal muscles are related internally to the endothoracic fascia and parietal pleura and externally to intercostal nerves and vessels.**





Subcostal Muscles (Subcostales):

These muscles cross 2-3 ribs between their attachments.

Extent:

- They are well developed only in the lower and posterior part of the thorax lying near the angles of the ribs.

Attachments:

- like those of the internal intercostal muscles.

Direction of fibers: ✗

- As internal intercostal; actually it is considered to be a part of the internal intercostal which is split off by the intercostal nerves and vessels.

Note

- **The subcostales, like innermost intercostals is related internally to fascia (endothoracic fascia) and parietal pleura and externally to the intercostal nerves and vessels.**

Transversus Thoracis (Sternocostalis):

This muscle connects the ribs to the sternum.

Extent:

- The muscle is present on the internal surface of the anterior part of the thoracic wall in line with the innermost intercostal.

Attachments:

- it arises from the lower half of the posterior surface of the body of sternum to be inserted by 5 slips into the lower border of the costal cartilages from the 2nd to the 6th.

Direction of fibers:

- upwards and laterally.

Note

- **The innermost intercostals, subcostales and sternocostalis muscles form together an incomplete inner layer which corresponds to transversus abdominis muscle.**

Nerve supply of muscles of The Thorax:

- All the above mentioned muscles are supplied by the adjacent intercostal nerves.

Action of muscles of The Thorax:

The external intercostal muscles

- are most active in inspiration (elevators of the ribs).

The internal and innermost intercostal muscles

- are most active in expiration (depressors of the ribs).

Subcostales muscle:

- probably depresses the ribs.

Transversus thoracis:

- draws down the costal cartilages to which it is attached.

Notes:

- The role of these muscles in producing movement of the ribs appears to be related mainly to forced respiration.
- The diaphragm is the primary muscle of inspiration.
- The intercostal muscles support the contents of the intercostal spaces preventing them from being drawn in during inspiration or bulged out during expiration.

Intercostal Nerves

- There are 12 pairs of thoracic spinal nerves.
- The ventral rami of these nerves form the intercostal nerves (the upper eleven) and the subcostal nerve (the 12th).
- Each intercostal nerve runs in the corresponding intercostal space while the subcostal nerve runs below the last rib.

The intercostal nerves are classified as:

- Typical intercostal nerves which have a similar course (T3 - T6).
- Atypical intercostal nerves which have different courses (T1, T2 and T7 - T11).

1. Typical intercostal nerves (T3-T6):

Course

- Each nerve emerges from the corresponding intervertebral foramen and runs in the posterior part of the intercostal space passing between the parietal pleura and the posterior intercostal membrane.
- From the angle of the rib to the costo-chondral junction, the nerve runs in the costal groove below the intercostal vein and artery (VAN) passing between the innermost intercostal muscle and the internal intercostal muscle.

Branches of a typical intercostal nerve

1. Rami communicantes:

- the intercostal nerve gives a white ramus communicans (preganglionic) to sympathetic trunk & receives a grey ramus communicans (postganglionic) from the sympathetic trunk.
- The gray ramus joins the nerve medial at the point at which the white ramus leaves it.

Note:

- **Grey rami communicantes are present in all spinal nerves while the white rami communicantes are present only in the thoracic and upper 2-3 lumbar nerves.**

2. Muscular branches:

- to the intercostal muscles and transversus thoracis.

3. A collateral branch:

- runs along the upper border of the rib below.

4. Pleural sensory branches

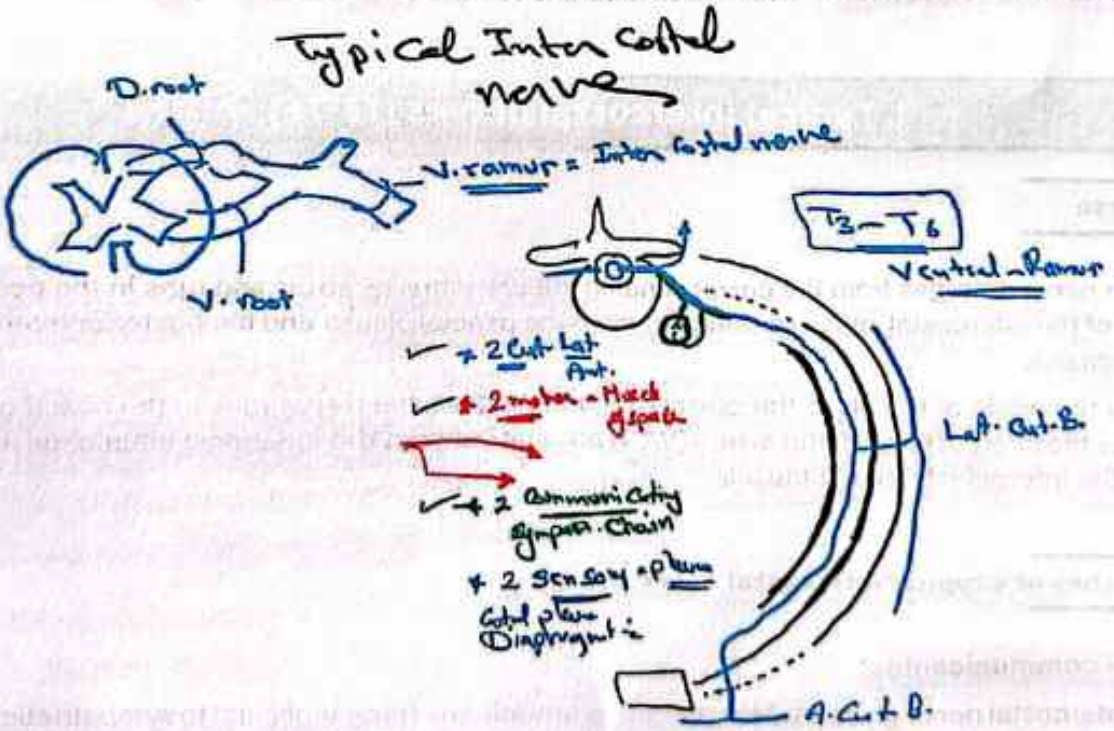
- go to the parietal pleura.

5. Lateral cutaneous branch:

- pierces the intercostal muscles and digitations of serratus anterior and emerges on the lateral side of the trunk dividing into an anterior and a posterior branch to supply the skin.

6. Anterior cutaneous branch:

- the terminal branch that appears near the sternum and divides into medial and lateral branches which supply the overlying skin.



2. Atypical intercostal nerves:

1. First intercostal nerve (T1):

- It divides unequally into a large branch that ascends across the neck of the 1st rib to share in the brachial plexus.
- The smaller 1st intercostal nerve runs in the 1st intercostal space and ends as the 1st anterior cutaneous nerve of the thorax.
- It gives no lateral cutaneous branch.

2. Second intercostal nerve (T2):

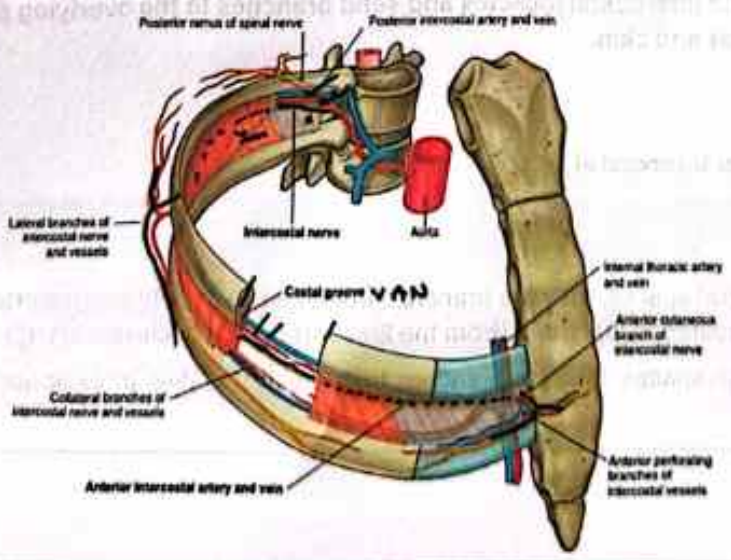
- Its lateral cutaneous branch is known as intercostobrachial nerve, which runs in the axilla communicating with the medial cutaneous nerve of the arm and supplies the skin of the floor of axilla and upper part of the medial side of the arm.

3. Lower 5 intercostal nerves (T1-T11):

- They pass between the costal slips of diaphragm to the anterior abdominal wall to supply muscles and skin.
- They also supply the parietal peritoneum.

Note:

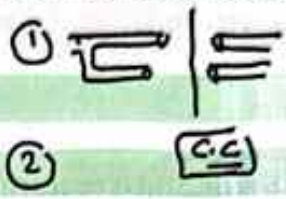
- The ventral ramus of T12 (subcostal) nerve also supplies the muscles and the skin of the anterior abdominal wall.



Arteries of the Thoracic Wall

I. The intercostal arteries:

- **The intercostal arteries course through the thoracic wall between the ribs.**
- **With the exception of the 10th and 11th intercostal spaces, each intercostal space is supplied by:**



A large posterior intercostal artery and its collateral branch .

A small pair of anterior intercostal arteries.

1. The anterior intercostal arteries:

They are 9 pairs for the upper 9 intercostal spaces.

- In the upper 6 intercostal spaces, they are branches from the internal thoracic artery.

In 7th 8th and 9th intercostal spaces

- they are branches from the musculophrenic artery which is one of the terminal branches of the internal thoracic artery .

The 10th and 11th intercostal spaces

- have NO anterior intercostal arteries because they are incomplete anteriorly.

- **In each space the 2 anterior intercostal arteries start from the anterior end of the space, run laterally then terminate at the costo-chondral junction by anastomosing with the posterior intercostal artery and its collateral branch of the same space.**
- **They supply the intercostal muscles and send branches to the overlying pectoral muscles, breast and skin.**

2. The posterior intercostal arteries:

Origin:

- In the 1st and 2nd spaces, they are branches from the superior intercostal artery which is a branch of the costocervical trunk (from the 2nd part of the subclavian artery).
- In the 3rd to 11th spaces, they are branches from the descending thoracic aorta.

Course:

- Each artery lies at first between the pleura and internal intercostal membrane as far as the costal angle, then passes between the internal intercostal and innermost intercostal muscles (at the neurovascular plane).
- In the costal groove it runs below its vein and above the intercostal nerve (VA)
- At the angle of the rib, it gives a collateral branch which runs along the upper border of the rib below.
- At the costo-chondral junction, the posterior intercostal artery and its coffateraf branch end by anastomosing with the anterior intercostal branches from the internal thoracic or musculophrenic artery.

Note:

- The 10th and 11th posterior intercostal arteries terminate in the muscles of the abdominal wall due to the absence of 10th and 11th anterior intercostal arteries.

ii. Internal thoracic artery

Origin:

- From the 1st part of the subclavian artery in the root of the neck.

Course:

- It descends vertically behind the upper 6 costal cartilages one cm from the lateral border of the sternum.

Termination:

- In the 6th intercostal space, it divides into musculophrenic and superior epigastric arteries.

Branches:

1. Pericardiacophrenic artery:

- a long slender branch accompanying the phrenic nerve to the diaphragm supplying twigs to pericardium, pleura and diaphragm.

2. Pericardial branches.

3. Anterior intercostal branches:

- two arteries for the upper 6 intercostal spaces.

4. Perforating branches:

- accompany the anterior cutaneous nerves
- The perforating branches of the 2nd, 3rd and 4th intercostal spaces supply the mammary gland in female and enlarge during lactation.

5. Musculophrenic artery:

one of the 2 terminal branches.

- It passes in ferolaterally behind the 7th to 9th costal cartilages.
- It traverses the diaphragm to supply it and the abdomin al muscles.
- It gives 2 anterior intercostal arteries for each of 7th, 8th and 9th intercostal spaces.

6. Superior epigastric artery:

is the 2nd terminal branch of the internal thoracic a.

- It descends between the costal & sternal origin of the diaphragm to enter the rectus sheath.
- It supplies the rectus abdominis and anastomoses with the inferior epigastric artery.
- It also supplies the abdominal skin and the diaphragm.

i. intercostal veins:

1. Anterior intercostal veins:

- Each of the upper 9 intercostal spaces is drained by 2 anterior intercostal veins.

Veins of the 7th, 8th and 9th spaces

- are drained into the venae comitantes of the musculophrenic artery

Those of the 6th, 5th, 4th , 3rd spaces

- are drained into the venae comitantes of the internal thoracic artery which unite to form the internal thoracic vein at the level of the 3rd costal cartilage.

The anterior intercostal veins of the upper 2 spaces drain directly into the internal thoracic vein which drains into the brachiocephalic vein.

2. Posterior intercostal veins:

- **They are 11 pairs; each runs posteriorly in th e costal groove above its artery (VAN).**
- **They have different arrangement and mode of drainage on both sides.**

a- On the right side:

The 1st right posterior intercostal vein

- ends in the right brachiocephalic vein.

The 2nd and 3rd right posterior intercostal veins

- unite to form right superior intercostal vein which ends in the arch of azygos vein.

The posterior intercostal veins from the 4th - 11th space

- ends separately in the azygos vein.

b- On the left side:

The 1st left posterior Intercostal vein

- ends in the left brachiocephalic vein.

The 2nd and 3rd left posterior intercostal veins

- unite to form left superior intercostal vein which ends in the left brachiocephalic vein.

The posterior intercostal veins from 4th to 8th space

- ends separately in the accessory (superior) hemiazygos vein which descends then turns to the right crossing the body of the 7th thoracic vertebra to joint the vena azygos.

The posterior intercostal veins from 9th to 11th space

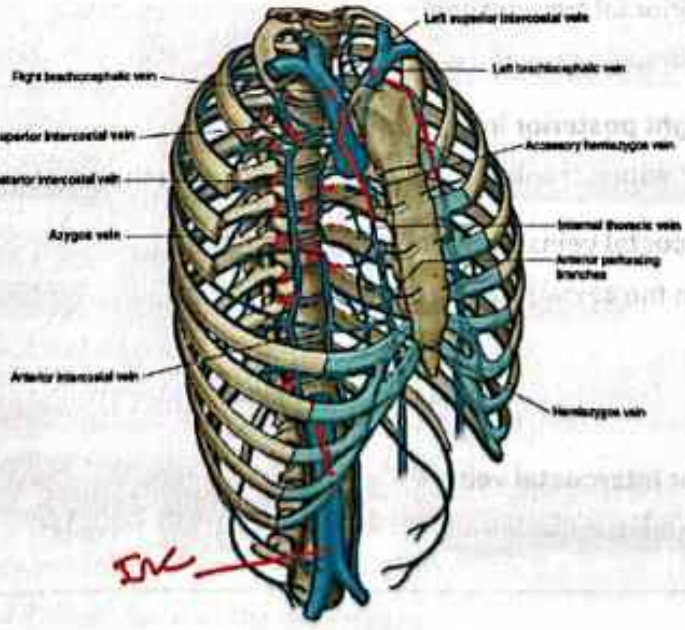
- ends separately in the hemiazygos (inferior hemiazygos) vein which arises from the back of the left renal vein, pierces the left crus of diaphragm to enter the thorax then crosses the body of the 8th thoracic vertebra to join the azygos vein.

Note:

- **The accessory hemiazygos and hemiazygos veins may be together by venous loop before crossing to the right side.**

ii. Internal thoracic vein:

- The venae comitantes of the superior epigastric and musculophrenic arteries unite at the anterior end of 6th intercostal space to form the venae comitantes of the internal thoracic artery.
- On each side, the 2 venae comitantes of the internal thoracic artery unite at the 3rd costal cartilage to form a single internal thoracic vein which ends in the corresponding brachiocephalic vein.



Diaphragm

- The diaphragm is a musculofibrous sheet that separates the thoracic cavity from the abdominal cavity.
- Its convex upper surface facing the thoracic cavity and its concave inferior (lower) surface is directed towards the abdominal cavrty.
- The central part of the diaphragm is depressed inferiorly than the lateral parts which are called cupolae

Origin

- The muscle fibres of the diaphragm arise from the oblique circumference of the thoracic outlet.
- Although it is a continuous sheet, the diaphragm can be considered to be formed of three parts; sternal, costal and lumbar (vertebral).

The sternal part:

- arises by 2 fleshy slips from the back of the xiphoid process.

The costal part:

- arises from the internal surfaces of the lower 6 costal cartilages and their adjoining ribs.

The lumbar (vertebral) part:

- arises from lumbar vertebrae by 2 crura (right crus and left crus) and by 2 aponeurotic arches, the medial and lateral arcuate ligaments on each side.

The Crura are thick fleshy bundles that taper inferiorly:

The right crus:

- is broader and longer than the left. It arises from the anterolateral surfaces of the bodies of the upper 3 lumbar vertebrae and intervertebral discs.

The left crus:

- arises from the anterolateral surfaces of the upper 2 lumbar vertebrae and intervertebral disc.

Note:

- ***Medial fibres of the right crus embrace the oesophagus where it passes through the diaphragm.***

The arcuate ligaments:

The median arcuate ligament:

- is an arch in the midline across the front of the aorta at the level of the disc between T 12 and L1 vertebrae connecting the medial tendinous margins of the crura.

The medial arcuate ligament:

is a tendinous arch in the fascia that covers the upper part of psoas major.

- **Medially,** It is continuous with the lateral margin of the corresponding crus, so it is attached to the side of the body of L1 or L2 vertebra.
- **Laterally,** it is attached ro the front Of the transverse process of L1 vertebra.

The lateral arcuate ligament:

a thickened band of fascia that arches across Upper part of quadratus lumborum muscle.

- **Medially** attached to the front of the transverse process of L1 vertebra
- **Laterally** attached to lower margin of the 12th rib near its midpoint.

The central tendon of the diaphragm

- **The fibers of origin of diaphragm converge to be inserted into a sreong aponeurotic tendon called the central tendon.**
- **It is semi lunar in shape and divided by slight indentations into 3 leaflets: 1 middle & 2 lateral.**



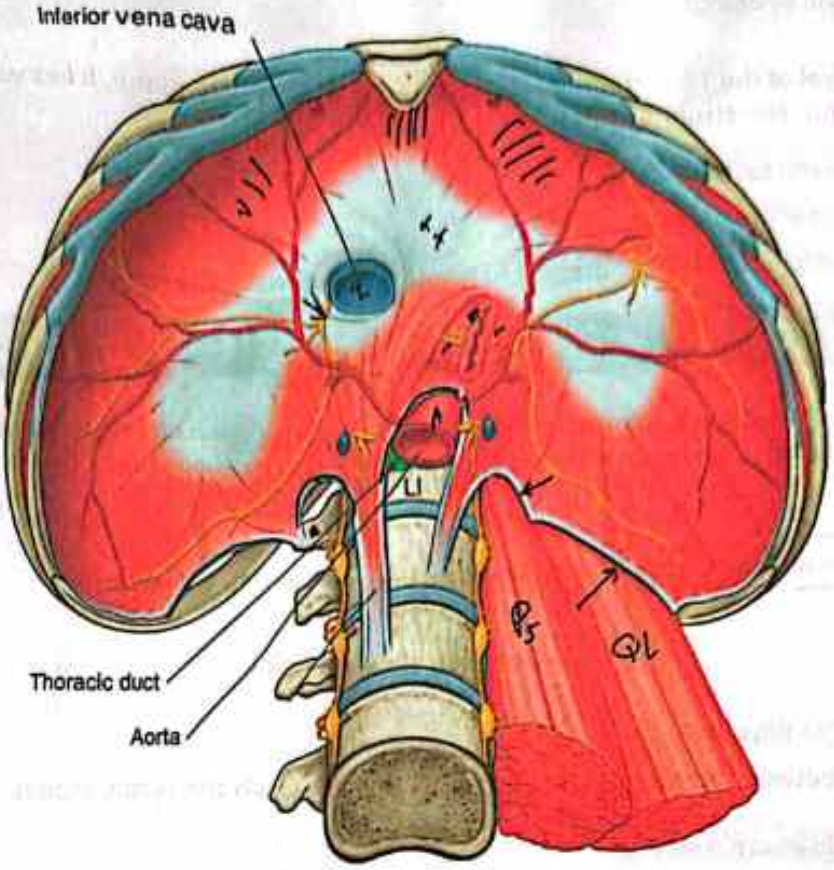
Relations of The diaphragm

Superiorly the diaphragm is related to:

- **The right cupola** is related to the right pleura and base of the right lung.
- **The left cupola** is related to the left pleura and base of the left lung.
- **The median part** of the central tendon Is related to the pericardium and the heart.

Inferiorly the diaphragm is related to:

- **The right cupola** is related to the right lobe of the liver, right kidney and right supra renal gland.
- **The left cupola** is related to the left lobe of the liver, fundus of the stomach, left kidney, left suprarenal gland and spleen.
- **The right cupola** reaches at a slightly higher level than the left cupola due to upward bulge of right lobe of the liver.



Major openings in The diaphragm

a- Aortic opening:

It is at the level of the lower border of the T12 vertebra, nearly in the median plane posterior to the median arcuate ligament. the structures passing through are:

- 1. Aorta (descending).
- 2. Thoracic duct.
- 3. Azygos vein.

b- Oesophageal opening:

It is at the level of the T10 vertebra one inch to the left of the median plane, within the fleshy fibres of the right crus which have a sphincteric action on the lower end of the oesophagus. the structures passing through are:

- 1. Oesophagus.
- 2. Gastric nerves (anterior and posterior).
- 3. Oesophageal branches of the left gastric vessels.

c- Vena caval opening:

It is at the level of the T8 vertebra one inch right to the median plane. It lies within the central tendon. The structures passing through are:

- 1. Inferior vena cava (I.V.C)
- 2. Branches of the right phrenic nerve.
- 3. Some lymph vessels from the liver to lymph nodes of the thorax.

Note:

- **The central tendon is adherent to the wall of the inferior vena cava thus keeping it always patent and its lumen is not affected by contraction of the diaphragm.**

Minor Foramina of The diaphragm and structures passing through

a- Between slips of origin:

1. Superior epigastric artery:

- passes between the sternal and the costal origins to reach the rectus sheath.

2. Musculo phrenic artery:

- pierces the diaphragm near the 9th costal cartilage

3. Lower 5 intercostal nerves:

- each passes between 2 slips of costal origin reach the anterior abdominal wall.

b- Behind the lateral arcuate ligament:

- 1. Subcostal vessels and nerve.
- 2. Quadratus lumborum muscle.

c- Behind the medial arcuate ligament:

- 1. Sympathetic trunk: continues from thoracic cavity to abdomina l cavity.
- 2. Psoas major muscle.

d- Through minute openings in the crura:

- 1. Greater and lesser splanchnic nerves via 2 openings in each crus.
- 2. Hemiazygos vein: pierces the left crus of diaphragm to enter the thoracic cavity.

Note:

- **The left phrenic nerve passes through the muscular part of the diaphragm anterolateral to the left part of the central tendon.**

Nerve supply of The diaphragm

Motor supply:

- Right and left phrenic nerves (C 3, 4, 5 mainly C4).
- These nerves pierce the diaphragm to supply it from its inferior surface.

Sensory supply:

- branches from phrenic and lower intercostal nerves.

Arterial supply

- Inferior phrenic arteries.
- Musculophrenic arteries.
- Pericardiacophrenic arteries.
- Twigs from lower 3 posterior intercostal arteries.
- Twigs from descending thoracic aorta.

Action of The diaphragm

It is the chief muscle of respiration.

- It is active during forced expulsive acts, e.g. coughing, vomiting, defecation, urination and parturition.

Movements of Respiration:

Respiration consists of two phases; inspiration and expiration during which an increase and a decrease of the capacity of the thoracic cavity occur respectively.

1. Inspiration:

The diaphragm is the major muscle of inspiration.

In Quiet inspiration: all the diameters of the thoracic cavity will increase

Vertical diameter

- is increased by contraction (descent) of the diaphragm.

Anteroposterior diameter

- is increased by contraction of intercostal muscles which raise the sternal ends of the downward-sloping ribs. Also the sternum will be pushed upward and forward increasing this diameter (pumphandle movement).

Transverse diameter

- Is increased by contraction of the intercostal muscles which elevate the middle of the shaft of the 7th and false ribs which lie in a lower level than their anterior and posterior ends (bucket-handles movement).

In forced inspiration:

There is a maximum increase in the capacity of the thoracic cavity. Every muscle that can raise the ribs is brought into action e.g.:

- Scalenus anterior and medius elevate the first rib.
- Sternocleidomastoid acts directly through the manubrium stemi and indirectly through the clavicle.
- The scapulae are fixed by the trapezius, levator scapulae and rhomboid muscles which enable the serratus anterior and pectoralis minor to pull up the ribs.
- Quadratus lumborum fixes the 12th rib so that the diaphragm can exert a more powerful downward push on the abdominal viscera.

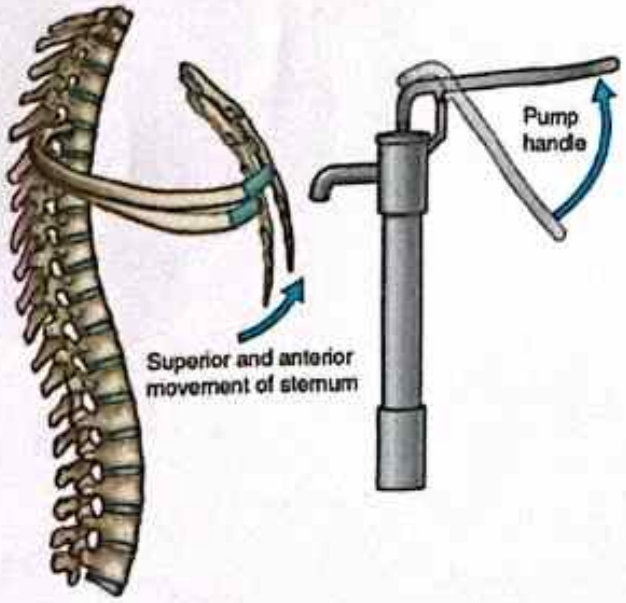
2. Expiration:

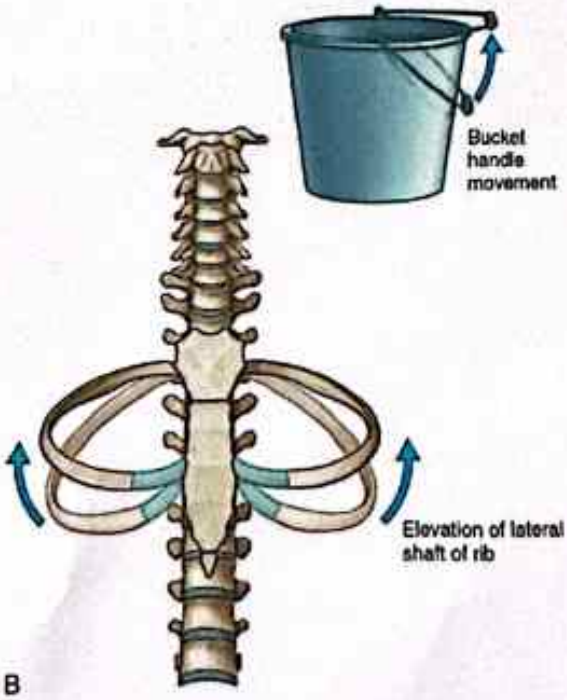
Quiet expiration is passive & achieved by:

- Elastic recoil of the lungs.
- Relaxation of the intercostal muscles and diaphragm.

Forced expiration:

- It is an active process during which the muscles of the anterior abdominal wall and latissimus dorsi contract strongly pushing the diaphragm upwards and drawing the lower ribs downwards and inwards.





B

Anatomy Of Pleura And Lung

The Pleura

- It is a completely closed serous sac which is invaginated on its medial side by the lung.
- It has 2 layers (visceral and parietal) separated by the pleural cavity.

To understand this, push your fist into an underinflated balloon. Your fist (lung) will be covered by a wall of the balloon (visceral pleura) and the outer wall of the balloon will be the parietal pleura, while the cavity of the balloon will be the pleural cavity.

- The pleural cavity is a potential, slit-like, space.
- The pleural cavity normally contains a small amount of tissue fluid, the pleural fluid, which covers the surfaces of the pleura as a thin film and permits the two layers to move on each other with the minimum of friction.



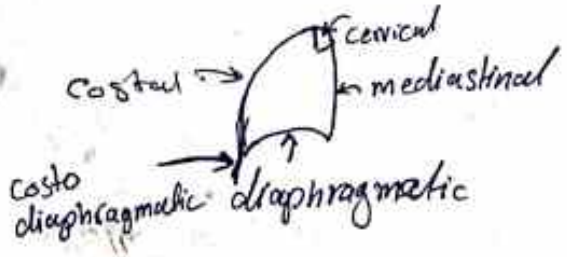
- The two layers become continuous with one another by means of a cuff of pleura that surrounds the structures entering and leaving the lung (e.g. bronchus, pulmonary vessels) that form the root of the lung.
- To allow for movement of the pulmonary vessels and large bronchi during respiration, this pleural cuff hangs down as a loose fold called the pulmonary ligament.

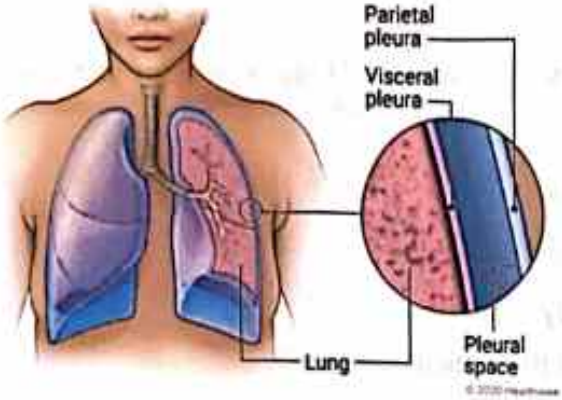
The visceral (pulmonary) pleura

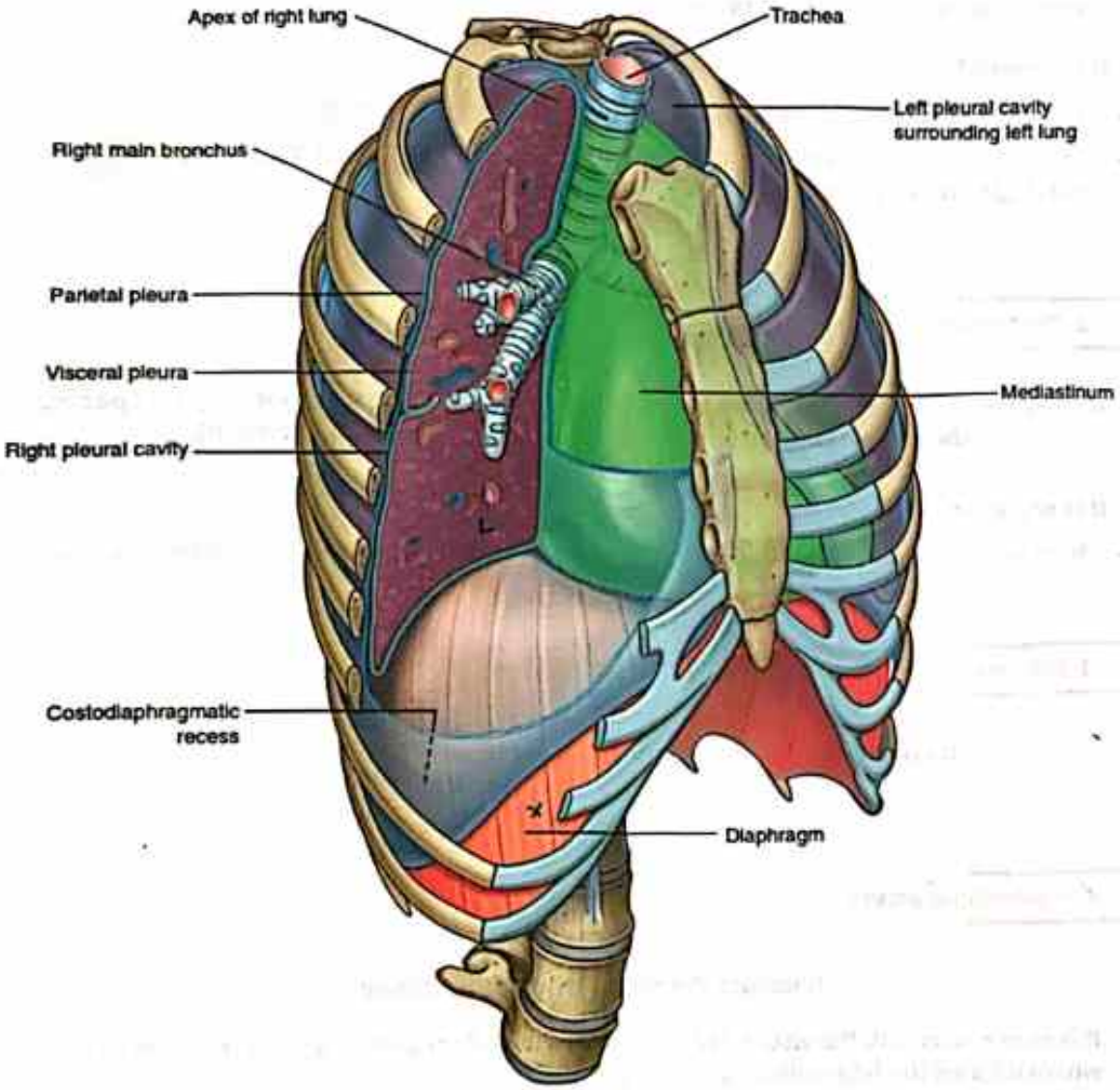
- It is adherent to all surfaces of the lung and extends into its fissures.
- Medially, around the hilum of the lung, and for a short distance below it, the visceral pleura leaves the surface of the lung to become continuous with the parietal pleura.

The parietal pleura

- For purposes of description, the parietal pleura is divided according to the region in which it lies or the surface that it covers into cervical, costal, diaphragmatic and mediastinal pleurae.




Parietal pleura
Visceral pleura
Lung
Pleural space


Apex of right lung
Trachea
Left pleural cavity surrounding left lung
Right main bronchus
Parietal pleura
Visceral pleura
Mediastinum
Right pleural cavity
Costodiaphragmatic recess
Diaphragm

1. Cervical pleura (dome of the pleura):

it extends over the apex of the lung and bulges up through the inlet of the thorax into the root of the neck.

It is crossed anteriorly

- by the subclavian artery.

It is related posteriorly

to structures in front of the neck of the 1st rib:

- the stellate ganglion
- superior intercostal artery
- ventral ramus of 1st thoracic nerve.

It is covered

- by a fan-shaped tough suprapleural membrane (Sibson's fascia).
- The latter is attached laterally to the inner border of the 1st rib and medially to the 7th cervical transverse process

2. Costovertebral (Costal) pleura:

it lines the inner wall of thoracic cavity (ribs, costal cartilages, the intercostal spaces, the sides of the vertebral bodies, and the back of the sternum).

It is separated

- from the thoracic wall by a thin layer of loose connective tissue called endothoracic fascia

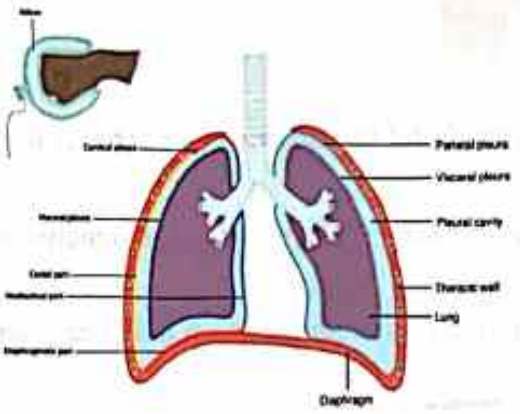
3. Diaphragmatic pleura

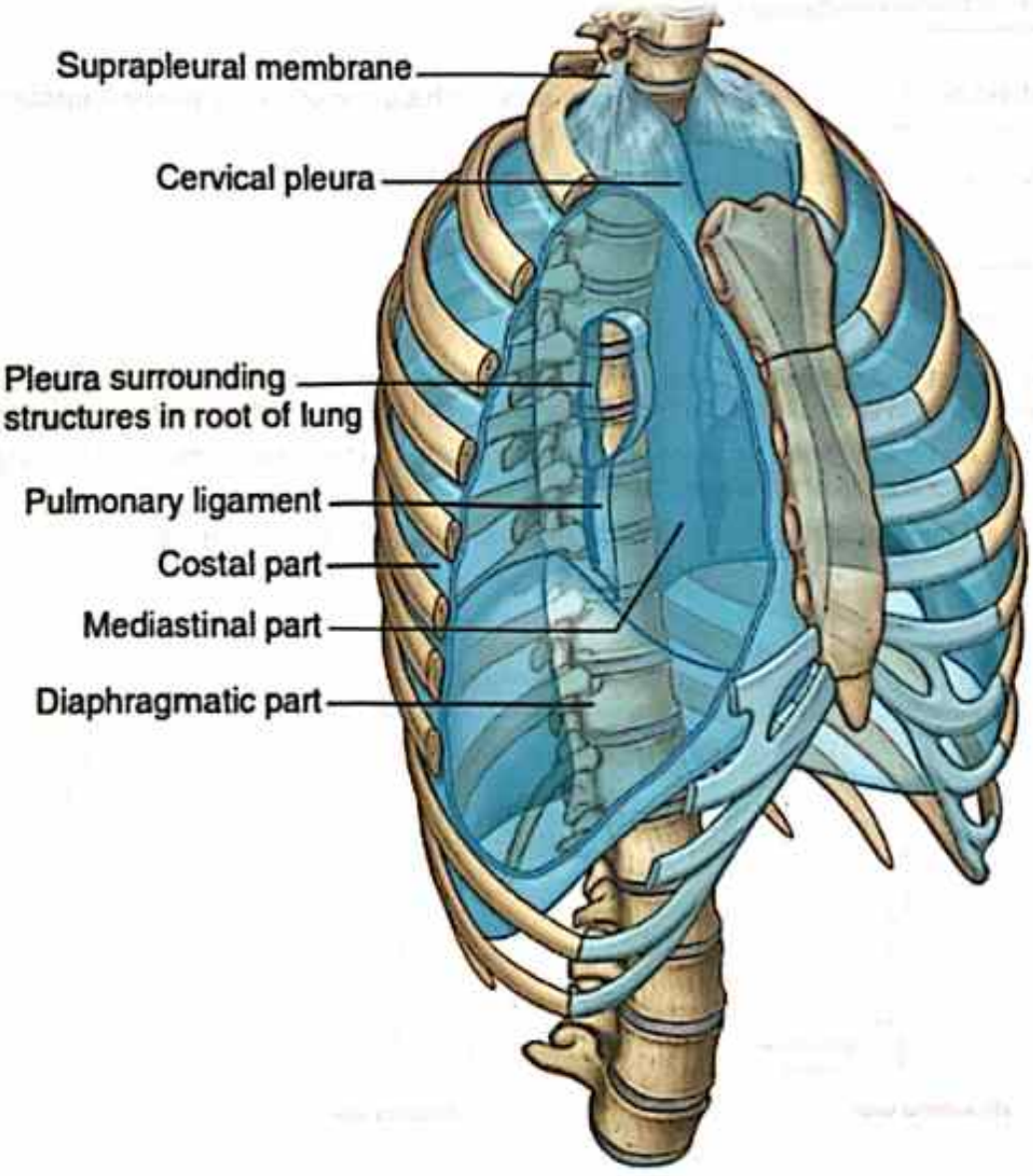
It covers the upper surface of its own half of the diaphragm.

4. Mediastinal pleura

It covers the side of the mediastinum.

- It is continuous with the visceral pleura around the hilum and for a short distance below it where it forms the fold called pulmonary ligament.


Parietal pleura
Visceral pleura
Pleural cavity
Thoracic wall
Lung
Diaphragm


Suprapleural membrane
Cervical pleura
Pleura surrounding structures in root of lung
Pulmonary ligament
Costal part
Mediastinal part
Diaphragmatic part

Pleural recesses

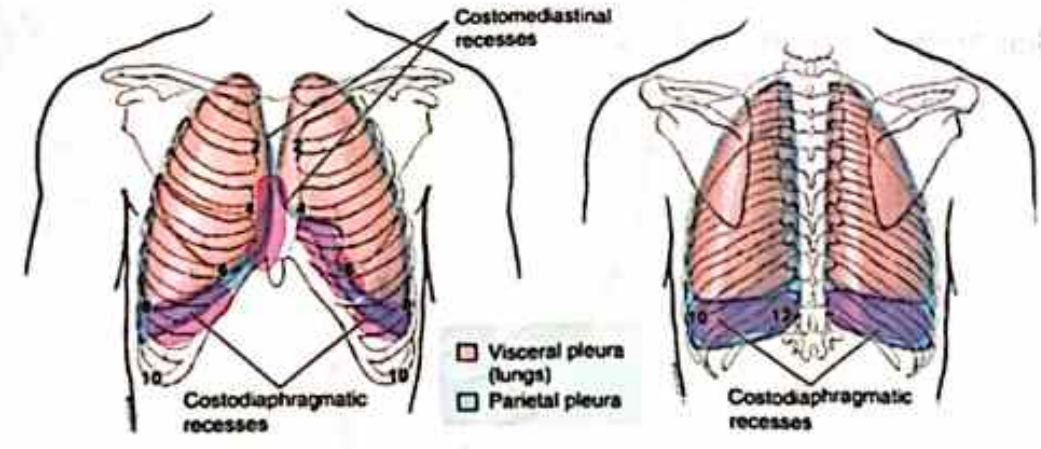
- **The lungs do not completely fill the anterior or posterior inferior regions of the pleural cavities.**
- **This results in recesses in which two layers of parietal pleura become opposed.**
- **Expansion of the lungs into these spaces usually occurs onfy during forced inspiration; the recesses also provide potential spaces in which fluids can collect and from which fluids can be aspirated.**

1. Costomediastinal recesses:

- Anteriorly, a costomediastinal recess occurs on each side where costal pleura is opposed to mediastinal pleura.
- The largest is on the left side in the region overlying the heart.

2. Costodiaphragmatic recess:

- The largest and clinically most important recesses are the costodiaphragmatic recesses, which occur in each pleural cavity be tween the costal pleura and diaphragmatic pleura.
- The costodiaphragmatic recesses are the regions between the inferior margin of the lungs and inferior margin of the pleural cavities.
- They are deepest after forced expiration and shallowest after forced inspiration.



(B) Anterior view

(C) Posterior view

Supply Of The Pleura

Nerve supply

The parietal pleura

- is supplied by somatic nerves and therefore it is sensitive to pain, temperature, touch, and pressure.

The costal & peripheral parts of diaphragmatic pleurae

- are supplied by intercostal nerves.

The mediastinal & central part of diaphragmatic pleurae

- is supplied by the phrenic nerve.

The visceral pleura

- is supplied by autonomic nerves derived from the anterior and posterior pulmonary plexuses and is sensitive to stretch.

Blood Supply

The parietal pleura

Is supplied by the vessels of the thoracic wall;

- intercostal, internal thoracic and musculophrenic vessels.

The visceral pleura

- is supplied by bronchial arteries and drained by bronchial veins.

Lymphatic Supply

The parietal pleura

- Lympahics join those in the thoracic wall draining into intercostal, parasternal, posterior mediastinal and diaphragmatic lymph nodes.

The visceral pleura

- Lympahics join those of the lung (bronchopulmonary lymph nodes).

Applied Anatomy

Inflammation of the parietal pleura (pleurisy) produces severe pain which is referred to:

Thoracic and abdominal wall

- if the costal and peripheral parts of the diaphragmatic pleurae are inflammed.

Lower neck and shoulder tip

- if the mediastinal and central part of the diaphragmatic pleurae are inflammed.

In certain diseases e.g. pulmonary tuberculosis (T.B) or chest injuries, the pleural cavity may be filled with;

Air

- pneumothorax.

Pus

- empyema.

Air and pus

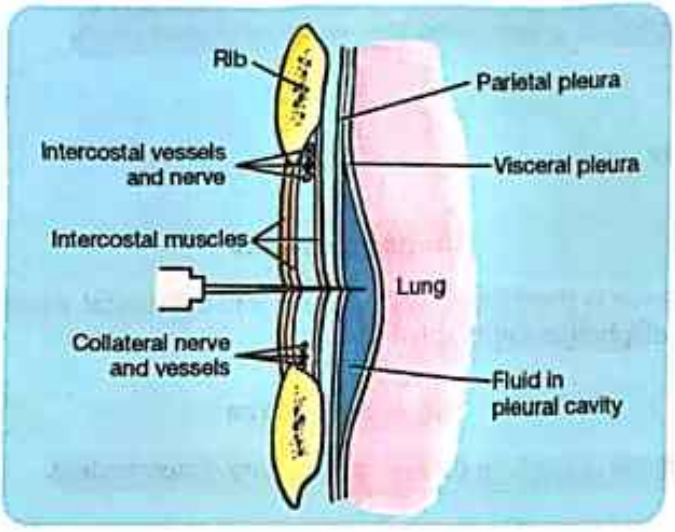
- pyopneumothorax.

Blood

- haemothorax.

Fluid (serous exudates)

- hydrothorax (pleural effusion).



The Lung

- **The lungs are the vital organs of respiration.**
- **They are situated on either side of the heart and other mediastinal contents.**
- **In the child, they are pink, but with age, they become dark and mottled because of the inhalation of dust particles that become trapped in the phagocytes of the lung. This is especially well seen in city dwellers and coal miners as well as smokers**

Shape

- Each lung resembles nearly half a cone having an apex, base, 3 borders and 2 surfaces.
- Each lung is divided by fissure(s) into 2 or 3 lobes

Apex

- **It is the rounded upper extremity that projects above the thoracic inlet into the root of the neck.**

Relations

Anterosuperiorly:

- It is related to supraplеural membrane outside the cervical pleura.
- The subclavian artery arches upand laterally grooving the anterior surface of the apex near its summit.

Posteriorly:

- It is related to the structures lying in front of the neck of the 1st rib

Arranged from medial to lateral these structures are:

- Stellate ganglion, superior intercostal vesseland ventral ramus of the 1st thoracic spinal nerve.

Base (diaphragmatic or inferior surface of lung)

It is concave resting upon the superior surface of the diaphragm.

- The diaphragm separates the right lung from the right lobe of the liver and the left lung from the left lobe of the liver, fundus of stomach and spleen.
- Due to presence of the liver, the diaphragm extends higher on the right than on the left and the concavity is deeper on the base of the right lung.

Borders of the lung

3 borders: anterior posterior & inferior

1- Anterior border:

It is a thin and sharp border that lies anteriorly between the costal and medial surfaces of the lung.

On the right:

- lies close to costomediastinal line of pleural reflection being almost vertical.

On the left:

- It is vertical down to the **4th costal cartilage.**
- Then it presents a deep notch laterally opposite the 4th and 5th intercostal spaces **(cardiac notch)** which leaves a part of the pericardium not covered by the lung **(bare area of pericardium)** and becomes directly related to the thoracic wall

The anterior border forms a tongue-shaped process below the cardiac notch known as the lingula.

2- Posterior border:

It is a thick and rounded border that lies posteriorly between the costal and medial surfaces of the lung.

It is related to

- the heads of the ribs and occupies the paravertebral gutter.

3- Inferior border:

It is thin and sharp anteriorly, laterally and posteriorly where it separates the base from the costal surface and extends into the costodiaphragmatic pleural recess.

- It is **rounded medially** where it separetes the base from the medial surface.

Surfaces of the lung

The lung has two surfaces, costal and mediastinal.

1. The Costal Surface

it is smooth and convex. It is related to the costal pleura which separates it from the ribs, costar cartilages and intercostal spaces

1. The Medial Surface

It is subdivided into 2 parts; posterior vertebral and antenor mediastinal parts.

- **The hilum of the lung** is a depression in the mediastinal part, through which the bronchi vessels, and nerves that form the root of the lung enter and leave the lung.

i. The mediastinal part (mediastinal surface)

- is deeply concave due to it relation to the heart **(cardiac impression).**
- Other structures in the mediastinum also produce impressions on the preserved lung.

ii. The vertebral part (vertebral surface)

- lies in contact with the sides of th thoracic vertebrae and intervertebral discs, posterior intercostal vessels, splanchnic nerves and sympathetic chain.

Lobes and fissures of the lung

The right lung

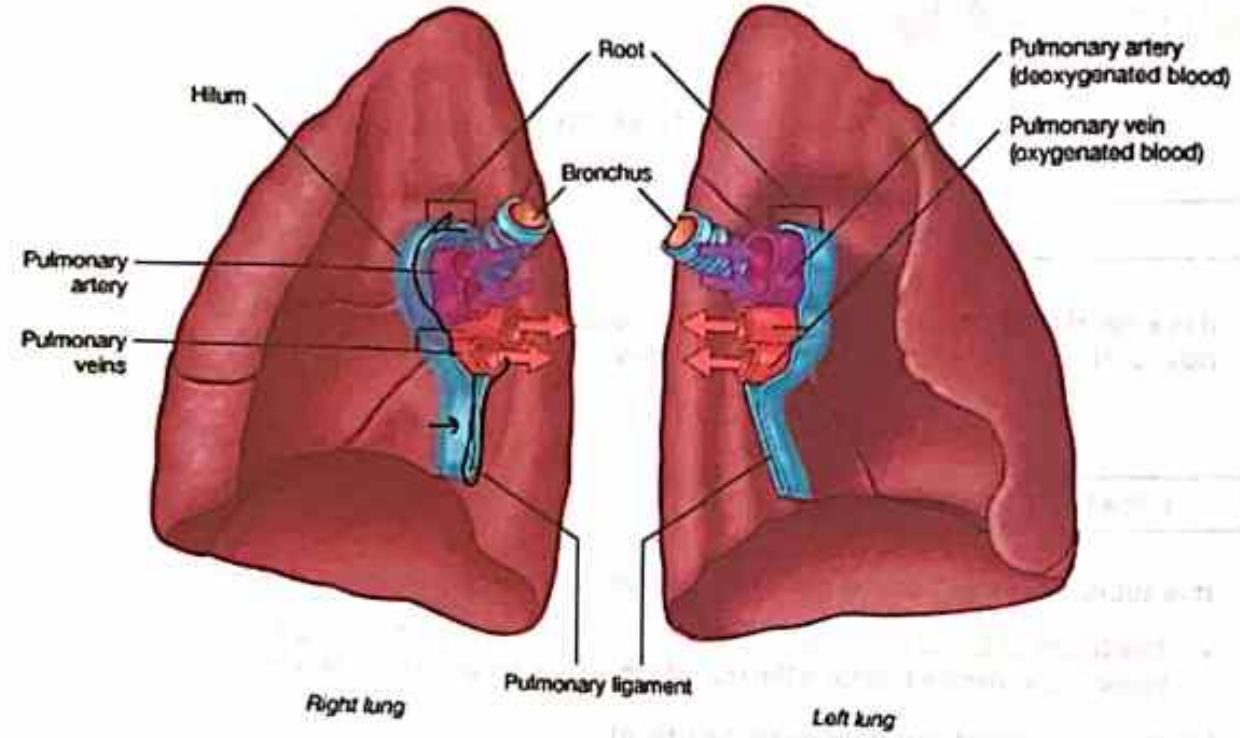
- is divided into 3 lobes (superior, middle and inferior) by 2 fissures, the oblique and horizontal (transverse) fissures.

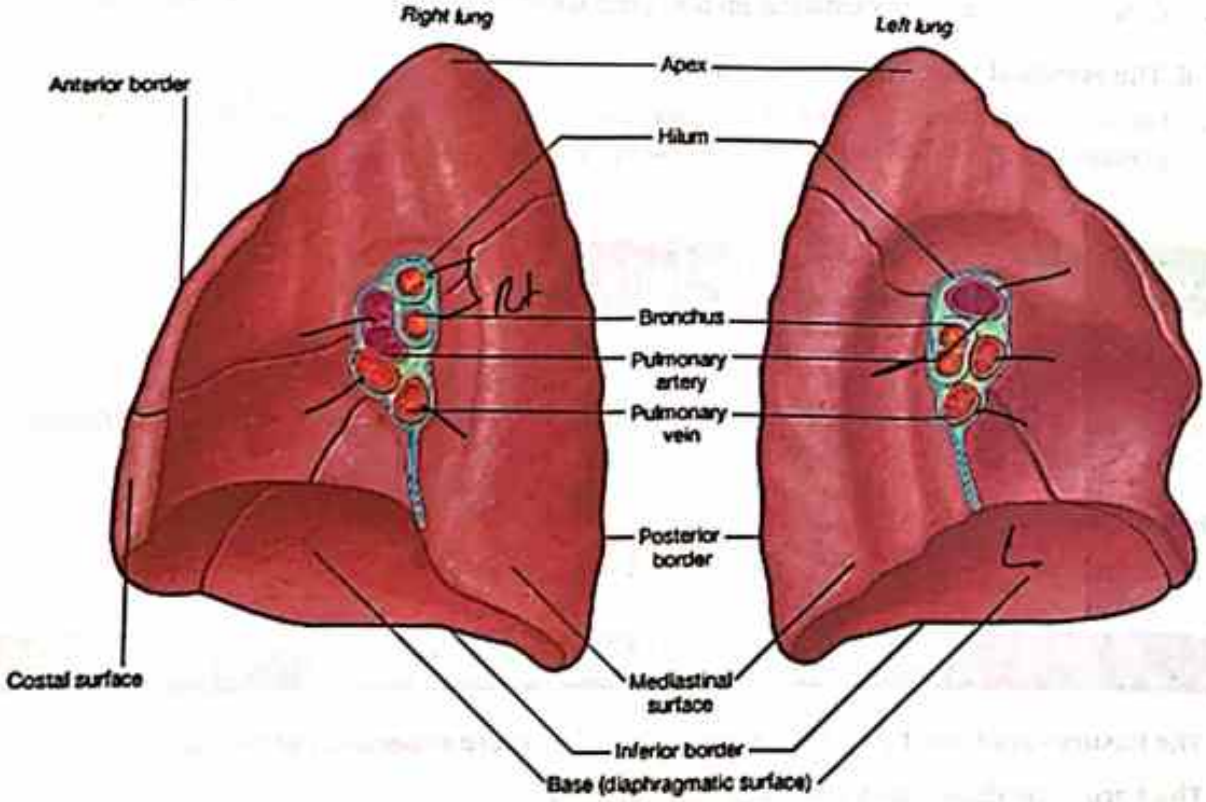
The left lung

- is divided into 2 lobes (superior and inferior) by the oblique fissure.

Note:

- **The fissures are lined by visceral pleura and allow more expansion of the lung.**
- **The horizontal fissure may be congenitally absent.**
- **Lung fissures may be partly or completely obliterated by inflammatory adhesions.**


Root
Hilum
Pulmonary artery (deoxygenated blood)
Pulmonary vein (oxygenated blood)
Bronchus
Pulmonary artery
Pulmonary veins
Pulmonary ligament
Right lung
Left lung


Right lung
Left lung
Anterior border
Apex
Hilum
Bronchus
Pulmonary artery
Pulmonary vein
Posterior border
Costal surface
Mediastinal surface
Inferior border
Base (diaphragmatic surface)

Relations of the mediastinal surface of the lung

- **As mentioned earlier, the mediastinal surface of the preserved lung presents grooves and impressions due to the pressure exerted by the mediastinal structures.**
- **These grooves and impressions differ in both lungs:**

1- Right Lung

In front of the hilum:

Pericardial impression:

- it is a wide concave area in front of the hilum.
- It is related to the **pericardium** enclosing the **right atrium** and part of **right ventricle.**

Inferior vena cava:

- it produces a short depression at the lower and posterior part of the pericardial area.

Superior vena cava:

- it is related to a **vertical groove** extending from upper part of the pericardial area just in front of the hilum.
- This groove is continuous superiorly with a narrower groove formed by the **right brachiocephalic vein.**

Area related to ascending aorta and remains of thymus gland:

- this is a small area located in front of the groove for **superior vena cava.**

The right phrenic nerve:

- Descends in the groove for **superior vena cava, pericardial area** and **groove for inferior vena cava.**

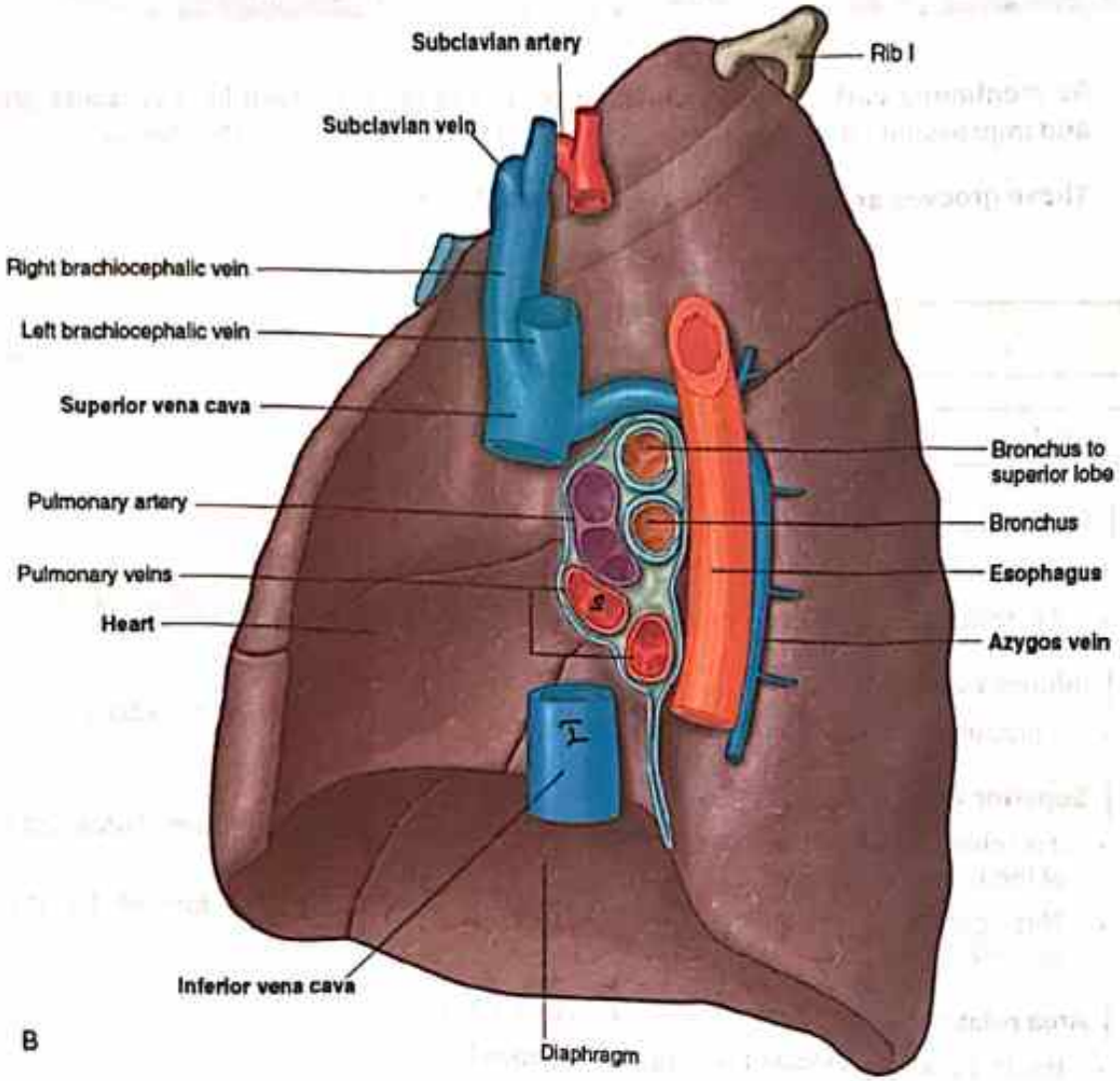
Above the hilum:

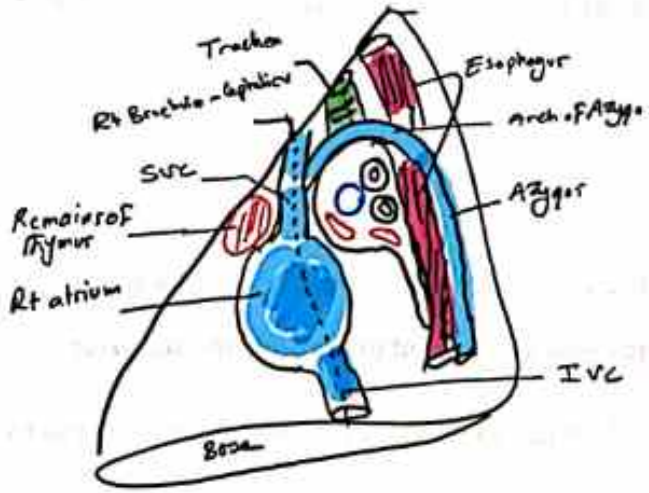
The arch of the azygos vein:

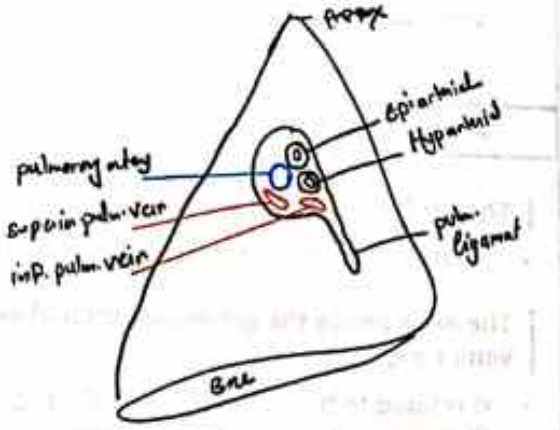
- forms a deep groove above the hilum that **joins the groove for the superior vena cava.**

The area above the groove for arch of azygos vein and behind the groove for superior vena cava

- is related to the **trachea and right vagus** (which lies on the right side of the trachea) and to **the esophagus behind the trachea.**


Subclavian artery
Rib I
Subclavian vein
Right brachiocephalic vein
Left brachiocephalic vein
Superior vena cava
Bronchus to superior lobe
Pulmonary artery
Bronchus
Pulmonary veins
Esophagus
Heart
Azygos vein
Inferior vena cava
Diaphragm
B


Trachea
Esophagus
Rt Bronchus - Epithelium
arch of Azygos
SVC
Azygos
Remains of thymus
Rt atrium
IVC
Base


Apex
Epiarterial
Hyparterial
pulmonary artery
Superior pulm. vein
Inf. pulm. vein
pulm. ligament
Base

2- Left Lung:

In front of the hilum:

Pericardial impression:

- it is a wide concave area in front of the hilum.
- It is deeper than that of the right lung as the heart and pericardium bulge more to the left side than to the right.
- It is related to the pericardium enclosing the left ventricle, left auricle and infundibulum of right ventricle.

Behind the hilum:

Descending aorta:

- it is related to a deep vertical narrow groove behind the hilum.

The esophagus:

- is related to a short groove in front of the lower part of the groove for the descending aorta.

Above the hilum:

The arch of the aorta

- Produces a wide deep groove above the hilum which becomes continuous posteriorly with the groove for the descending aorta.
- The floor of the groove is also related to the structures on the left side of the arch; **left vagus nerve, left superior intercostal vein** and **left phrenic nerve.**

Two vertical grooves

extend upwards from the groove for the aortic arch;

- **the anterior one** is related to **left common carotid artery,**
- **while the posterior one,** which is sharply defined, is related to the **left subclavian artery.**

The esophagus

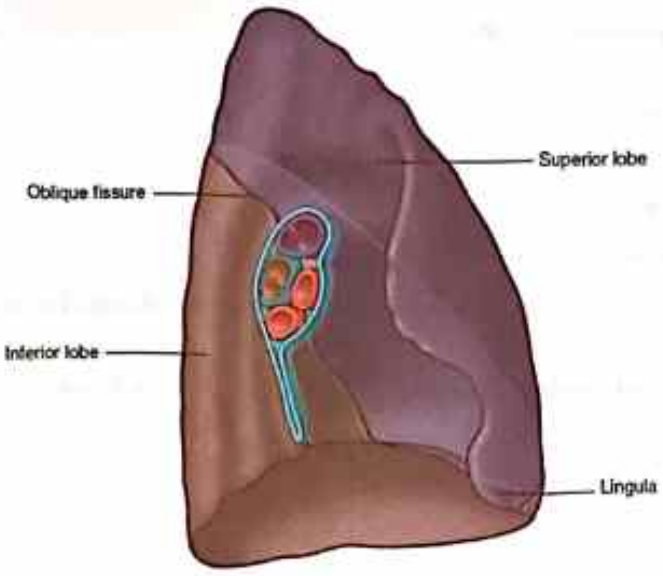
- The area above the groove for the arch of aorta and behind that for subclavian artery is related to the esophagus, together with the thoracic duct on its left side and **left recurrent laryngeal** nerve in front of it.

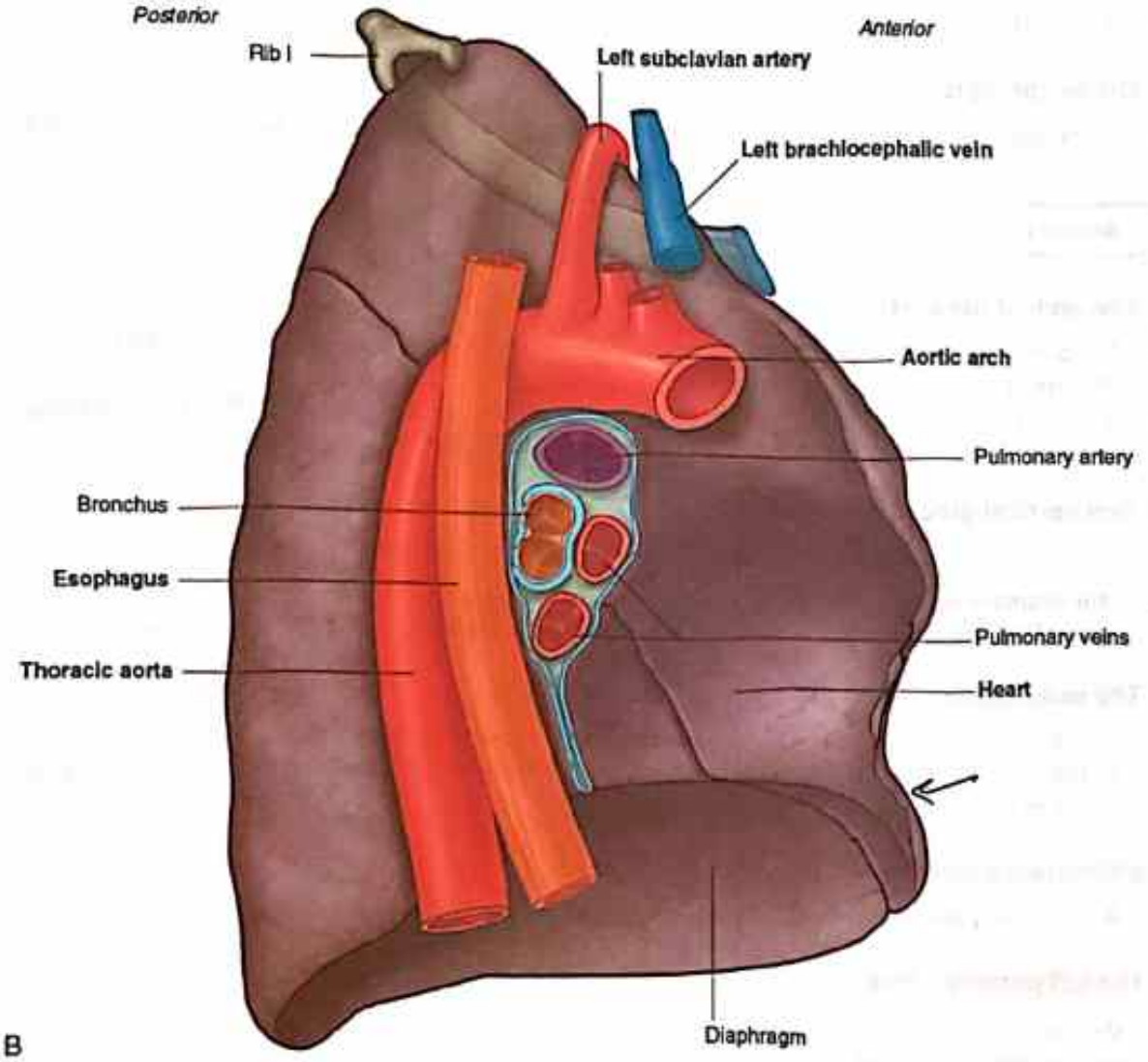
pulmonary trunk and remains of thymus gland.

- A small area located in front of the aortic arch groove and above the pericardial impression.

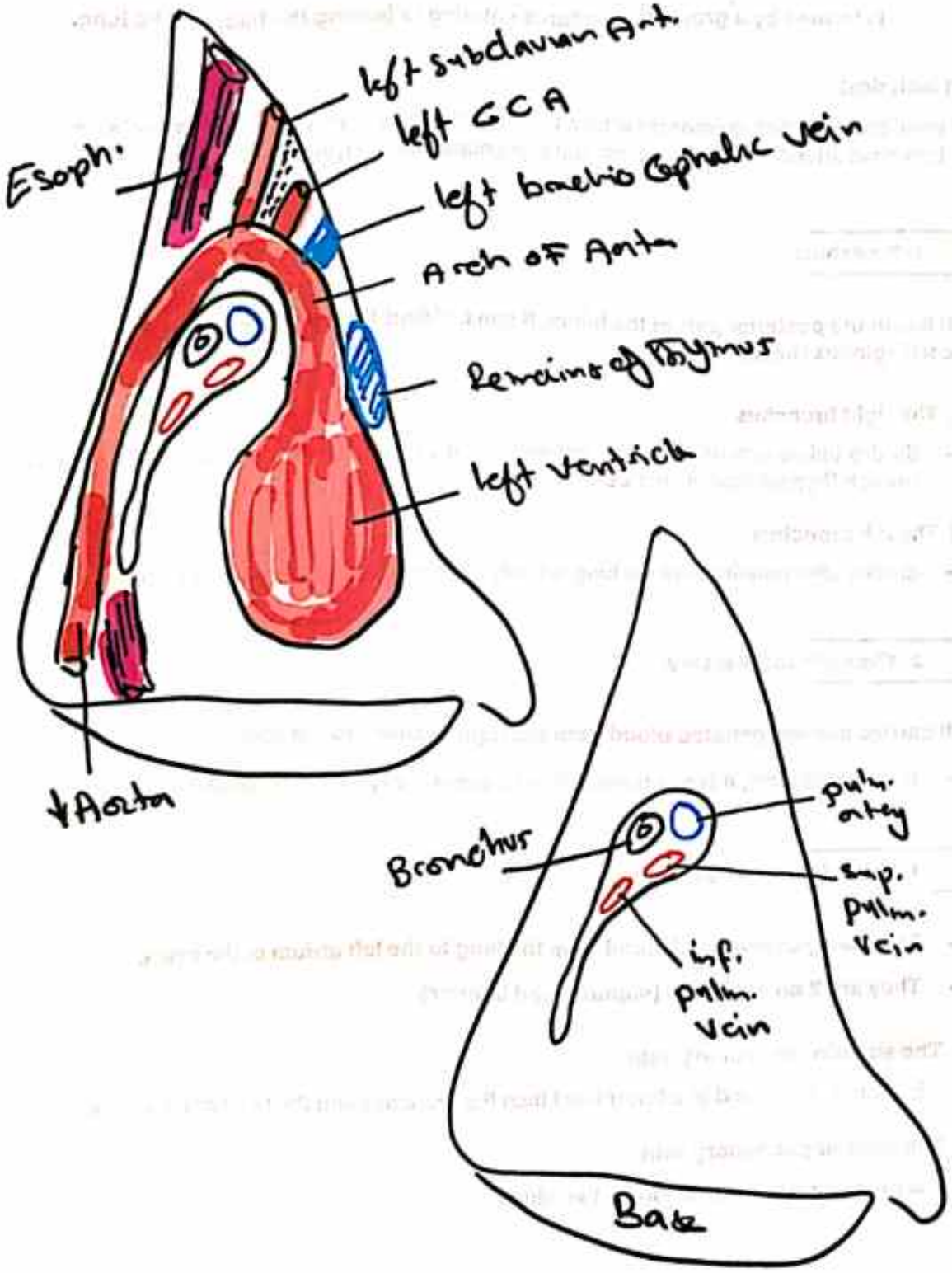
The left phrenic nerve

- descends in relation to the groove for left common carotid, the groove for aortic arch and the pericardial impression.


Superior lobe
Oblique fissure
Interior lobe
Lingula


Posterior
Anterior
Rib I
Left subclavian artery
Left brachiocephalic vein
Aortic arch
Pulmonary artery
Bronchus
Esophagus
Pulmonary veins
Thoracic aorta
Heart
Diaphragm

B


left subclavian Art.
left CCA
Esoph.
left brachiocephalic vein
Arch of Aorta
Remains of Thymus
left ventricle
Aorta
pulm. artery
Bronchus
sup. pulm. vein
inf. pulm. vein
Base

Root of the lung:

Is formed by a group of structures entering or leaving the hilum of the lung.

It includes;

- principal bronchus, pulmonary artery, pulmonary veins, pulmonary auto nomic plexus, bronchial arteries, bronchial veins, and bronchopulmonary lymph nodes.

1- Bronchus:

It lies in the posterior part of the hilum. It can be identified by the presence of cartilaginous rings in its wall.

The right bronchus

- divides before entering into the lung into upper branch **(eparterial bronchus)** and lower branch **(hyparterial bronchus).**

The left bronchus

- divides after entering into the lung, so only a **single bronchus** is present in the hilum.

2- The pulmonary artery:

It carries non-oxygenated blood from the right ventricle to the lung.

- **In the right lung,** it lies anterior and in between the eparterial and hyparterial bronchi.

3- The pulmonary veins:

- **They carry oxygenated blood from the lung to the left atrium of the heart.**
- **They are 2 on each side (superior and inferior).**

The superior pulmonary vein

- is more anterior and at a lower level than the bronchus and the pulmonary art ery.

The inferior pulmonary vein

- is the most inferior structure in the hilum.

4- Bronchial vessels:

They supply oxygenated blood to the bronchial wall, lymph nodes and stroma of the lung and visceral pleura.

- They are slender vessels and lie on the posterior aspect of the bronchus and its branches.

The bronchial arteries

- **Two arteries in the left lung** (branches from the **descending aorta**)
- **One artery in the right lung** (branch of **left superior bronchial artery** or the **posterior intercostal artery** of the **3rd right** intercostal space)

The bronchial veins

- **2 veins on each side** that join the **azygos vein** (right lung) and the **hemiazygos** or **left superior intercostal vein** (left lung).
- They carry blood from the walls of larger bronchi.

5- Pulmonary autonomic plexuses:

The anterior and posterior pulmonary plexuses lie in the anterior and posterior parts of the hilum.

They are formed of

sympathetic fibers

- From **2nd - 5th sympathetic ganglia**

parasympathetic fibers

- **From the vagus**

They supply the bronchial tree, lung substance and visceral pleura.

6- The bronchopulmonary lymph nodes:

- They lie between the main structures of root.

7- Pleural covering:

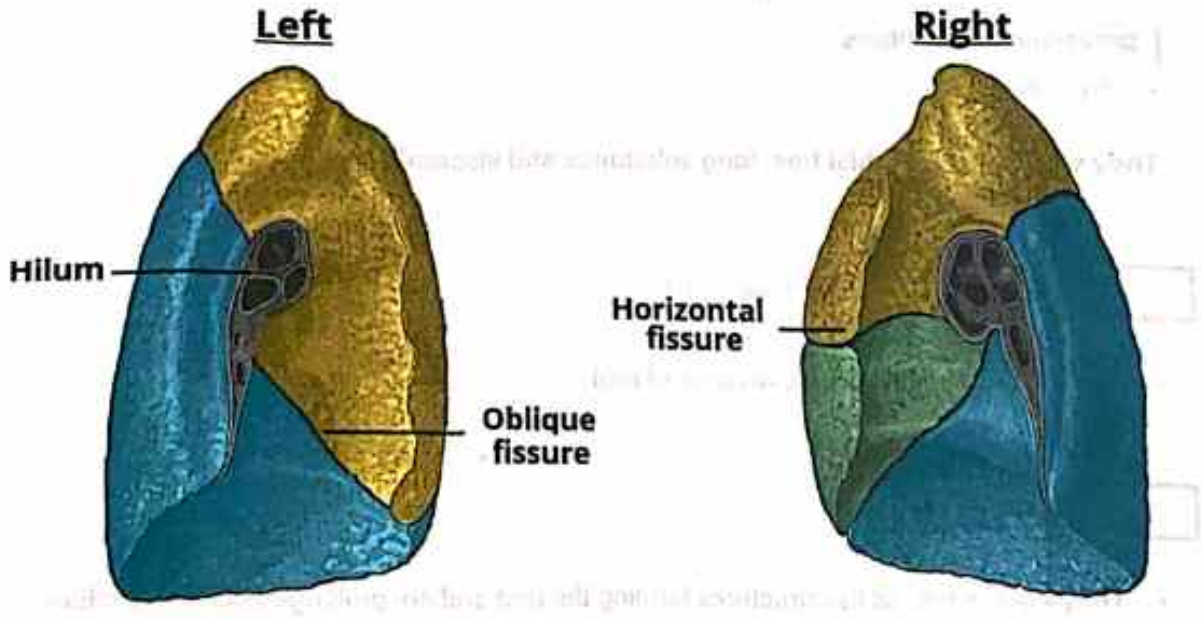
- The pleura invests all the structures forming the root and are prolonged below the hilum as the **pulmonary ligament.**

How to identify the side of a given lung?

Hold the lung so that

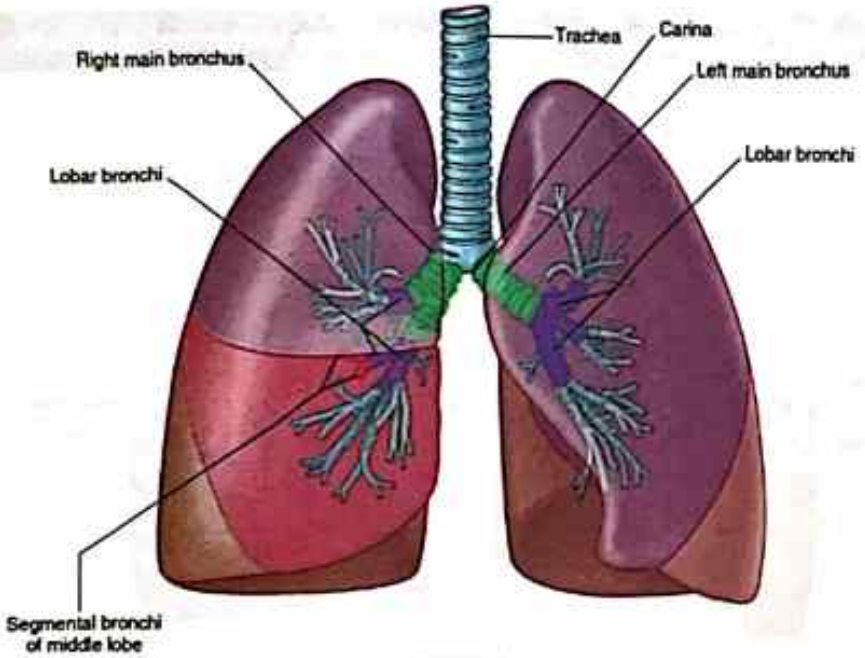
- **The apex is directed upwards.**
- **The hilum is directed medially.**
- **The thin sharp anterior border is directed anteriorly.**

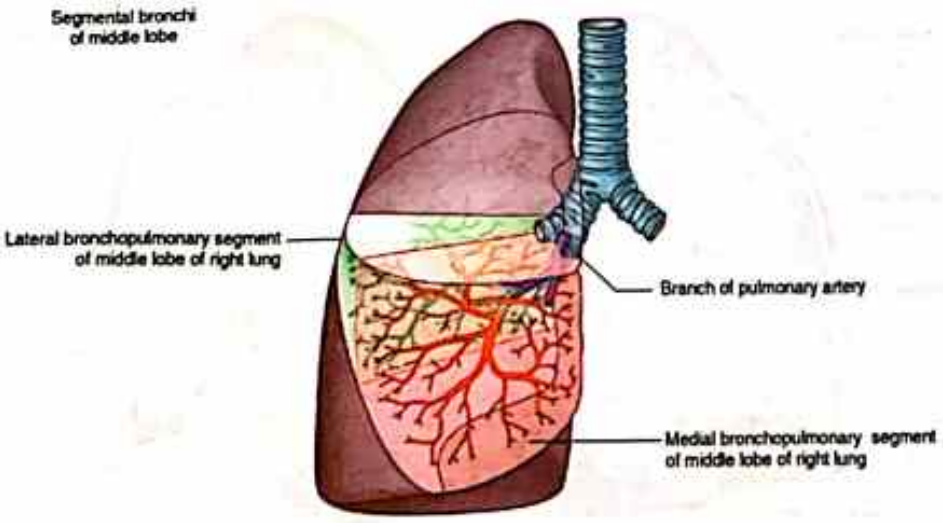
	Right Lung	Left Lung
Size	Larger	Smaller
Length	Shorter	Longer
Width	Longer	Narrower
Anterior border	Straight	Presents cardiac notch and lingula
Lobes	Three lobes	Two lobes.
Hilum	Contains two bronchi; eparterial and hyparterial	Contains only one bronchus



Bronchial tree:

- The trachea bifurcates into right and left principal (primary) bronchi.
- The Principal bronchus divides into a superior and an inferior lobar (secondary) bronchus.
- Each lobar bronchus passes to a lobe of the lung and gives off branches called segmental (tertiary) bronchi.
- The segmental bronchus passes to a structurally and functionally independent unit of a lung lobe called a bronchopulmonary segment.





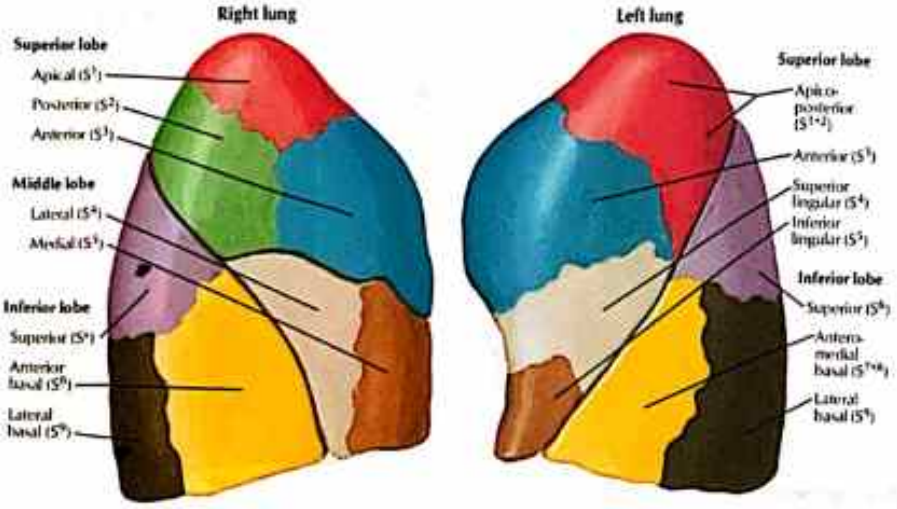
Broncho Pulmonary segments:

The anatomic, functional, and surgical unit of the lung (i.e . is a subdivision of a lung lobe).

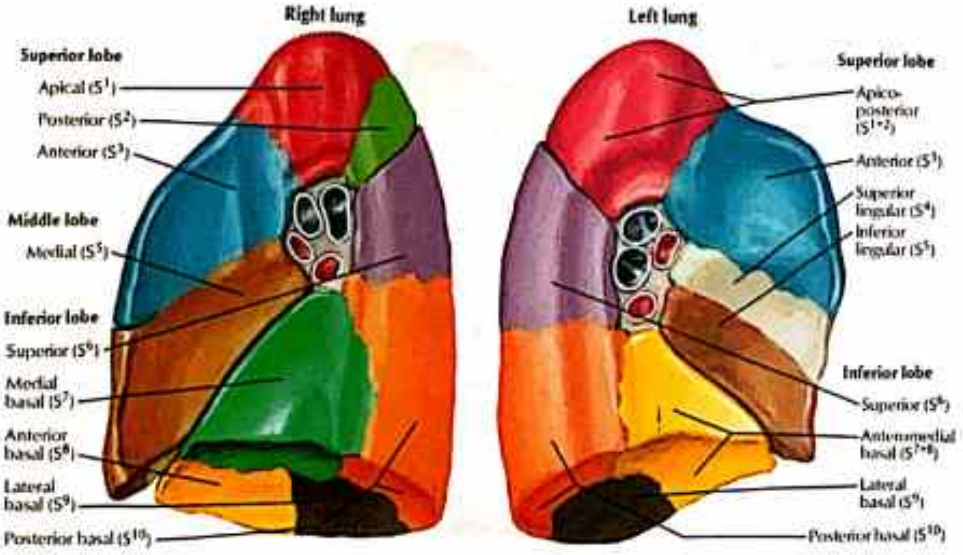
- It is pyramid shaped, its apex is toward the lung root and the base is directed towards the surface of the lung.
- It has a segmental bronchus, a segmental artery (a branch of the pulmonary artery carrying deoxygenated blood), lymph vessels, and autonomic nerves
- It is surrounded by connective tissue.
- Veins (carrying oxygenated blood) lie in the connective tissue betwe en adjace bronchopulmonary segments (intersegmental).

- Note; Because it is a structural unit a diseased segment can be removed surgically.

Lateral views



Medial views



Supply Of The lung

Nerve supply

The lungs and the visceral pleura are innervated by the autonomic anterior and posterior pulmonary plexuses (see before).

The parasympathetic fibers

- **Motor** to the smooth muscles of bronchi (**bronchoconstrictor**)
- **Secretory** to the mucous glands of the bronchial tree.

The sympathetic fibers are:

- Bronchodilators and inhibitory to the mucous gland of the bronchial tree.

Afferent fibers mediating sensory impulses from the bronchial mucous membrane and from stretch receptors in the alveolar walls travel in both sympathetic and parasympathetic components of the pulmonary plexuses to reach the central nervous system.

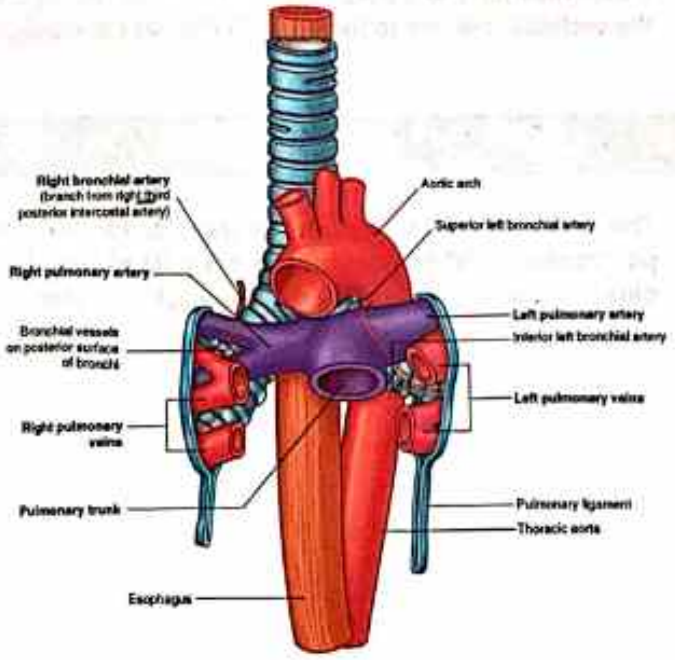
Blood Supply

The bronchi, the connective tissue of the lung, and the viscral pleura

- are supplied by **bronchial arteries** and drained by **bronchial veins.**

Lymphatic Supply

- All lymph vessels from lung drain into the **bronchopulmonary lymph nodes.**



Surface Anatomy Of Pleura

1- The cervical pleura:

- is represented by a curved line from the **sterno-clavicular joint** to the **junction of** the medial and middle thirds of the clavicle.
- Its summit is **2.5 cm** above the medial third of the clavicle.

2- The anterior margin of the pleura:

- It extends from the **summit of cervical pleura** to a point behind the **sterno-clavicular joint** then downwards and medially to the midline behind the sternal angle.

From the sternal angle, the right and left anterior margins of the pleura descend in contact with each other to the level of the 4th costal cartilage below which they differ:

On the right side:

- The anterior margin of the pleura descends to the **back of the xiphisternal joint.**

On the left side:

- The anterior margin of the pleura deviates laterally and descends at a distance **2-2.5 cm** from left margin of the sternum to the **6th costal cartilage.**

3- The inferior margin of the pleura:

- It begins behind the **6th costal cartilage**, then descends inferolaterally to cross the **8th rib** in the midclavicular line, the **10th rib** in the midaxillary line and ascends to cross the **12th rib** to end **2 cm lateral to 12th thoracic spine.**

4- The posterior margin of the pleura:

- It extends from the medial end of the **inferior margin** (2 cm from the T12 spine) upwards along the **vertebral column** to the spine of the **7th cervical vertebra.**

Note:

- **The deviation of the anterior margin of the left pleura results in direct contact of the pericardium and heart with the left part of the sternum and parts of 5th and 6th costal cartilages with no pleural covering that facilitates the puncture of the pericardial cavity.**

Surface anatomy of the lungs

1- The apex of the lung

- lies 3-4 cm above the level of the 1st costal Cartilage.
- Its summit is about 1.5 cm above the medial 1/3 of the clavicle.

2- The anterior border:

- Extends from the apex to the sterno-clavicular joint then downward and medially to the midline behind the sternal angle (like that of the pleura).

From the sternal angle the right and left borders of the lung descend vertically close to the median plane to the level of the 4th costal cartilage below which they differ.

On the right:

- the anterior border of the lung continues descending vertically down to the 6th costal cartilage.

On the left:

- The anterior border of the lung passes laterally for 3.5 cm and inferiorly to the 6th costal cartilage presenting the cardiac notch.

3- The inferior border:

- The inferior border of the lung is 2 ribs higher than that of the pleura.
- It crosses the 6th rib in the midclavicular line, the 8th rib in the midaxillary line
- and crosses the 10th rib to end 2 cm lateral to the 10th thoracic spine.

4- The posterior border:

- It extends from the medial end of the inferior border (T10 spine) upwards along the vertebral column to the spine of 7th cervical vertebra.

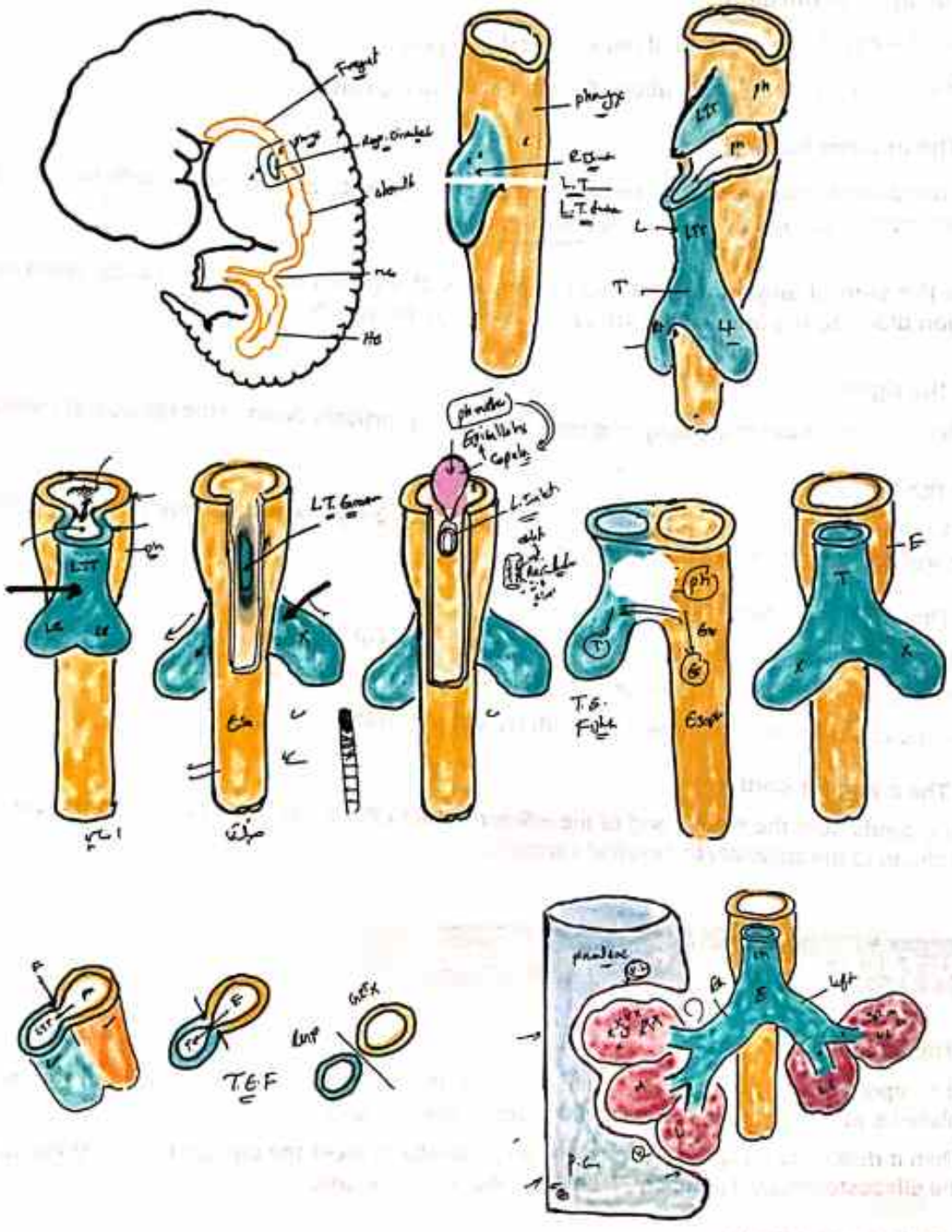
Surface anatomy of the fissures

1- The oblique fissure:

- It is represented by a line that begins opposite the 3rd thoracic spine 2 cm from the median plane i.e. at the level of the medial end of the scapular spine.
- Then it descends downwards, forwards and laterally to meet the inferior border of the lung at the 6th costochondral junction 7.5 cm from the median plane.

2- The horizontal fissure:

- It begins at the anterior border of the right lung opposite the lower border of the 4th costal cartilage and extends horizontally and laterally to join the oblique fissure at the midaxillary line.


T.E.F

3. Development of the laryngotracheal tube & lung buds

It begins to develop around the middle of the 4th week

Development:

- The **laryngeo-tracheal groove** appears as a median longitudinal groove in the floor (i.e., the ventral wall) of the **primitive pharynx** (the pharynx is he cranial most part of the foregut).
- The 2 margins of the laryngeo-tracheal groove (called the tracheaesophageal folds or ridges) fuse together in a caudo-cranial direction forming the tracheo-esophageal septum which separates the lumen of the primitive pharynx into **2 parts:**

a. Dorsal part

- will give the pharynx & esophagus.

b. Ventral part

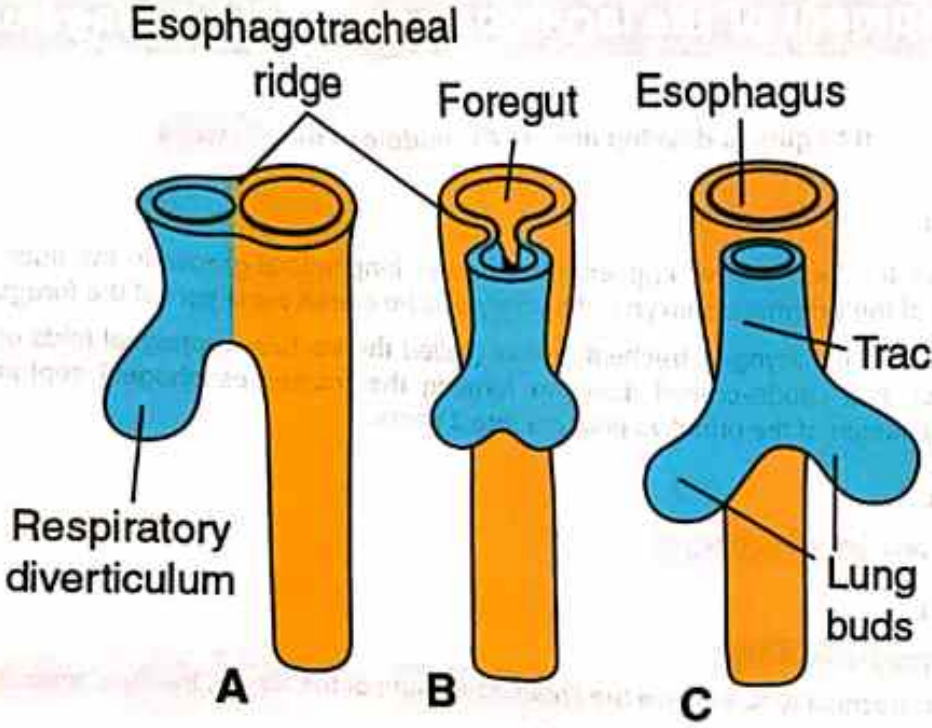
- called laryngeo-tracheal tube.
- The laryngeo-tracheal tube will give the lining epithelium of the larynx, trachea, bronchial tree and alveoli.

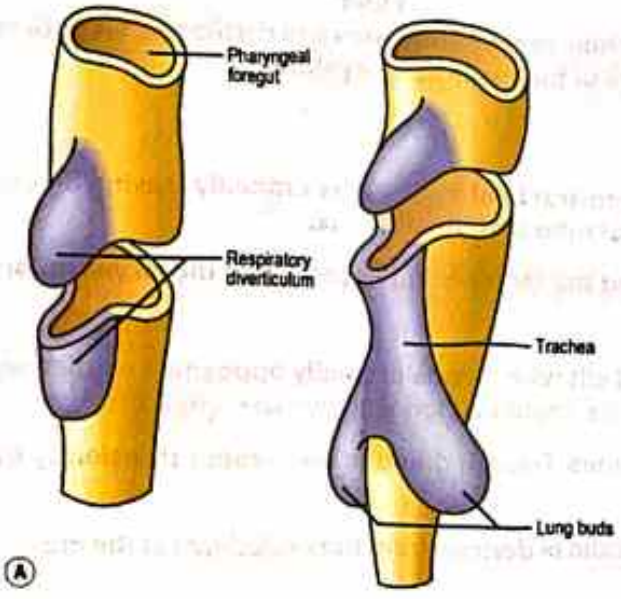
The surrounding splanchnic mesoderm forms the cartilages, smooth muscles connective tissue and blood vessels of the respiratory system.

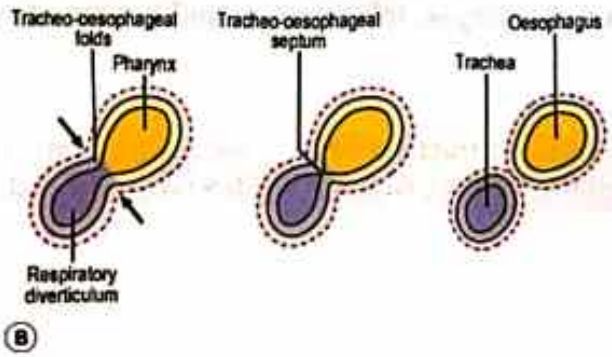
- **Fusion of the laryngeo-tracheal folds stops cranially leaving a communication between the laryngeo-tracheal tube and the pharynx.**
- **This opening is called the laryngeal inlet (aditus to the larynx or laryngeal orifice).**

- **The inlet is a vertical slit which ends cranially opposite a median elevation in the floor of the pharynx, called the copula which will give the epiglottis.**
- **Later, the inlet becomes T-shaped and is obliterated transiently between the 8th - 10th weeks.**
- **The laryngeal epithelium is derived from the endoderm of the cranial part of the laryngeo-tracheal tube.**
- **During recanalization of the larynx, folds appear and become the vocal cords (true & false).**

- **The caudal end of the laryngeo-tracheal tube grows down into the splanchnic mesoderm on the ventral aspect of the foregut, where it divides into right and left lung buds.**


Esophagotracheal ridge
Foregut
Esophagus
Trachea
Respiratory diverticulum
Lung buds
A
B
C


Pharyngeal foregut
Respiratory diverticulum
Trachea
Lung buds
A


Tracheo-oesophageal folds
Pharynx
Respiratory diverticulum
Tracheo-oesophageal septum
Oesophagus
Trachea
B

The lung buds undergo the Following

A- Division:

- Each lung bud forms a main bronchus. The right bud is wider and more vertical than the left. Each main bronchus divides into secondary (lobar) bronchi (3 on the right side & 2 on the left side).
- The 2ry. bronchi divide giving tertiary (segmental) bronchi; each becomes surrounded by a mass of splanchnic mesoderm forming a broncho-pulmonary segment.
- Repeated dichotomous division by the 6th Month will give 17 orders of branches ending in terminal bronchioles.
- Further division continues for some time after birth (up to 8 years) till the respiratory bronchioles and alveoli are formed after 7 additional orders of divisions (i.e., total of 24 orders).

B- The surrounding splanchnic mesoderm

- will give the cartilaginous plates, smooth muscles, connective tissue & blood capillaries (bronchial & pulmonary).

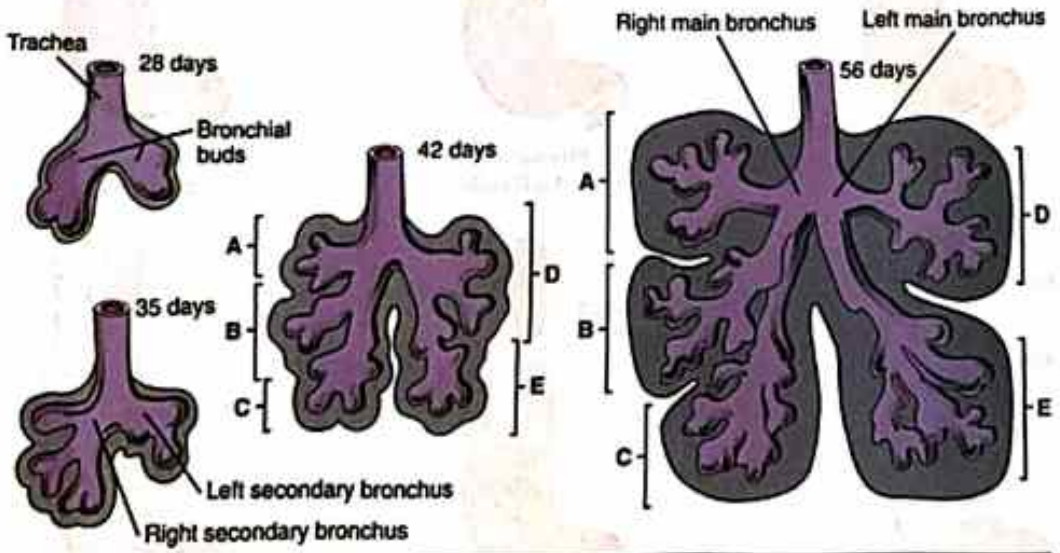
C- Invasion of the pleura:

- Each lung grows laterally around the foregut and invaginates the part of the intraembryonic coelom, (i.e. the pericardio-peritoneal canal) which gives the pleural sac.

The visceral pleura is derived from the splanchnic mesoderm

The parietal pleura is derived from the somatic mesoderm.

Therefore, the visceral pleura is supplied by autonomic nerevs and is insensitive whereas the parietal pleura is supplied by somatic nerves and is sensitive to pain.



A. Right upper (superior) lobe
B. Right middle lobe
C. Right lower (inferior) lobe
D. Left upper (superior) lobe
E. Left lower (inferior) lobe

Anomalies

(1) Esophageal atresia & tracheo-esophageal fistula:

Atresia = obliteration & fistula = abnormal communication.

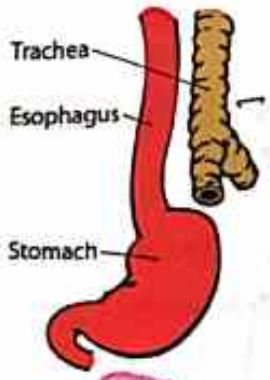
- The tracheaesophageal fistula results from incomplete fusion of the trachea-esophageal folds.
- Four varieties may occur, the most common is atresia of the upper portion of esophagus & fistula of its lower portion with the trachea
- impossible feeding & aspiration of milk into the lungs. (Before birth, it causes polyhydramnios).

(2) Surfactant deficiency

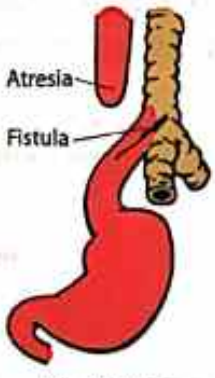
- hyaline membra ne disease (HMD) which is the major cause of respiratory distress syndrome (RDS).
- Thyroxin and cortisone increase surfactant production.

(3) Other lung problems

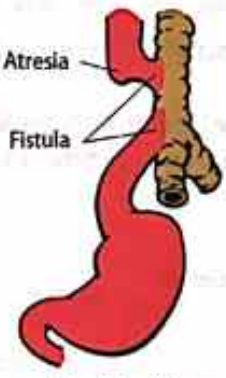
- include agenesis, hypoplesia (small lung), variation in lobe number, accessory lung, ectopic lung lobes (from the trachea or esophagus) and congenital lung cyst.



Normal Anatomy



Atresia with distal Fistula



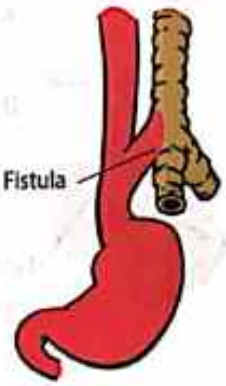
Atresia with double Fistula



Atresia with proximal Fistula

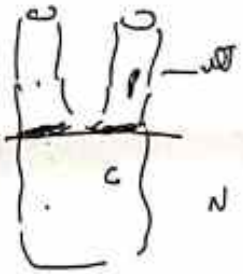


Atresia



Fistula

4. Development of the diaphragm



It develops from 4 embryonic sources:

- 1. Septum transversum.
- 2. Pleuroperitoneal membranes.
- 3. Dorsal mesentery of the esophagus.
- 4. Lateral body walls

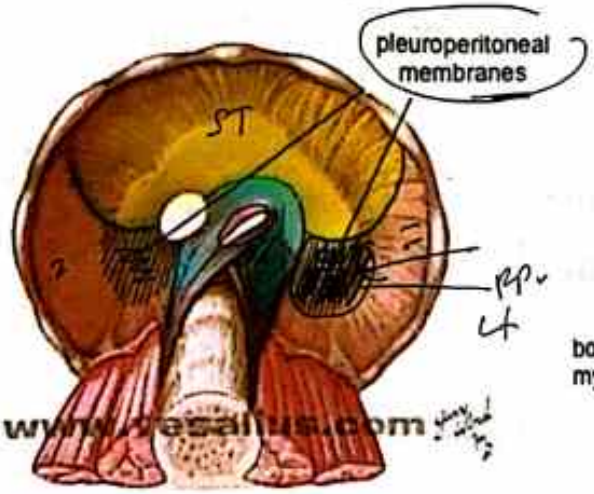
After folding of the embryo, the septum transversum grows dorsally & separates the heart from the liver.

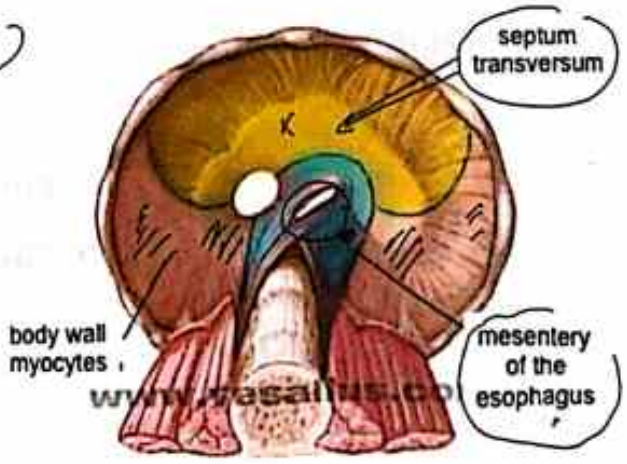
- A large part of the liver is embedded in the **septum transversum** during its early development.
- The **septum transversum** is located caudal to the pericardial cavity & separates it from the peritoneal cavity.
- The **septum transversum** expands & fuses with the mesenchyme ventral to the esophagus and the pleuroperitoneal membranes.



Positional changes and innervation of the diaphragm:

- During 4th week of development, the septum transversum lies opposite 3rd - 5th cervical somites.
- During 5th week, the myoblasts from these somites migrate into the developing diaphragm, bringing their nerve fibers with them.
- Consequently, the phrenic nerves that supply motor innervation to the diaphragm arise from the ventral rami of 3rd, 4th, & 5th cervical spinal nerves.





Anomalies

Congenital diaphragmatic hernia (CDH):

- A posterolateral defect of the diaphragm: Herniation of the abdominal contents into the thoracic cavity.
- This condition is usually unilateral & results from failure of formation or fusion of pleura-peritoneal membrane with the other 3 parts of the diaphragm.
- The defect is more common on the left side (due to earlier closure of the right pleura-peritoneal opening).

Pockdalic HERNIA

